MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATIO REPORT FOR DAF BASIN, COAGULANT BASIN, PIT, SLUDGE BASIN, VESSEL FOUNDATION**
- 49W1 Raw Water Treatment Plant

Document Number: **MD-216-49W1-EG-CV-CAL-1006**
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1. INTENT

this calculation report illustrates the design process of DAF basin, coagulant basin, pit, sludge basin and vessel foundation in Ma'aden Mansourah Massarah Gold Project located in Saudi Arabia.

2. SCOPE

This calculation report is in accordance with SBC-301 for loads and forces, SBC-304 for concrete structures design and project design criteria. The analysis is carried out with Dubal RFEM 5.21 program.

3. MATERIAL PROPERTY

Material properties are defined in Project design criteria MD-216-1000-EG-ST-CRT-1001.

Cast in-situ concrete compressive strength at 28 days:

- C30 f_{ck} = 30 N/mm^2 for slab on grade
- C35 f_{ck} = 35 N/mm^2 for all the other structures

Precast concrete: C45 f_{ck} = 45 N/mm^2

Reinforcing bars ASTM A615- Grade 60, yield strength f_{yk} = 420 N/mm^2

4. SOIL BEARING CAPACITY

Vertical modulus of subgrade reaction K_s = 33,750kN/m^3 and horizontal modulus of subgrade reaction K_h = 155,000 kN/m^2 is applied at 0.6m below FGL is given in geotechnical report MD-216-3000-EG-CV-RPT-1008-C01, BH08.

Recommended allowable bearing capacity $\sigma_{s,max}$ = 375kN/m^2. Therefore, the contact stress results shall not exceed the limit.

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5. MICROFLOATION BASIN 49W1-TNK-002

5.1. Geometry of the structure

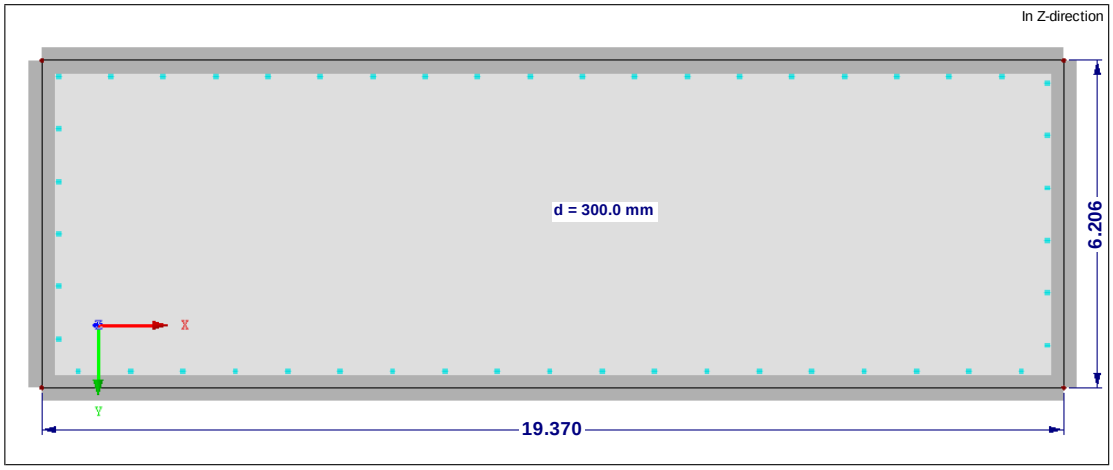


Figure 1 calculation model in RFEM

5.2. Load calculations

5.2.1. Design loads for microflotation basin

Loads for microflotation basin are based on drawing number FD074M001. All Design Loads are indicated in SBC-301. Basic Load Cases are:

1	Dead load (D)
2	Operational load (F)

5.2.1.1. Dead load D_1

Includes self-weight of the foundation and basin dry weight.

basin dry weight = 5200kg/ (13752mm*6206mm) =0.61kN/m^2

5.2.1.2. Operational load D2

Operational load uniformly distributed on the slab is given as 12kN/m^2.

5.2.2. Load combinations for microflotation basin

Ultimate Strength Design - Ultimate Load Combinations: SBC_301, Ch. 2.3)

Load Combination	Expression

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101	1,40 D_1
102	1,40 D_1+ 1,70 D_2

Allowable Stress Design - Serviceability Load Combinations: SBC_301, Ch. 2.4)

Load Combination	Expression
201	D_1
202	D_1+D_2

5.3. Analysis results

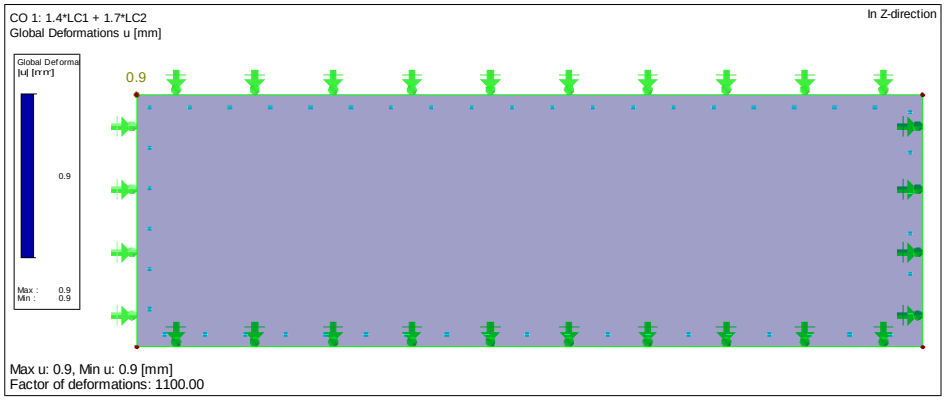


Figure 2 Vertical displacement u_z

$u_z = 0.9mm < u_{z,lim} = 50mm$, ok!

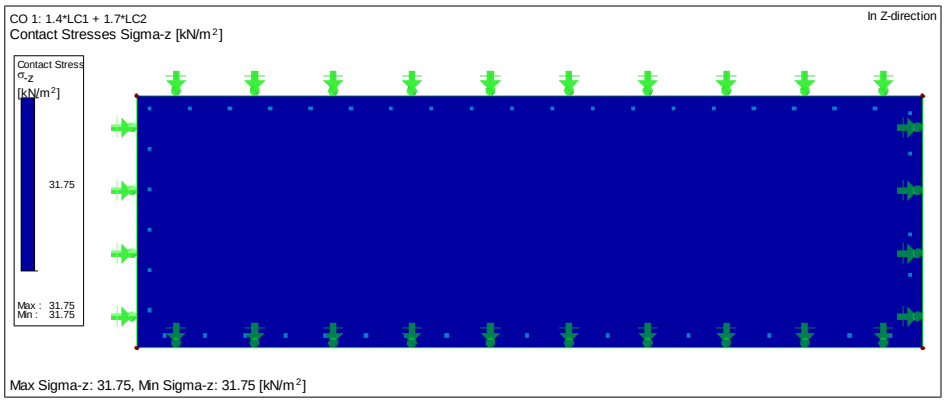


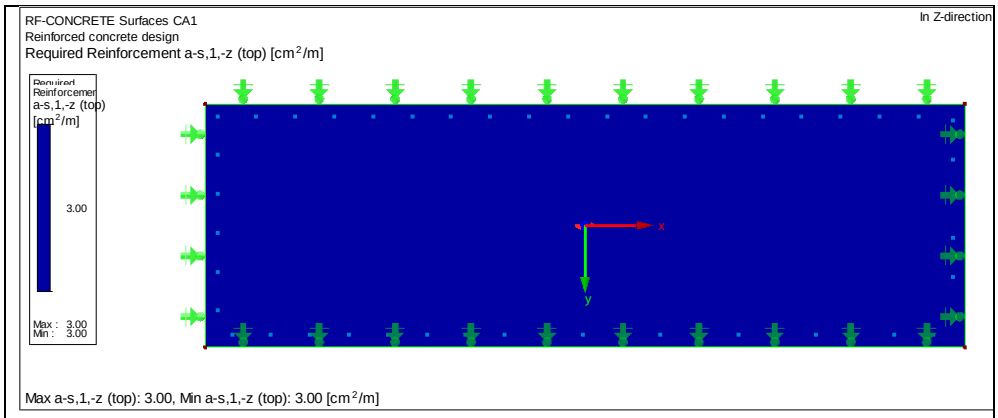
Figure 3 Soil contact stress

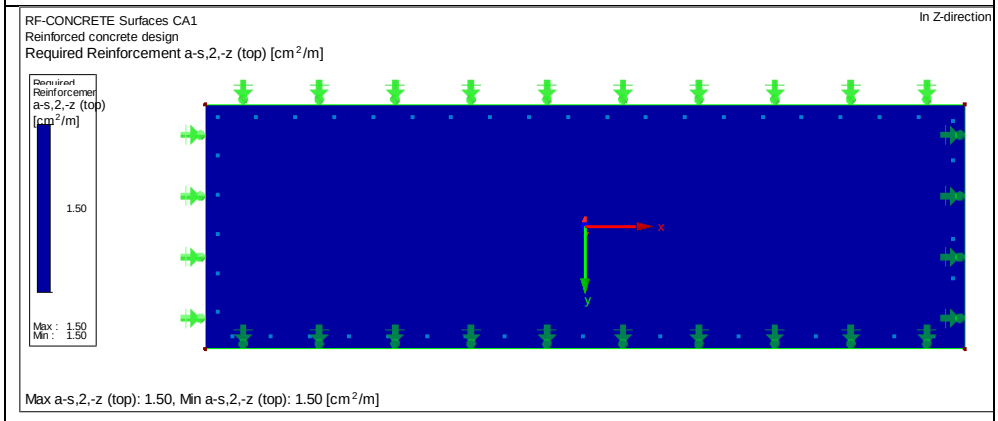
$\sigma_s < \sigma_{s,max} = 375kN/m^2$, ok!

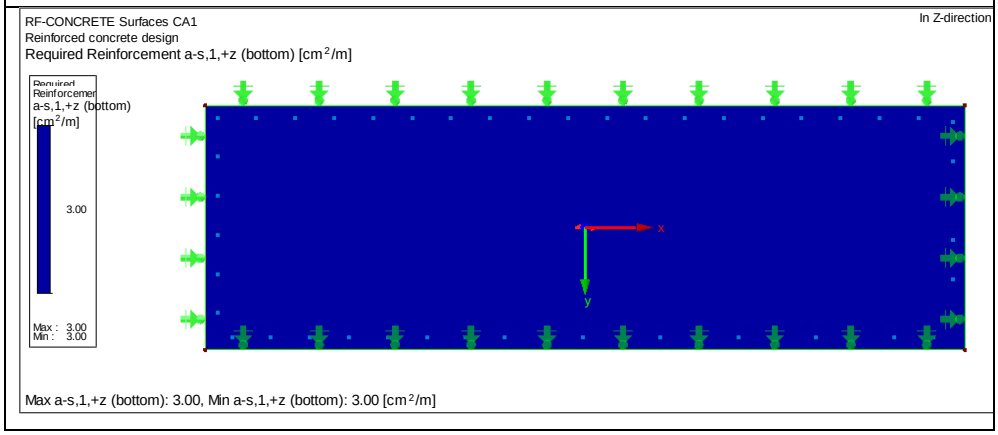
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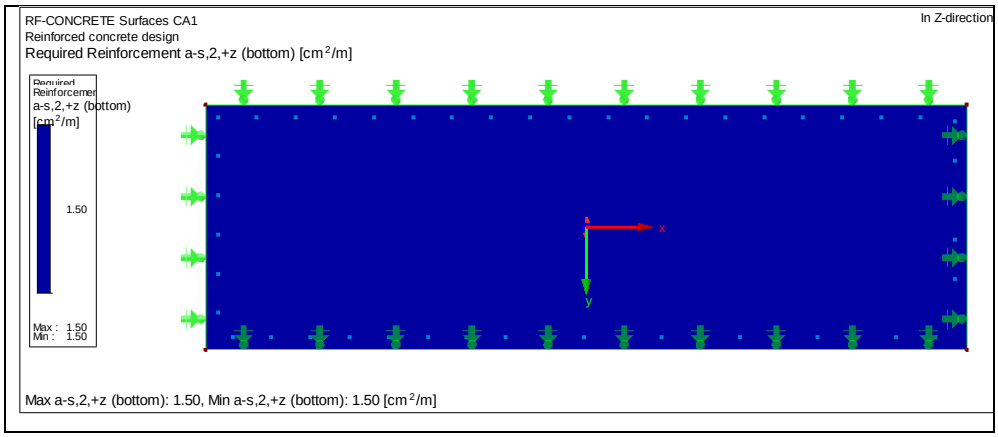


Figure 4 Required reinforcement results

$A_{s,min}$ = 5,40cm^2/m to control shrinkage and temperature effects.

Provide T12/200mm, $A_{s,provided}$ = 5,65 cm^2/m.

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6. COAGULANT MIXING BASIN 49W1-TNK-001

6.1. Geometry of the structure

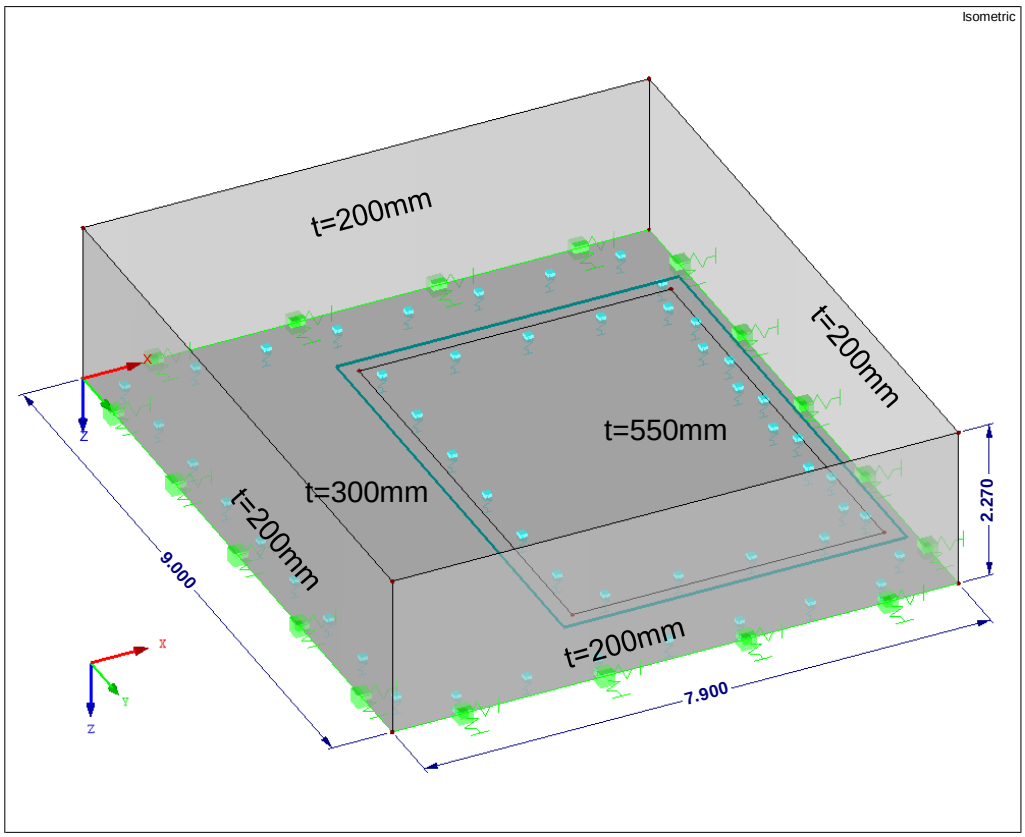


Figure 5 Calculation model in RFEM

coagulant mixing basin has side wall thickness t=200mm and slab max. thickness 550mm and min. thickness 300mm.

6.2. Load calculations

6.2.1. Design loads for coagulation tank

Loads for coagulant mixing basic are based on drawing number P06204M002. All Design Loads are indicated in SBC-301. Basic Load Cases are:

6.2.1.1. Dead load D_1

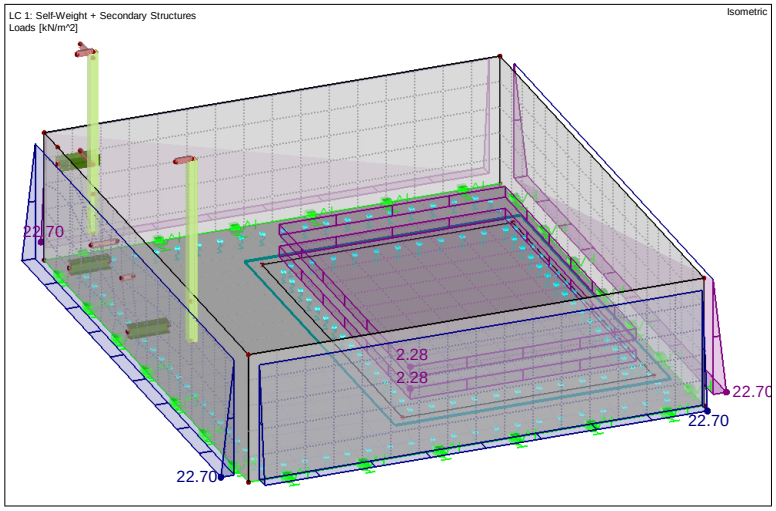
Includes self-weight of the foundation and dry weight of coagulant mixing basin.

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dry weight 6500kg/(4511mm*6327mm) = 2.28kN/m^2. Max pressure at bottom of the side walls $p_{s.max} = C_p \gamma_s h$ = 0.5*20kN/m^3*2.27m = 22.7kN/m^2



6.2.1.2. Operational load D2

Operational load uniformly distributed on the slab is given as 33kN/m^2.

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Pipes and cabletrays are placed on steel supports. The loads have been calculated using their exact weight as point- or lineloads according to the associated span. Maximum full weight of pipes given by the process designers:

DN 15 1.8 kg/m

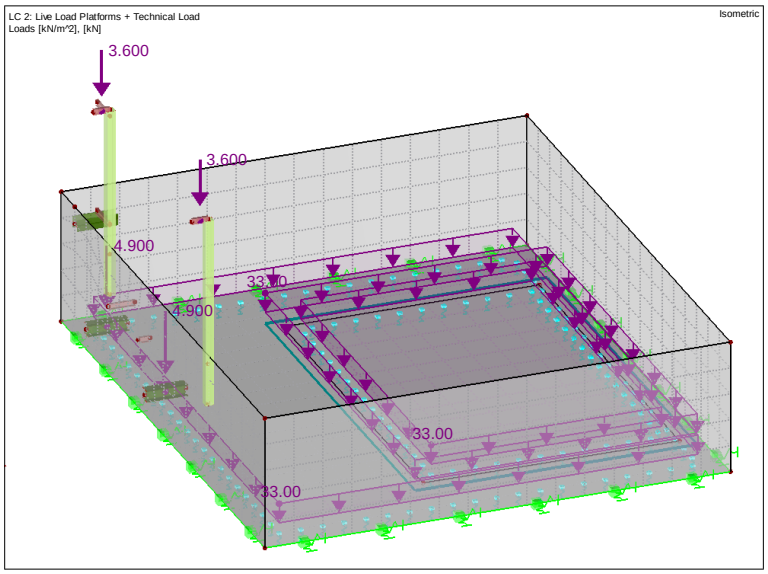
DN 50 + tray8.14 + 70 kg/m(Applied pointload: 0.7814 kN/m x 4.5 m = 3.6 kN)

DN 300 162.8 kg/m (Applied pointload: 1.628 kN/m x 3 m = 4.9 kN)

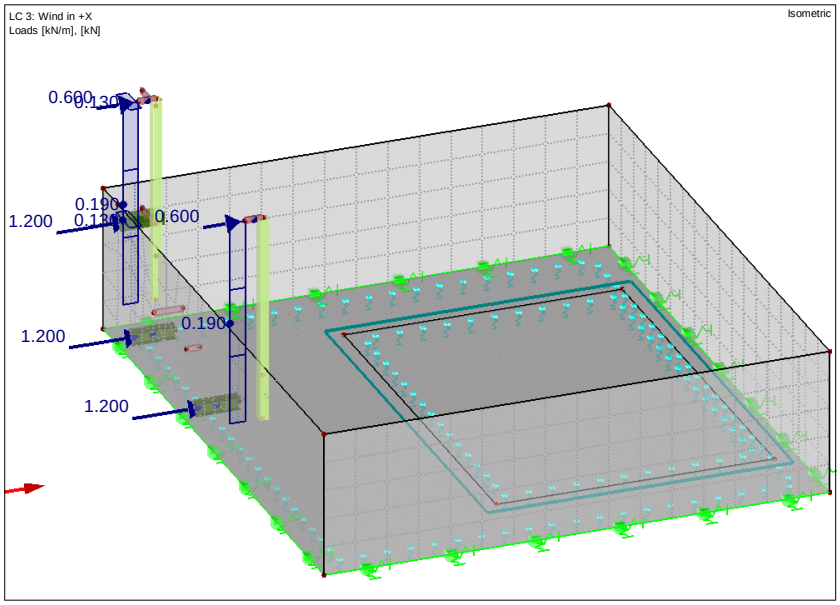
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6.2.1.3. wind load in X+

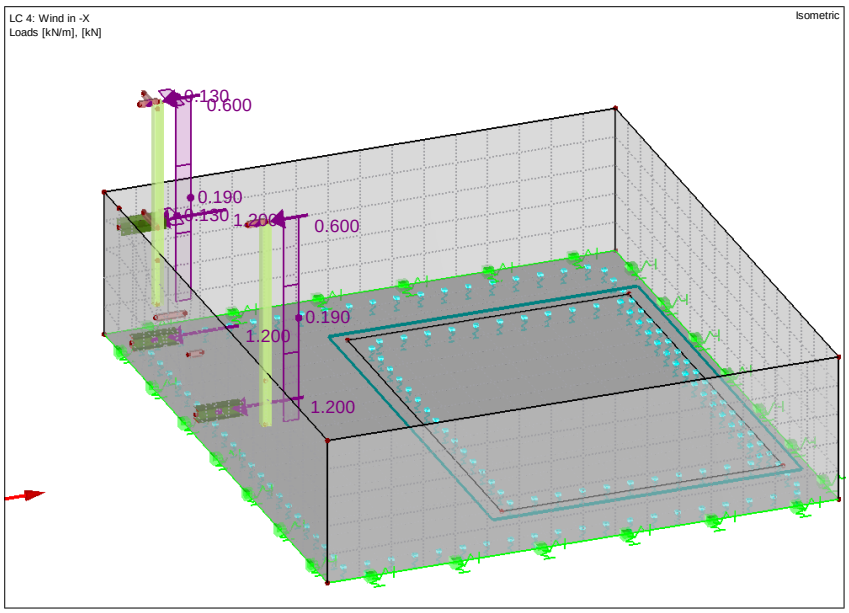


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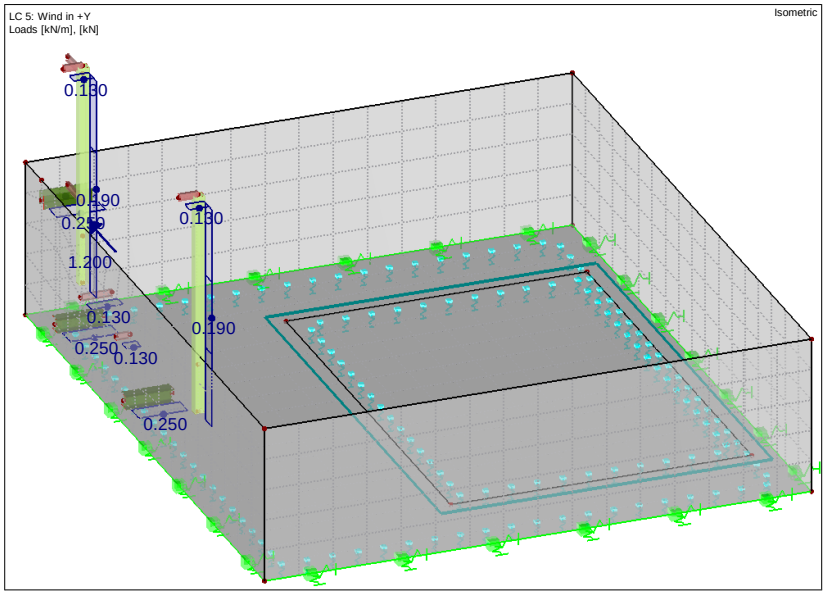
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6.2.1.4. wind load in X-



6.2.1.5. wind load in Y+

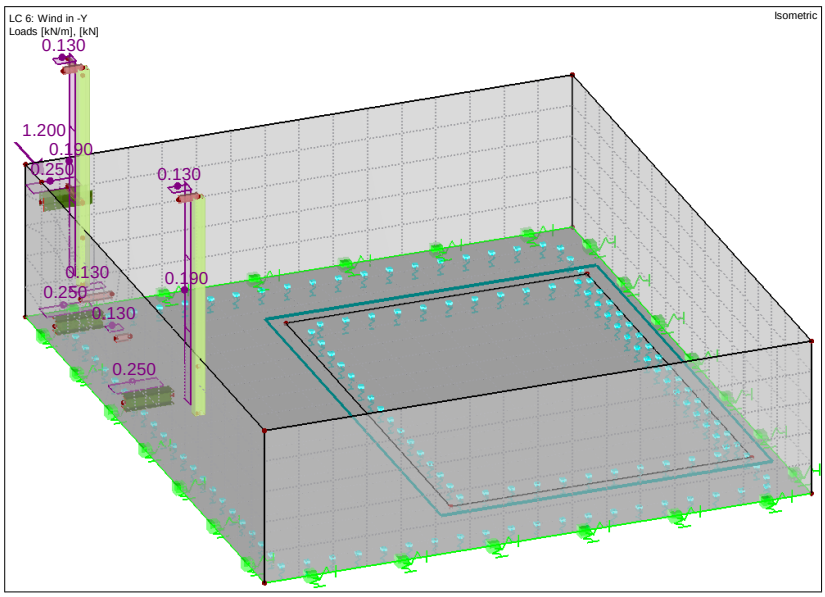


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6.2.1.6. wind load in Y-



6.2.2. Load combinations for coagulation tank

Ultimate Strength Design - Ultimate Load Combinations: SBC_301, Ch. 2.3)

Load Combination	Expression
101	1,40 D_1 + W
102	1,40 D_1 + 1,70 D_2 +W

Allowable Stress Design - Serviceability Load Combinations: SBC_301, Ch. 2.4)

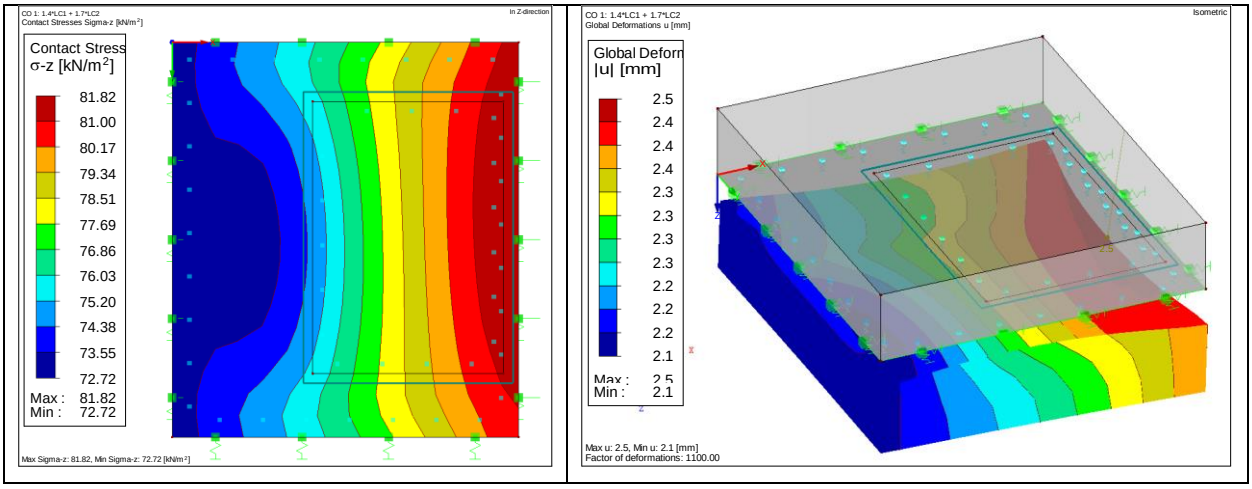
Load Combination	Expression
201	D_1
202	D_1 +D_2 + W

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6.3. Analysis results



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Figure 6 Soil contact stress σ_s and global deformation u_z

$\sigma_s < \sigma_{s,max} = 375kN/m^2$ (MD-216-3000-EG-CV-RPT-1008) and $u_z < u_{z,max} = 25mm$ (according to and MD-101-SMEM-EG-ST-CRT-00001), , ok!

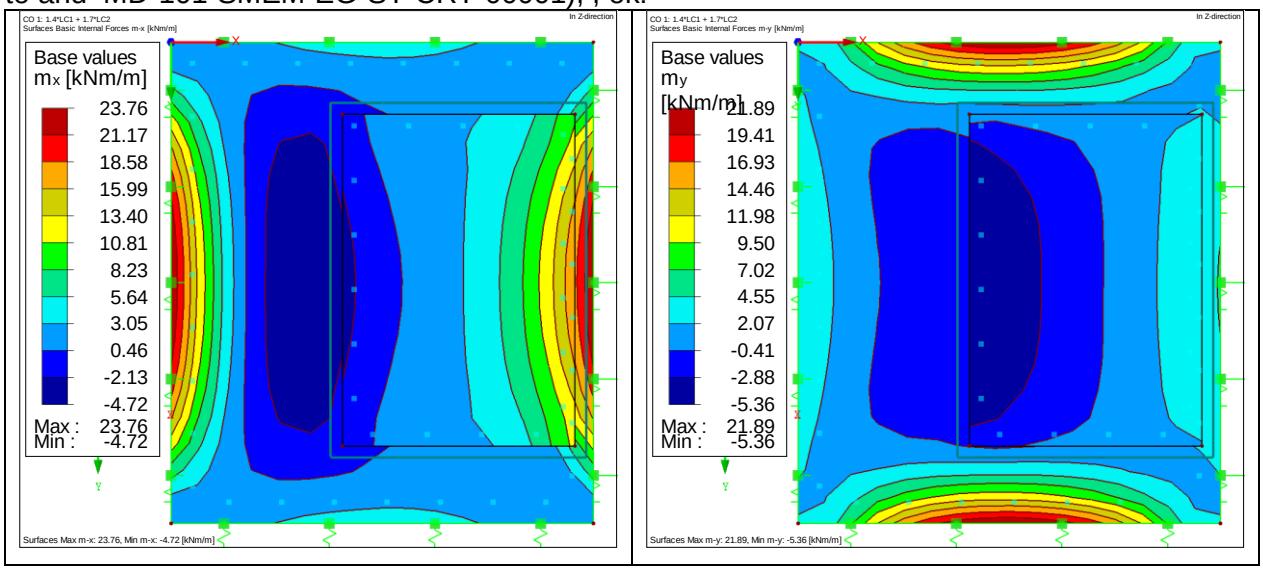


Figure 7 Bending moment results for slab

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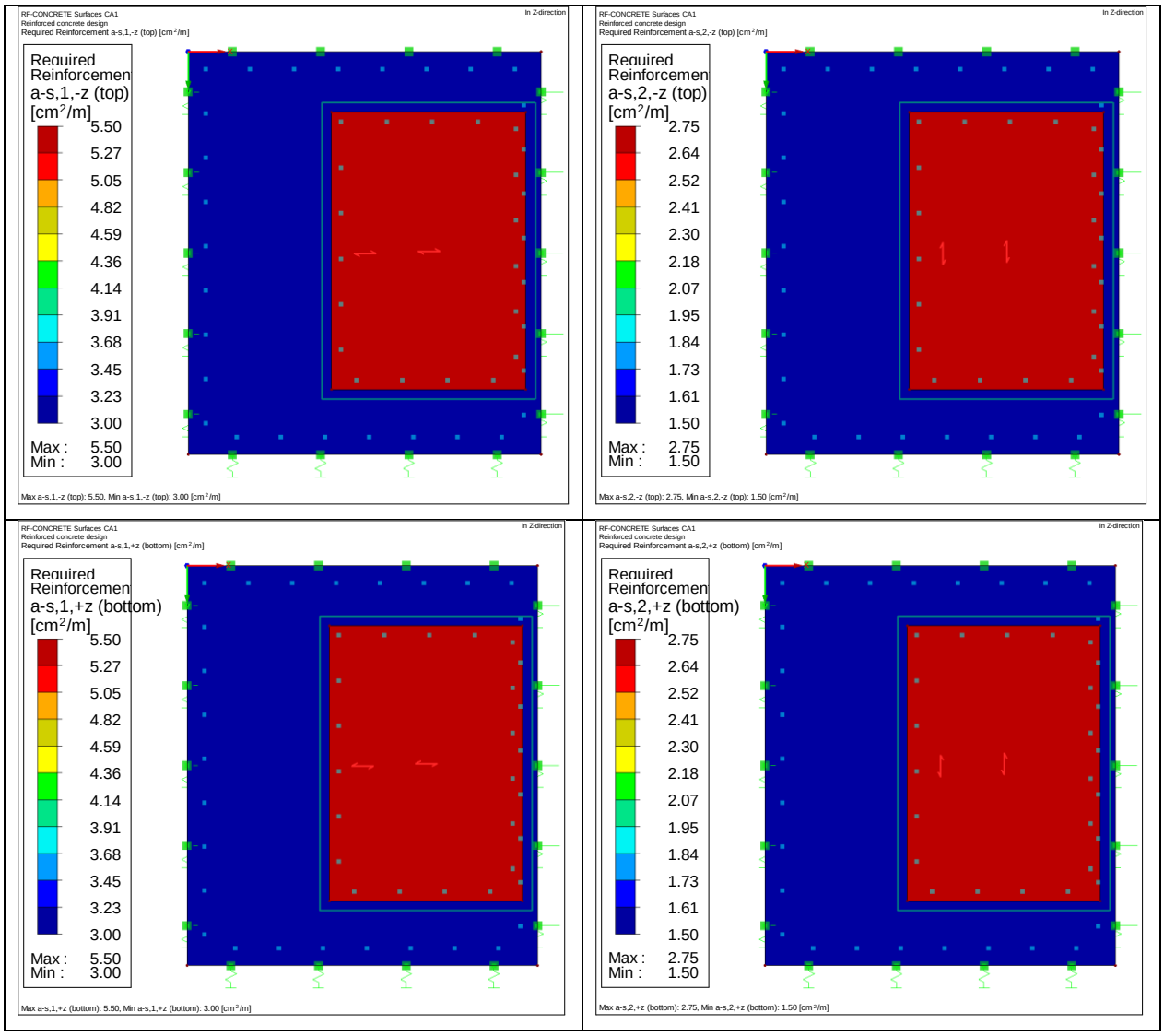


Figure 8 Required reinforcement

As,min = 5,40 cm^2/m for thickness t=300mm, provide T12/200mm.$A_{s,provided}$=5,65 cm^2/m

As,min = 9.90 cm^2/m for thickness t=550mm, provide T12/200mm as top reinforcement $A_{s,provided}$=5.65 cm^2/m and T16/200mm as bottom reinforcement $A_{s,provided}$=10,05 cm^2/m

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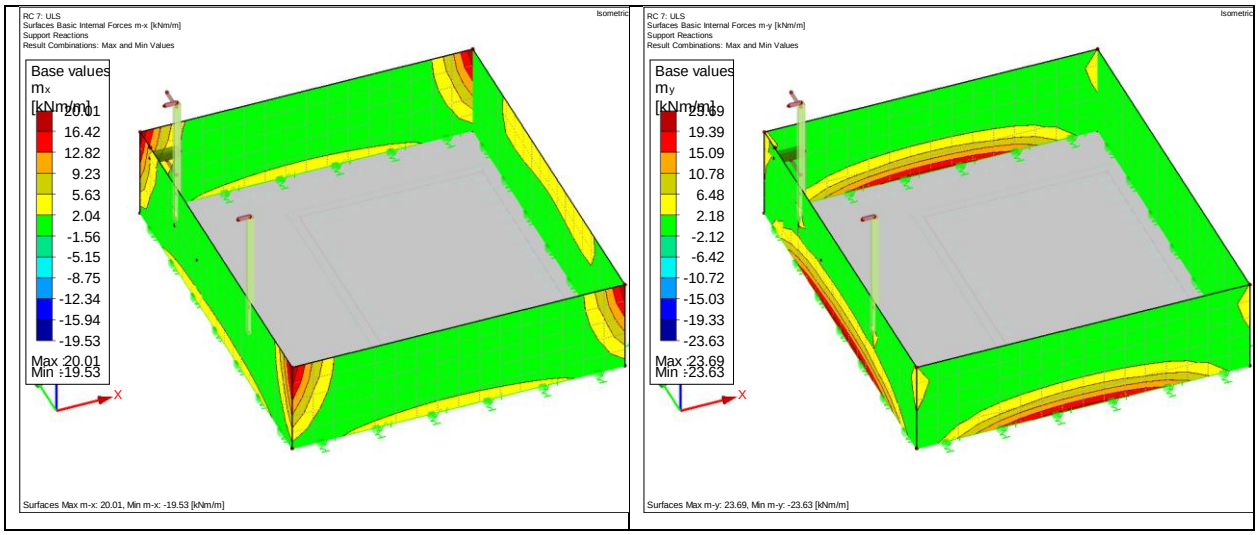


Figure 9 Bending moment result for side walls t=200mm

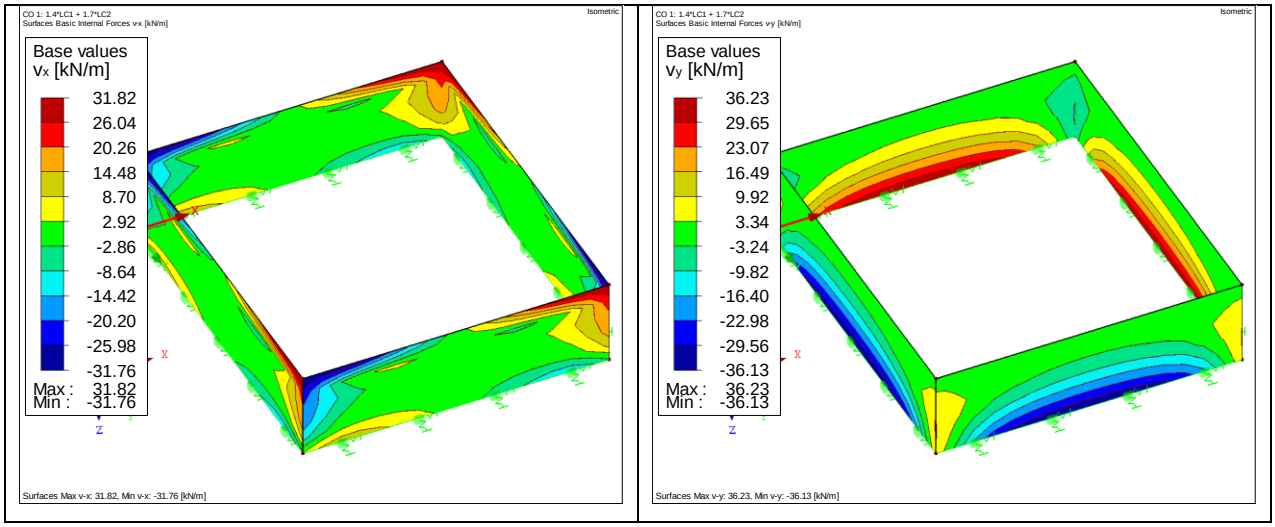


Figure 10 Shear force result

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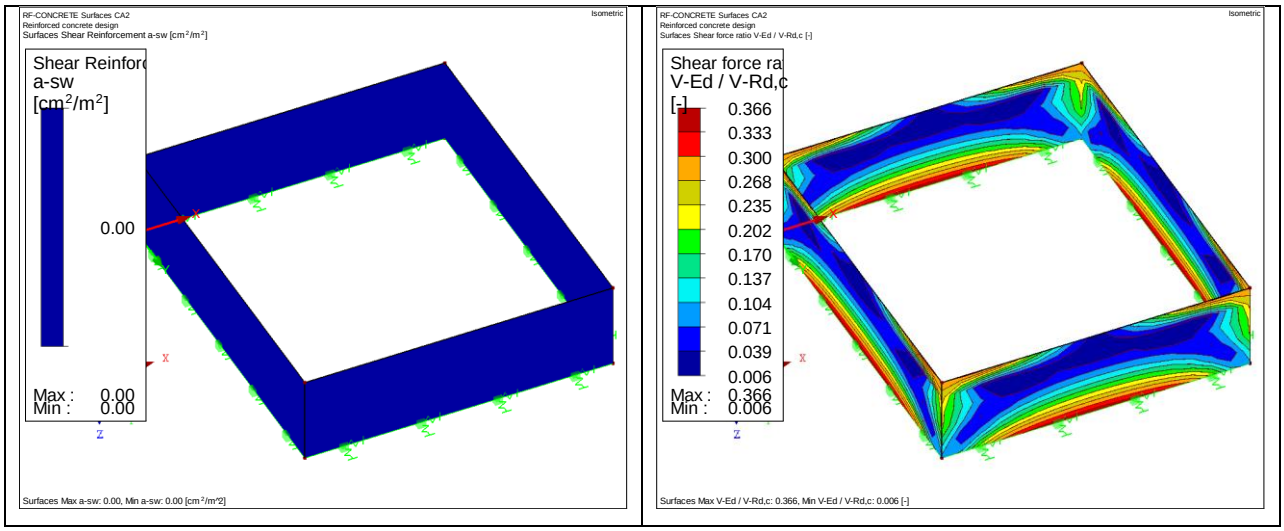
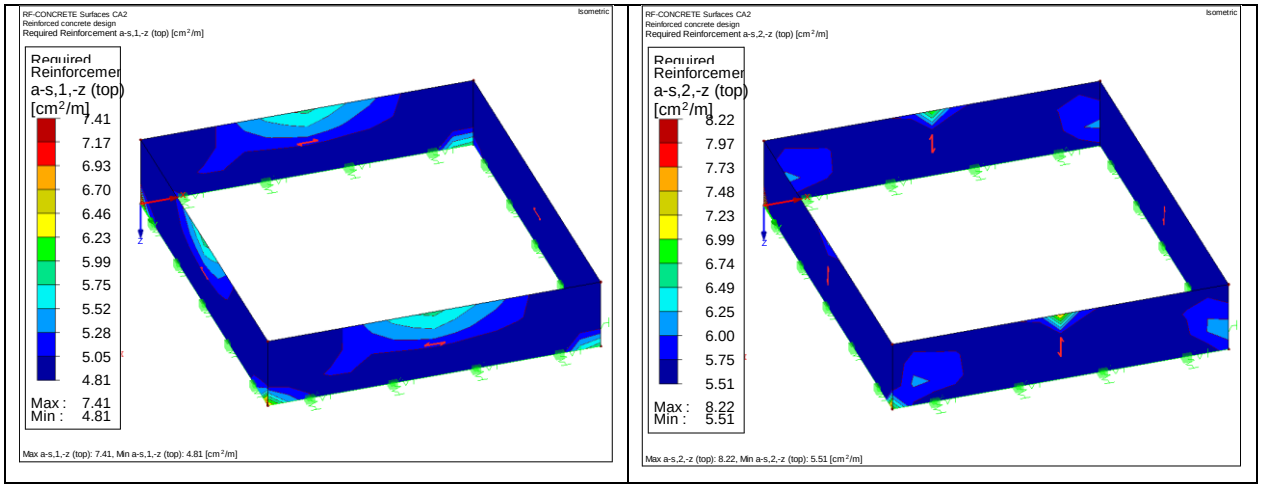


Figure 11 Required shear reinforcement (left) and $V_{Ed}/V_{Rd,c}$ (right)

Utility ratio = applied shear force/ design shear resistance without shear reinforcement

= $V_{Ed}/V_{Rd,c} < 1.0$, no shear reinforcement needed



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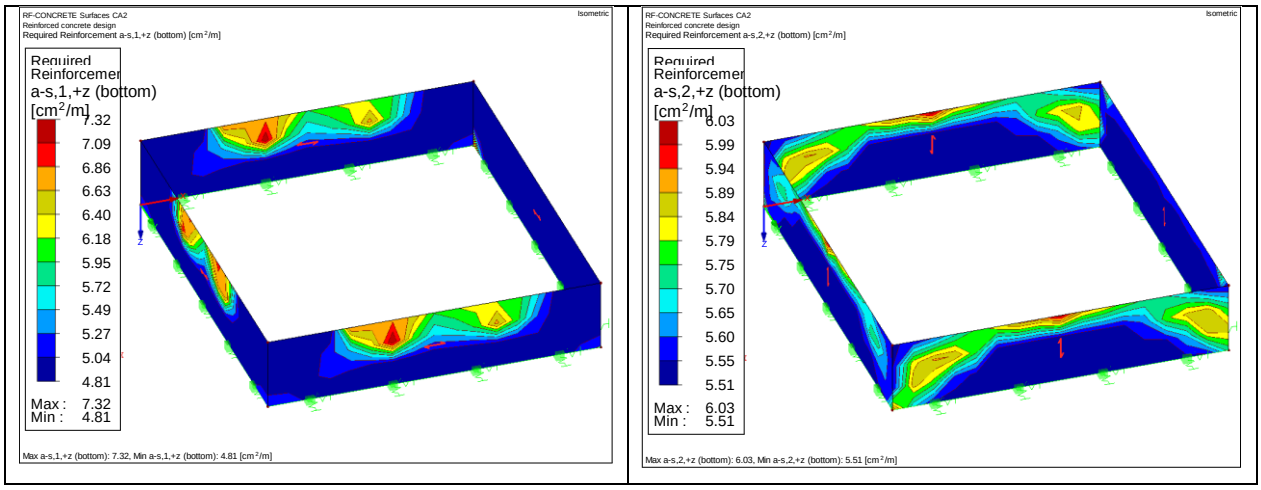


Figure 12 Required reinforcement for walls

Provide T12/150mm, $A_{s,provided}$ = 7,54 cm^2/m.

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7. PRECAST CONCRETE PIT DESIGN

7.1. Geometry of the structure

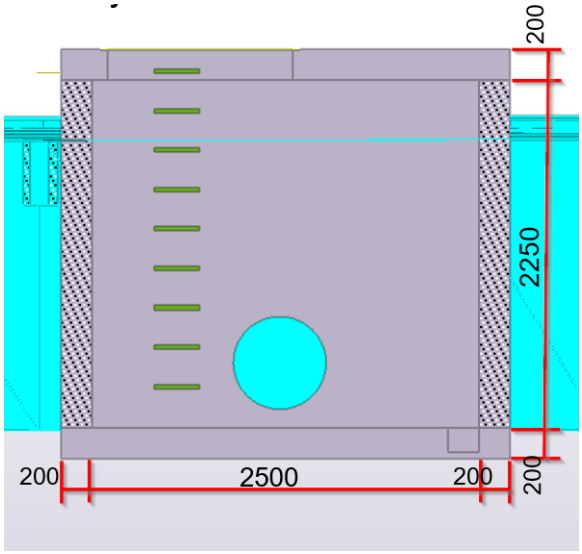


Figure 13 Geometry of the pit

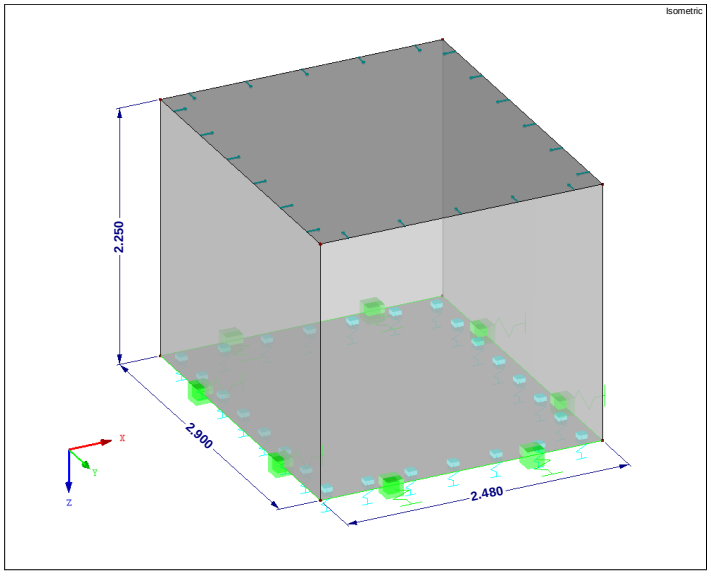


Figure 14 Calculation model in RFEM

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7.2. Load calculations

7.2.1. Design loads for pit

All Design Loads are indicated in SBC-301. Basic Load Cases are:

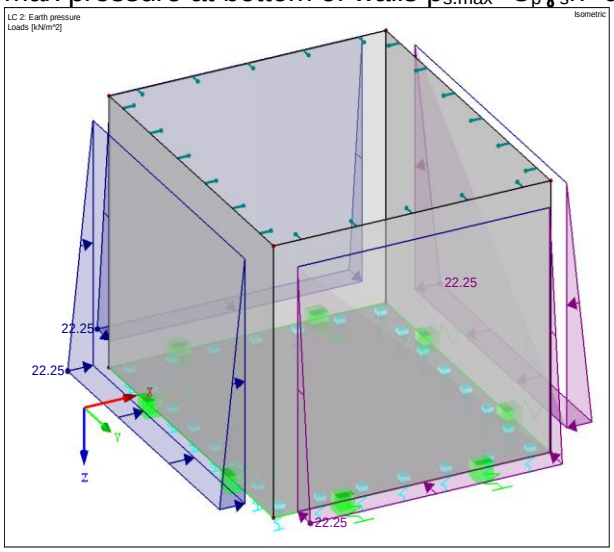
1	Dead load (D_1)
2	Soil pressure (D2)
3	Surcharge pressure due to road traffic D3

7.2.1.1. Dead load D_1

Includes self-weight of the pit.

7.2.1.2. Soil pressure D_2

max pressure at bottom of walls $p_{s.max}=C_p\gamma_s h=0.5*20kN/m^3*2.225m=22.25kN/m^2$



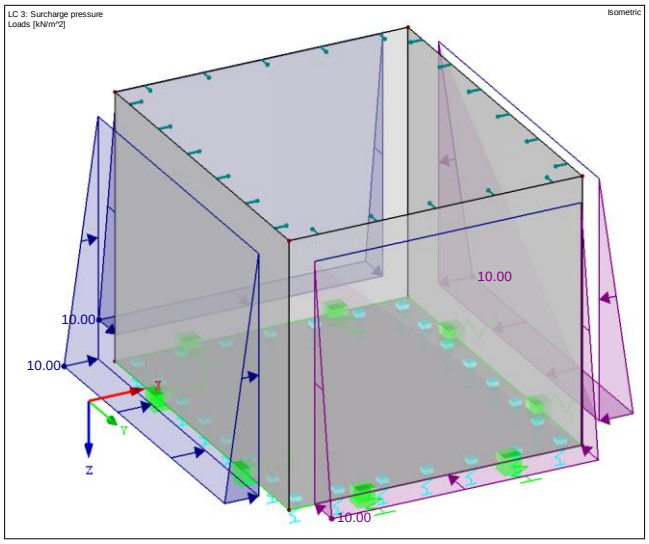
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7.2.1.3. Surcharge pressure due to road traffic D_3

max pressure at bottom of walls $p_{s.max}=C_p p=0.5*20kN/m^2=10kN/m^2$



7.2.2. Load combinations for pit

Ultimate Strength Design - Ultimate Load Combinations: SBC_301, Ch. 2.3)

Load Combination	Expression
101	1,40 D_1
102	1,40 D_1+ 1,70 (D_2 + D_3)

Allowable Stress Design - Serviceability Load Combinations: SBC_301, Ch. 2.4)

Load Combination	Expression
201	D_1
202	D_1+ (D_2 + D_3)

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7.3. Analysis results

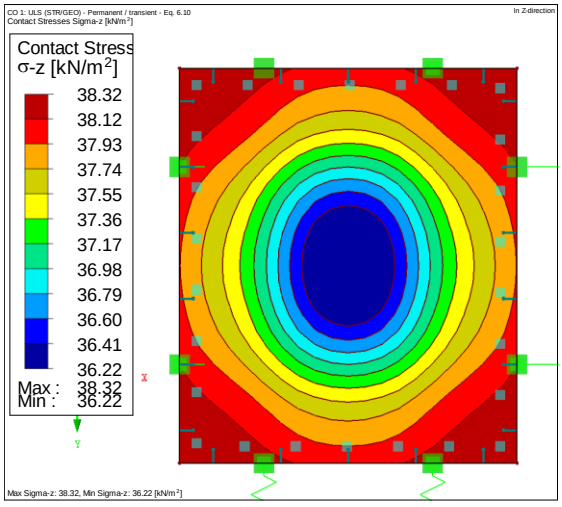


Figure 15 Soil contact pressure

base pressure $\sigma_s < \sigma_{s,max} = 375kN/m^2$, ok!

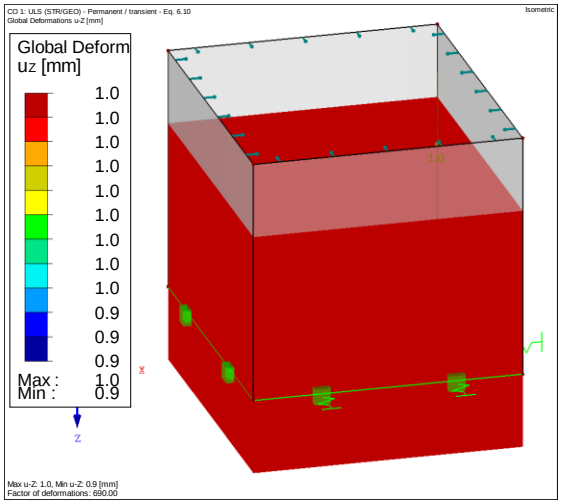


Figure 16 Vertical displacement uz

C01

$u_z = 1mm < u_{z,lim} = 25mm$ (in MD-101-SMEM-EG-ST-CRT-00001), ok!

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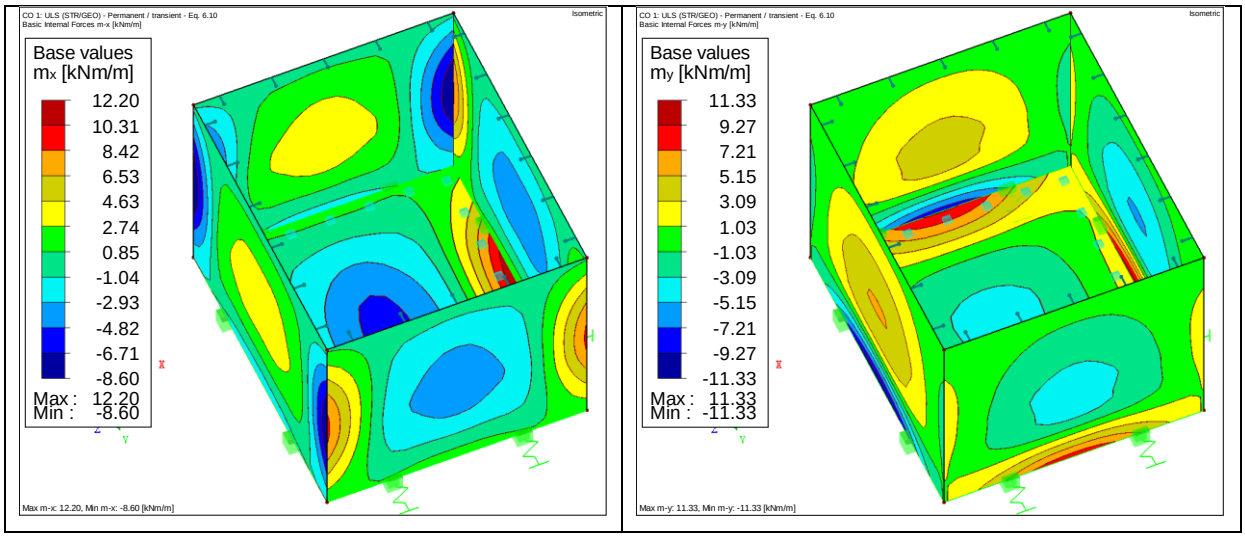


Figure 17 Bending moment results

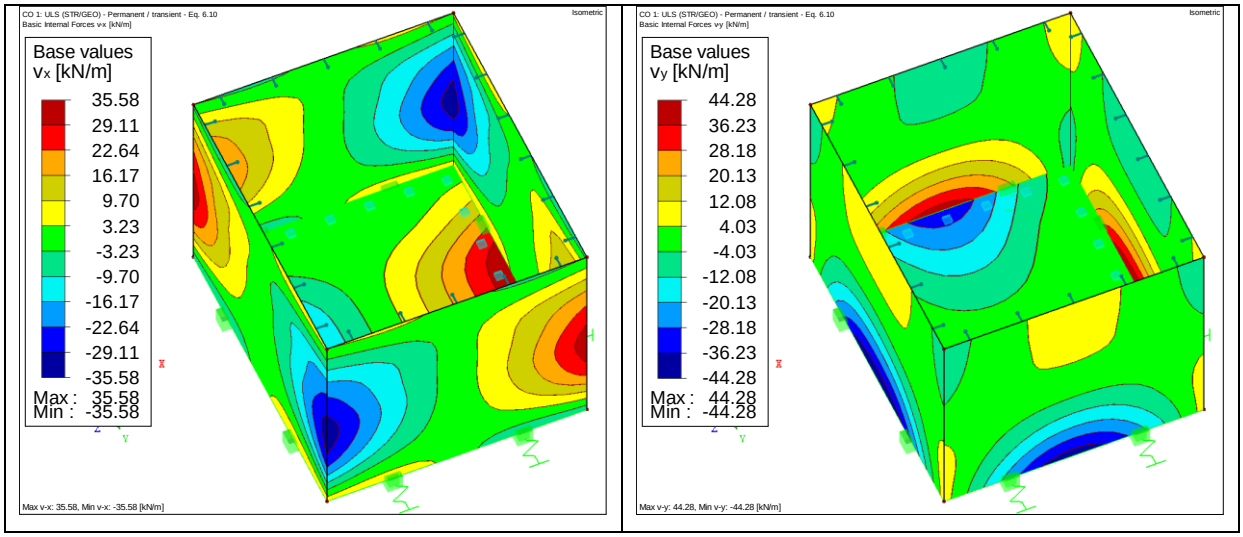


Figure 18 Shear force results

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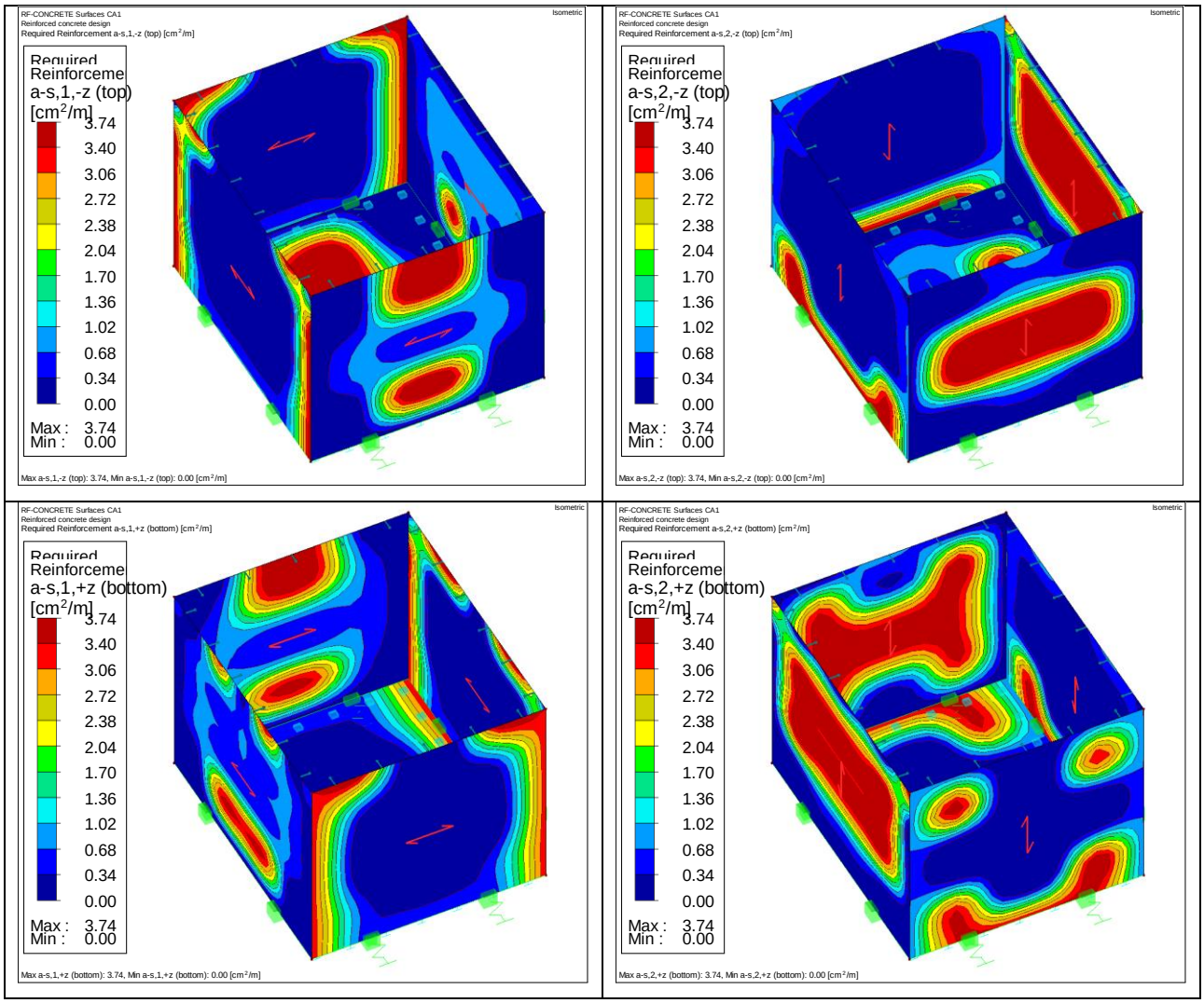


Figure 19 Required reinforcement for pit

Provide T12/200mm, $A_{s,provided} = 5.65 m^2/m$

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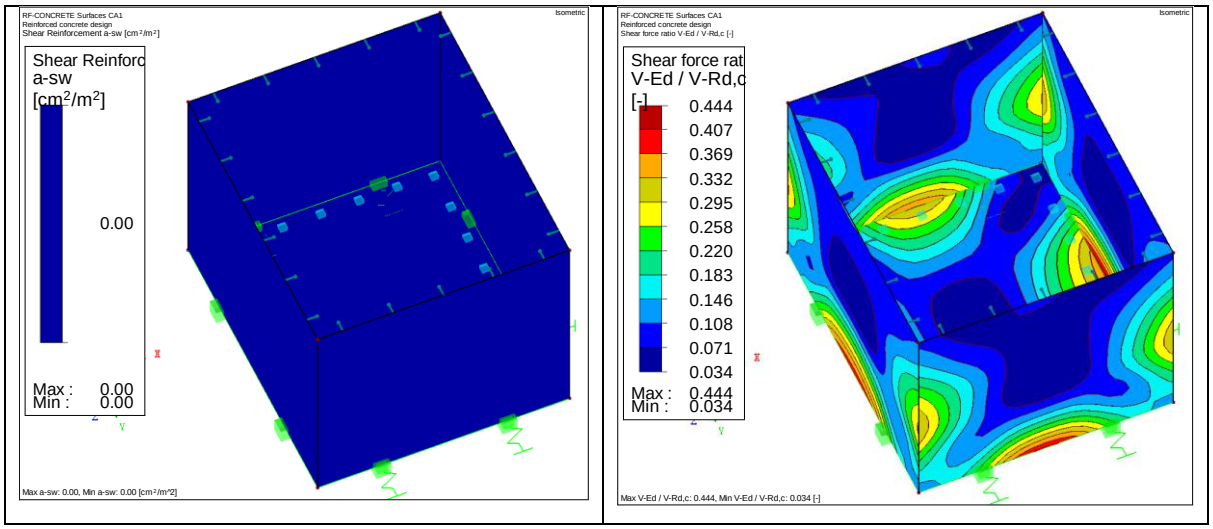


Figure 20 Required shear reinforcement (left) and $V_{Ed}/V_{Rd,c}$

utility ratio =applied shear force/ design shear resistance without shear reinforcement = $V_{Ed}/V_{Rd,c} < 1.0$, no shear reinforcement needed

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8. BACKWASH STAIRCASE FOUNDATION

All design loads are results from staircase in MD-216-49W1-EG-ST-CAL-1002. Input load values are given in Attachment 2. Basic load cases include:

1	Self-weight of foundation and back soil weight (D)
2	Live load (L)
	Wind load in ±X and ±Y direction (W)
4	Temperature ±25 (T)
5	Seismic load (E)

For strength calculation:

Load Combination	Expression
101	1,40 D
102	1,40 (D +T) + 1,70 L
103	1,20 D + 1,00 L
104	1,20 D + 0,80 W
105	1,20 D + 1,60 W + 1.00 L
106	1,20 D + 1,00 E + 1,00 L
107	0,90D + 1,60 W
108	0,90 E + 1,00 E

For allowable stress calculation:

Load Combination	Expression
201	D
202	D + L + T
203	D + W
204	D + 0,70 E
205	D + 0,75 W + 0,75 L
206	D+ 0,75 (0,70 E) + 0,75 L
207	0,60 D + W
208	0,60 D + 0,70 E

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8.1. Structure geometry

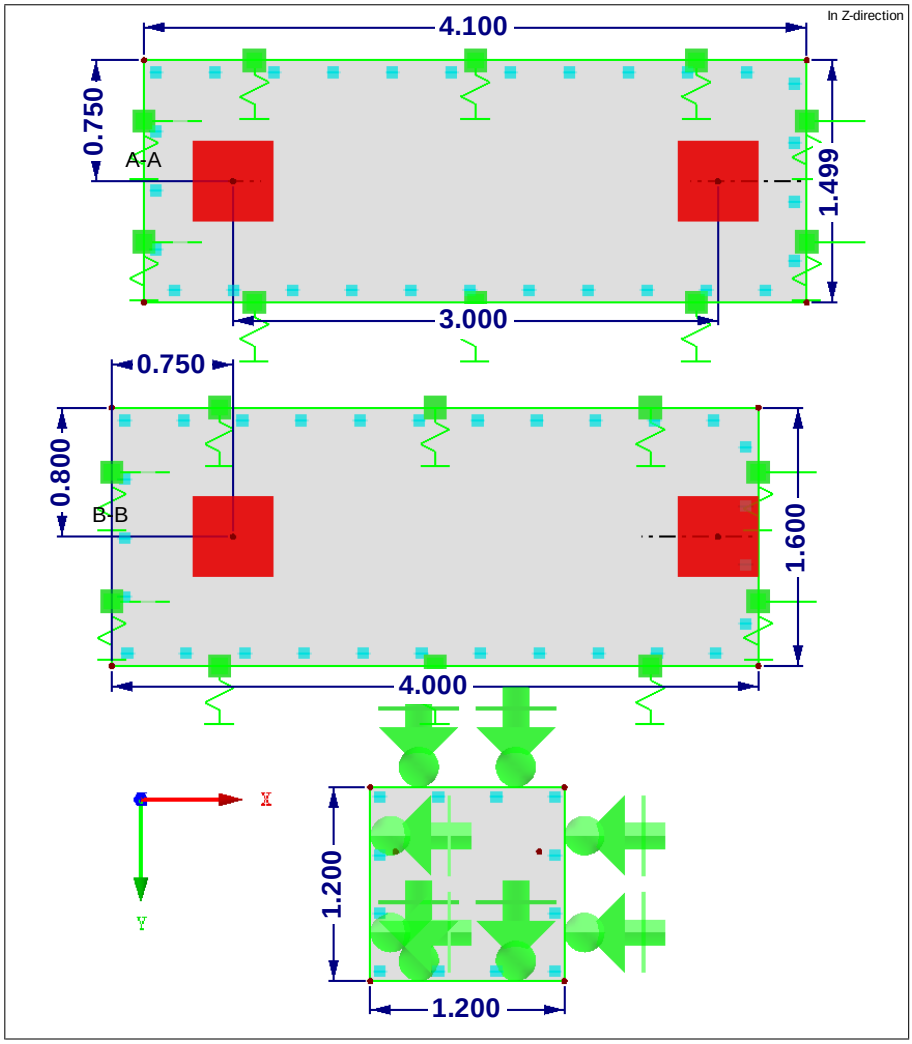


Figure 21 Calculation model in RFEM

For combined footings, the pedestal dimension 500mm*500mm, height 1.150m. The pad footing thickness 400mm and stair landing thickness 200mm.

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8.2. Load calculations

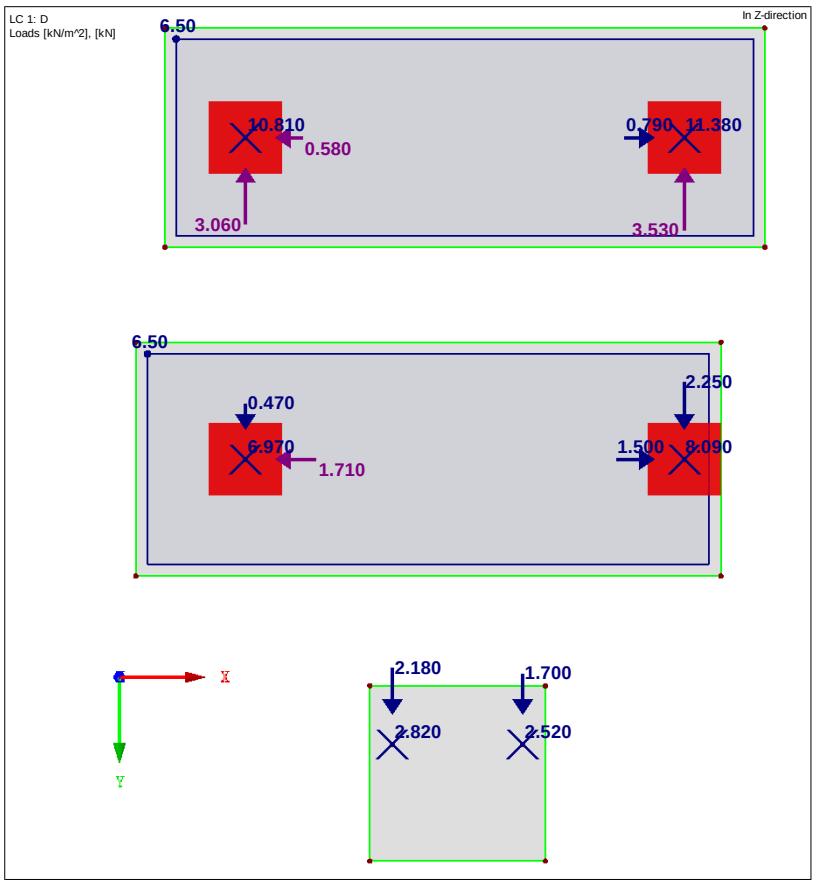


Figure 22 Dead load

Dead load includes self-weight and backfill soil 0.4m. Backfill soil weight applied as surface load on pad footing:

(4.1m*1.5 -2*0.5m*0.5m) * 18kN/m^3 *0.4m /(4.1m*1.5m) = 6.5kN/m^2

and

(4m*1.6m-2*0.5m*0.5m) * 18kN/m^3 * 0.4m/(4m*1.6m) = 6.5kN/m^2

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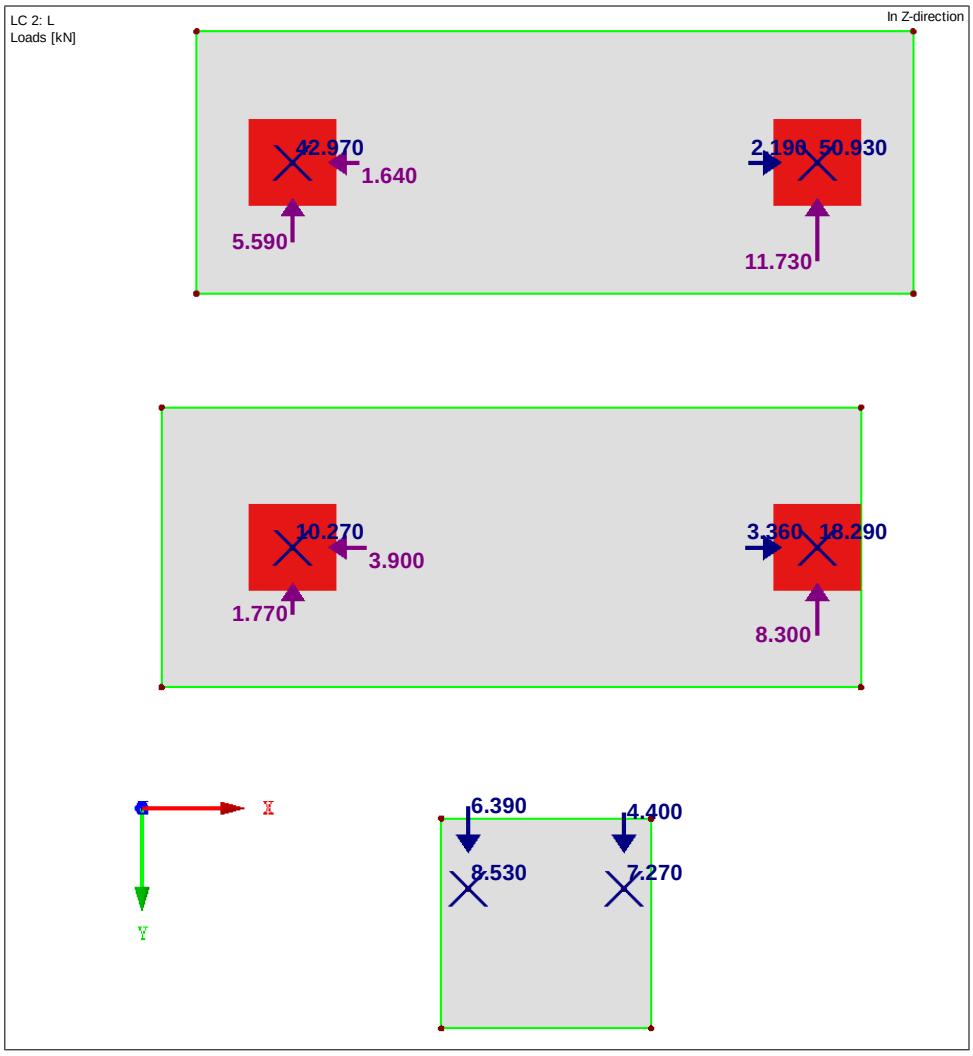


Figure 23 Live load

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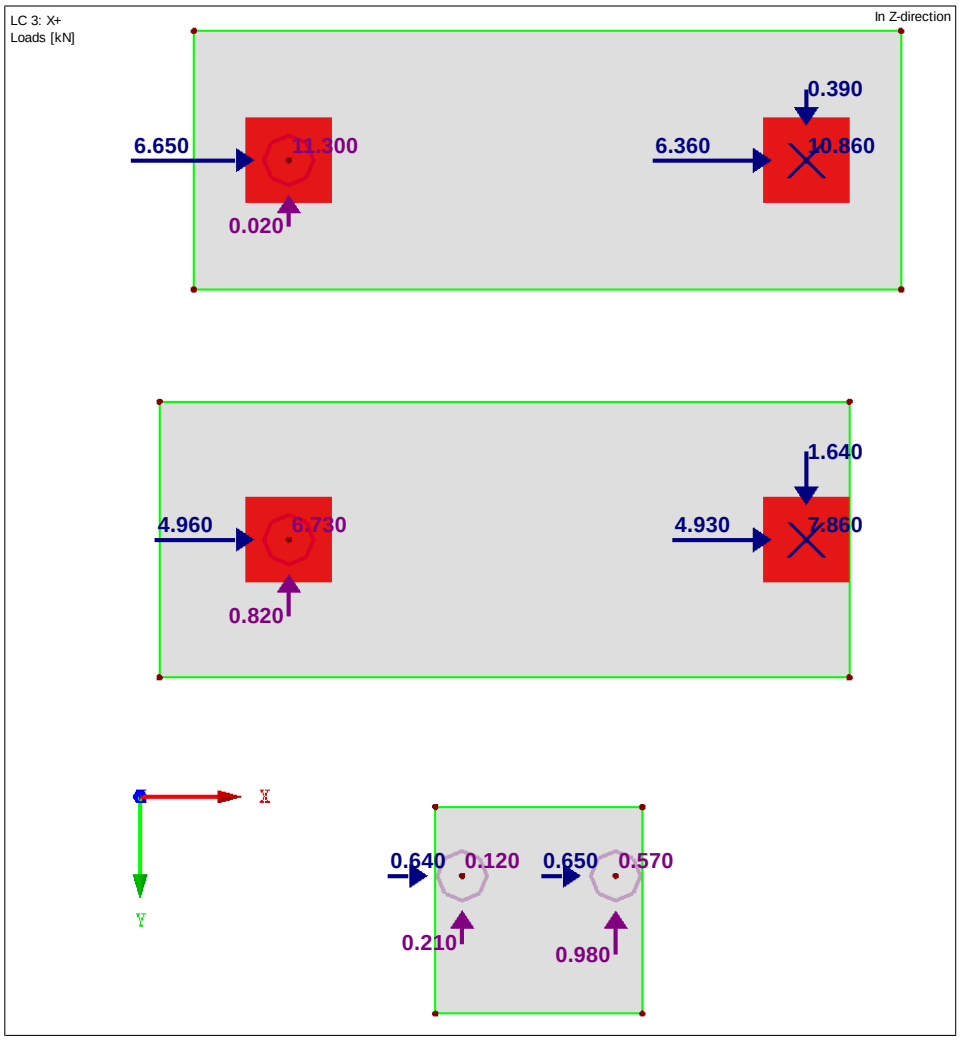


Figure 24 Wind load x+

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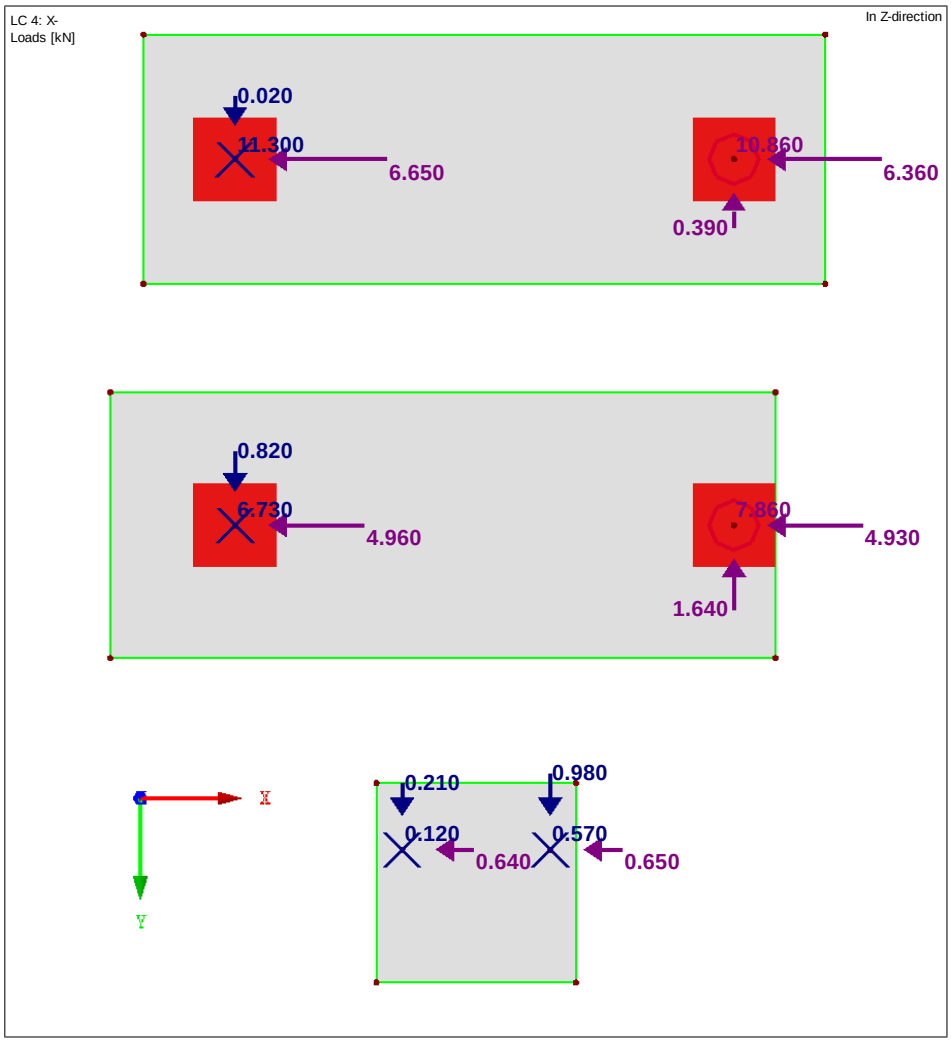


Figure 25 Wind load x-

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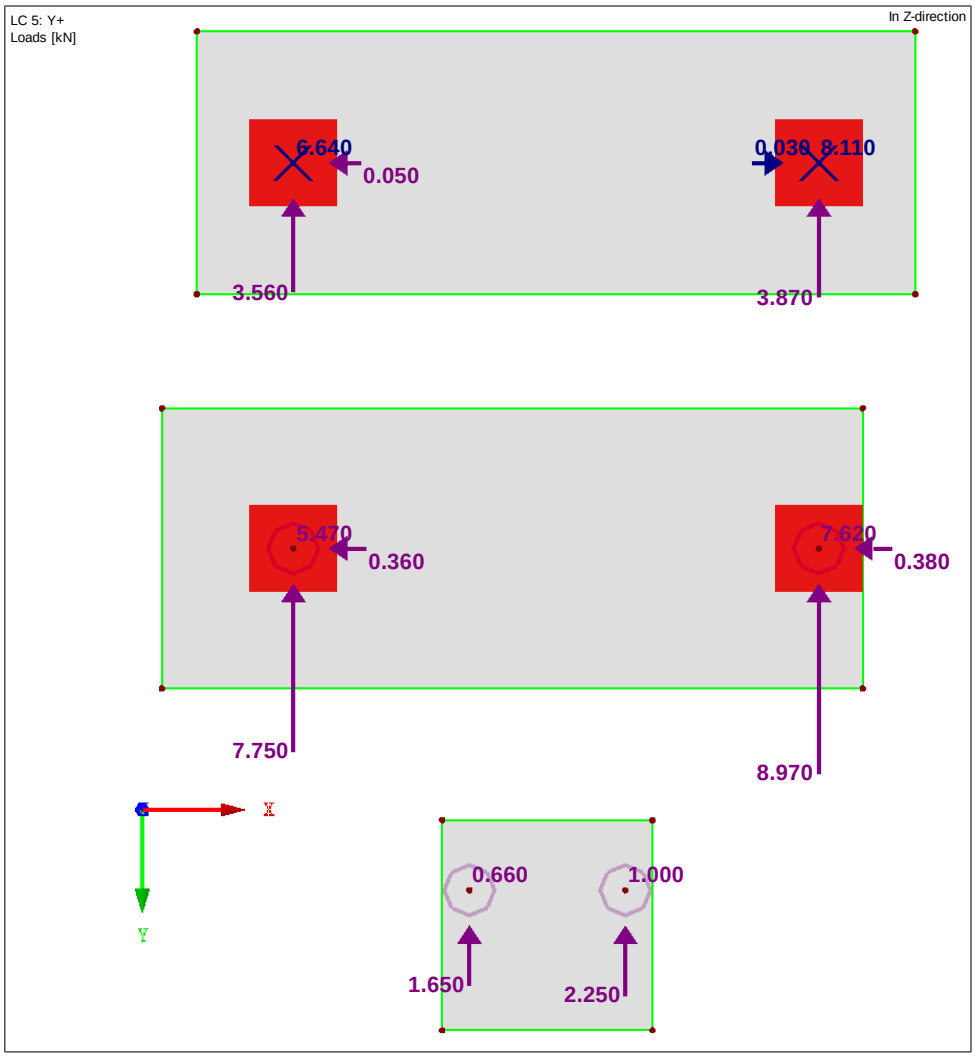


Figure 26 Wind load y+

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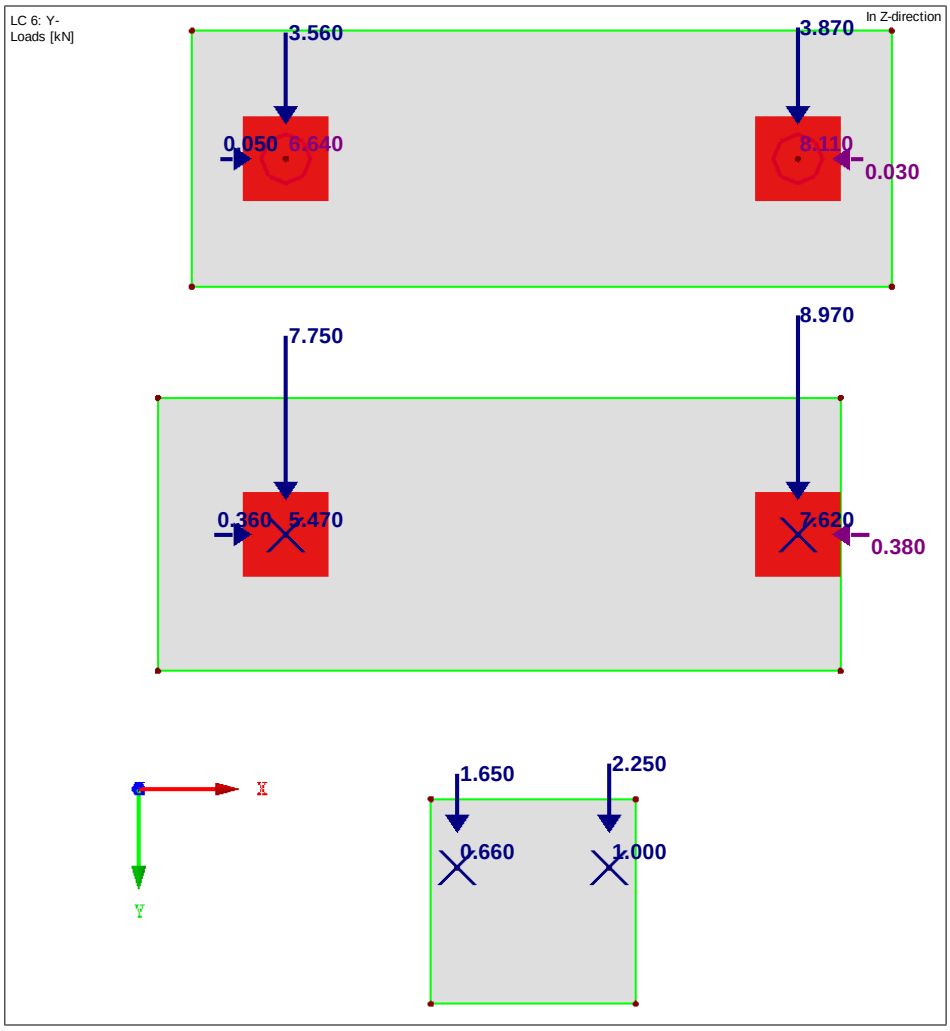


Figure 27 Wind load y-

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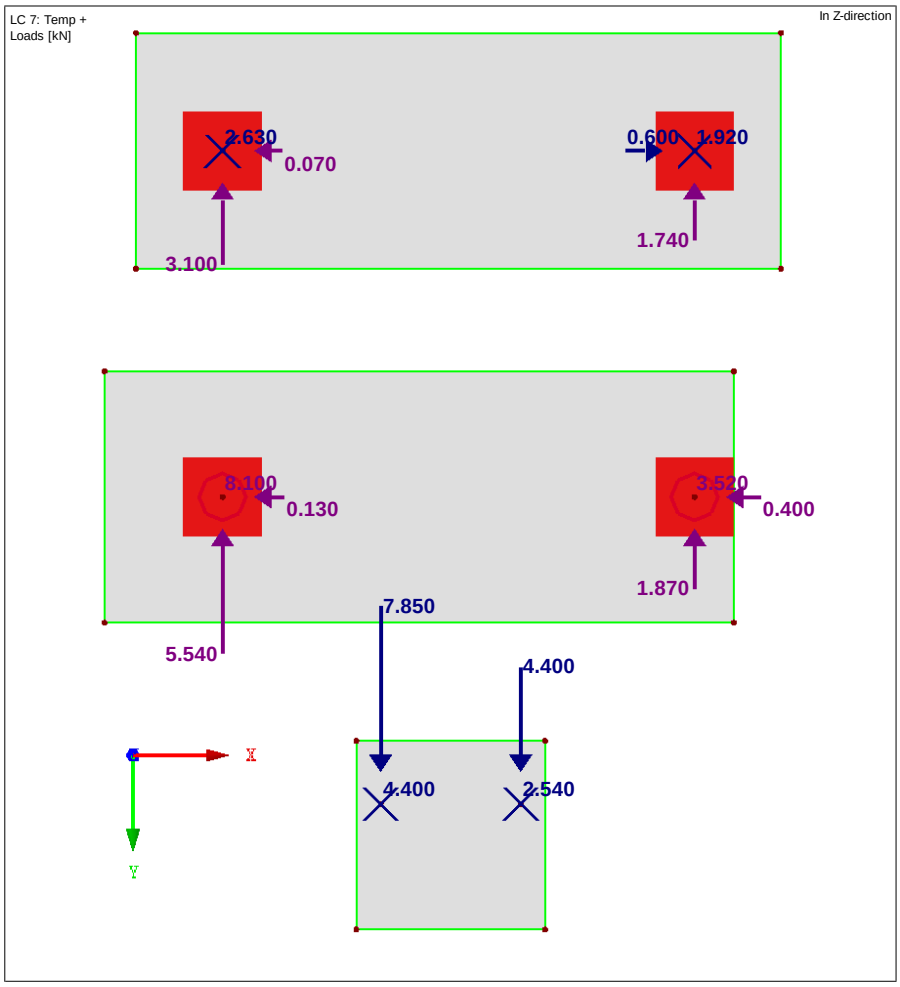


Figure 28 Temperature +25

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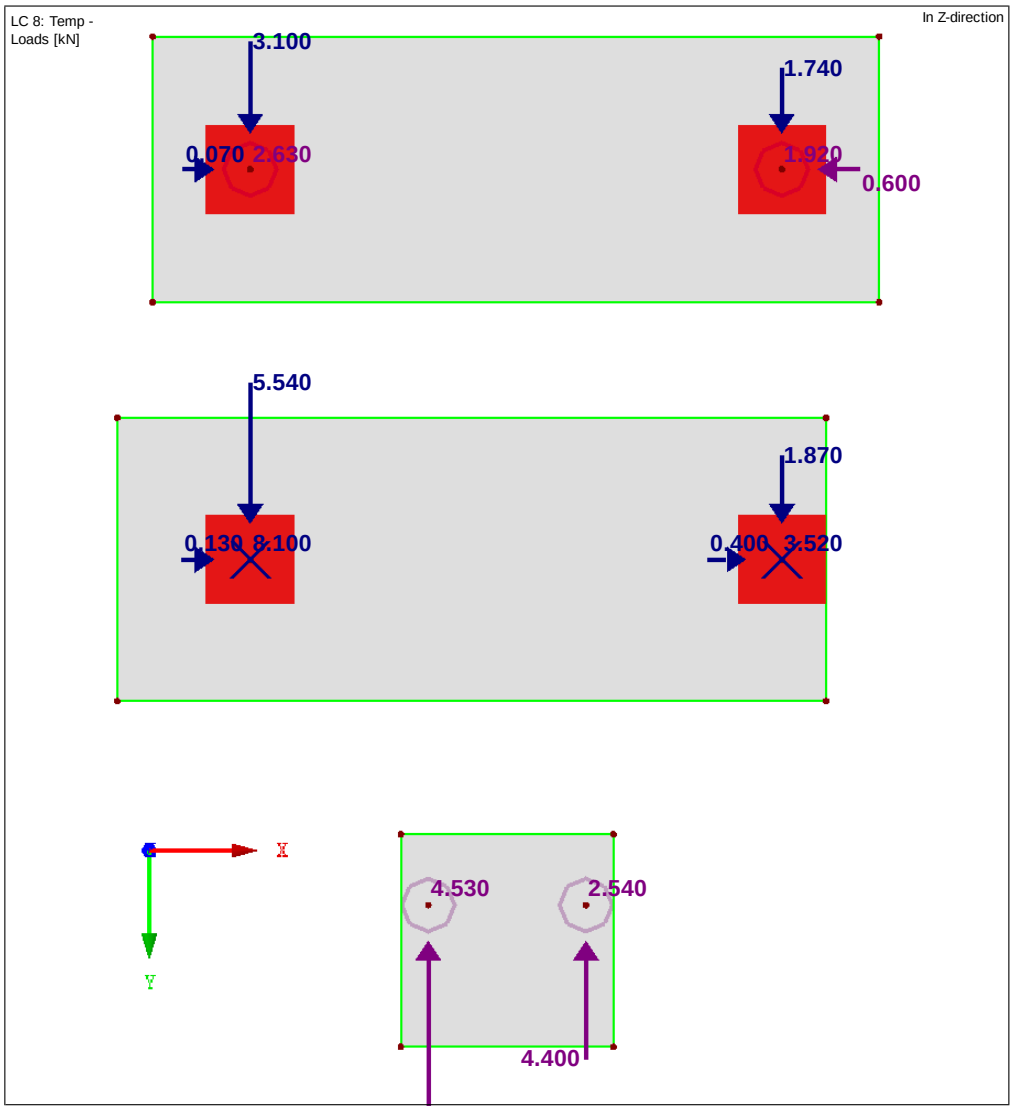


Figure 29 Temperature -25

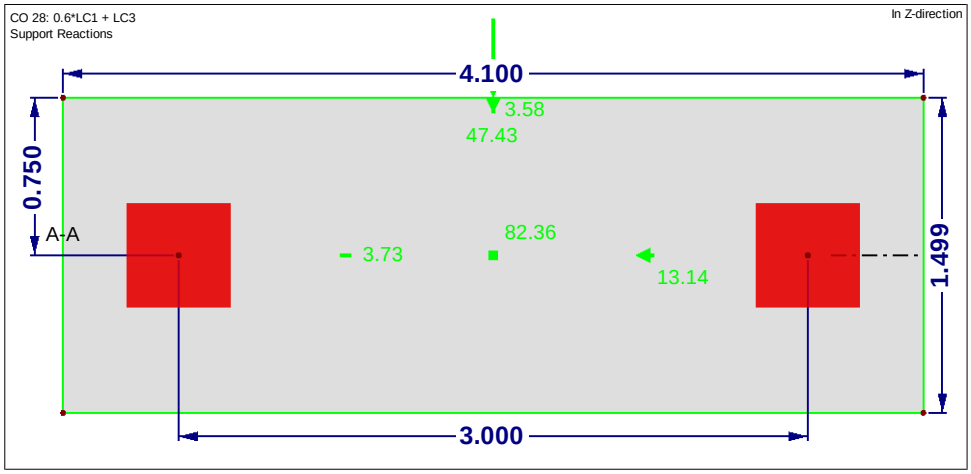
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8.3. Stability check

8.3.1. north footing

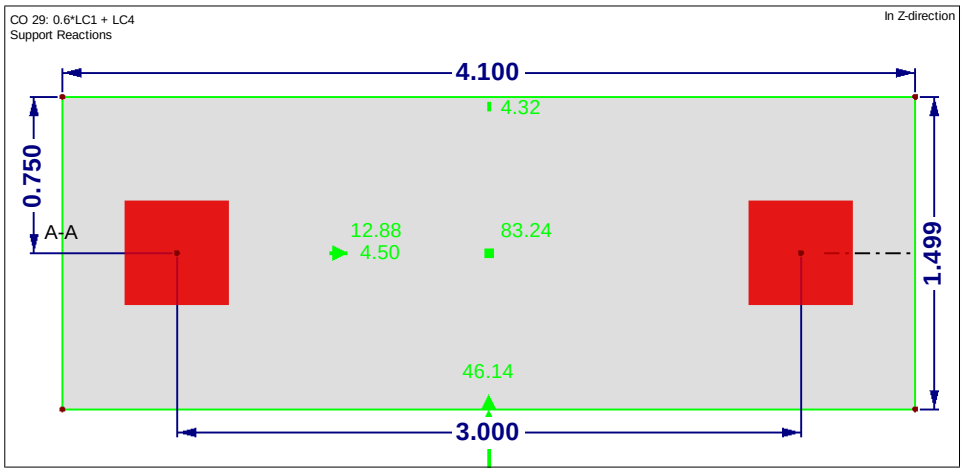


OVERTURN:	My=	47.43	kNm	overturn moment				
	Fz=	82.36	kN	total vertical load				
	e=	0.576	m	eccentricity				
	B/2=	2.05	m					
	F.O.S=	3.560	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	13.14	kN	lateral force in x				
	Fy=	3.58	kN	lateral force in y				
	Hd=	13.62	kN	sliding force				
	Rd=	32.944	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	2.419	>1.50	Factor of safety against sliding				

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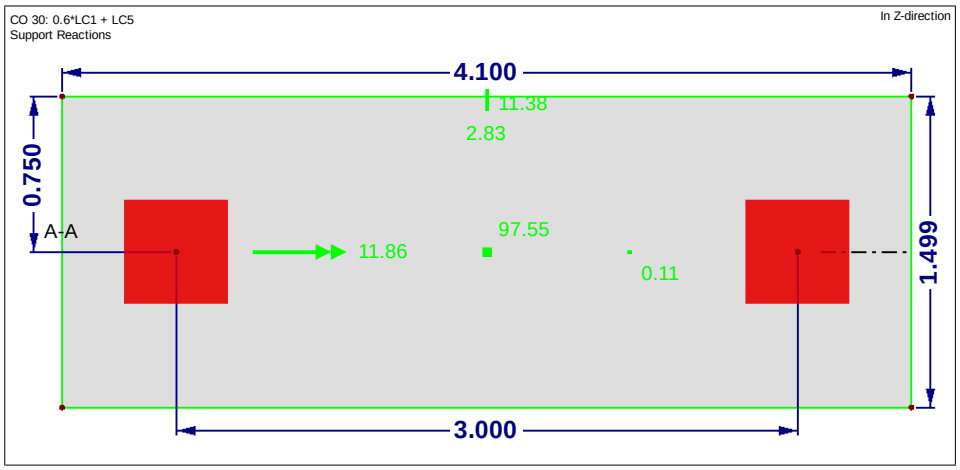


OVERTURN:	My=	46.14	kNm	overturn moment			
	Fz=	83.24	kN	total vertical load			
	e=	0.554	m	eccentricity			
	B/2=	2.05	m				
	F.O.S=	3.698	>1.50	Factor of safety against overturning			
SLIDING:	Fx=	12.88	kN	lateral force in x			
	Fy=	4.32	kN	lateral force in y			
	Hd=	13.59	kN	sliding force			
	Rd=	33.296	kN	resistance to sliding			
	F.O.S=	2.451	>1.50	Factor of safety against sliding			

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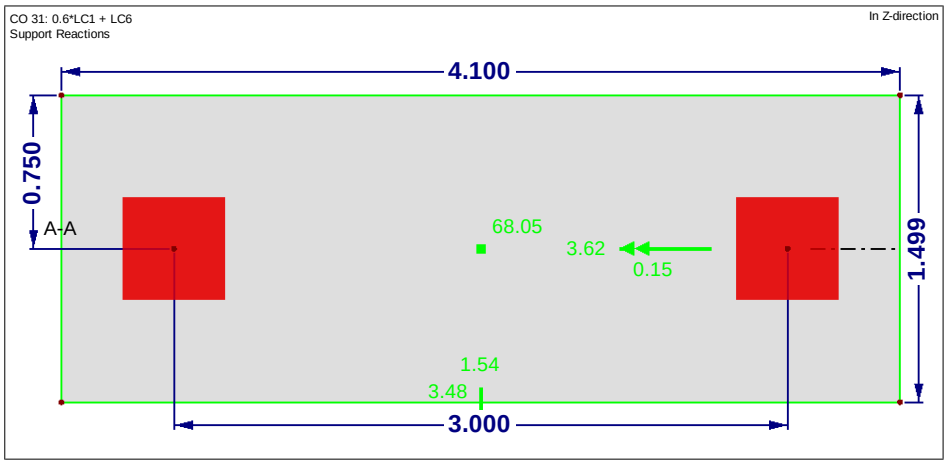


OVERTURN:	Mx=	11.86	kNm	overturn moment		
	Fz=	97.55	kN	total vertical load		
	e=	0.122	m	eccentricity		
	B/2=	0.75	m			
	F.O.S=	6.169	>1.50	Factor of safety against overturning		
SLIDING:	Fx=	0.11	kN	lateral force in x		
	Fy=	11.38	kN	lateral force in y		
	Hd=	11.38	kN	sliding force		
	Rd=	39.02	kN	resistance to sliding		
	F.O.S=	3.429	>1.50	Factor of safety against sliding		

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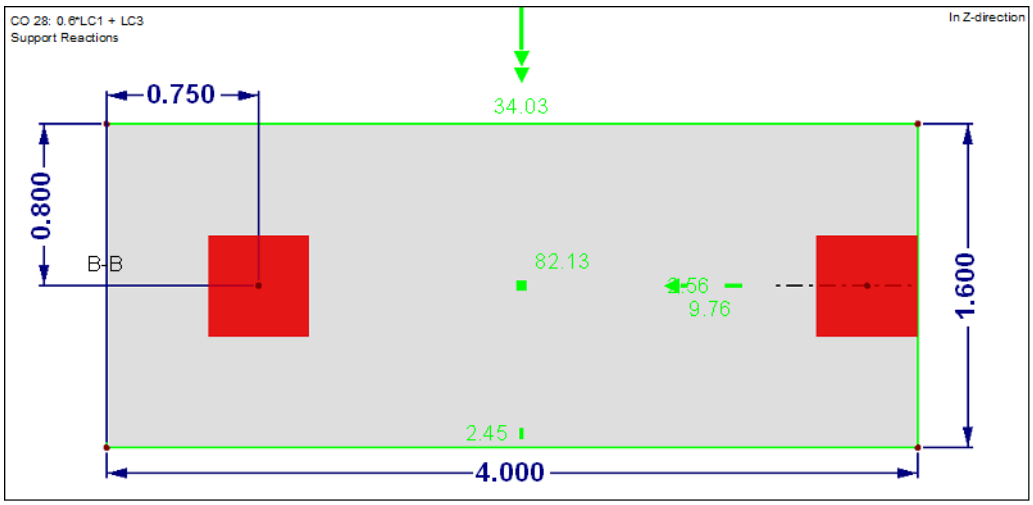
OVERTURN:	Mx=	3.62	kNm	overturn moment		
	Fz=	68.05	kN	total vertical load		
	e=	0.053	m	eccentricity		
	B/2=	0.75	m			
	F.O.S=	14.099	>1.50	Factor of safety against overturning		
SLIDING:	Fx=	0.15	kN	lateral force in x		
	Fy=	3.48	kN	lateral force in y		
	Hd=	3.48	kN	sliding force		
	Rd=	27.22	kN	resistance to sliding		
	F.O.S=	7.815	>1.50	Factor of safety against sliding		

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8.3.2. south footing

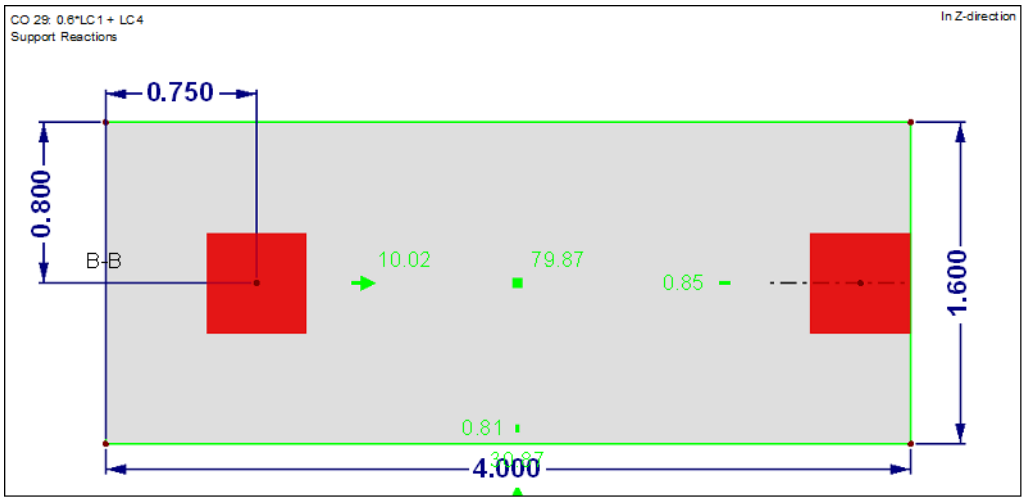


OVERTURN:	My=	34.03	kNm	overturn moment				
	Fz=	82.13	kN	total vertical load				
	e=	0.414	m	eccentricity				
	B/2=	1.954	m					
	F.O.S=	4.716	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	9.76	kN	lateral force in x				
	Fy=	2.45	kN	lateral force in y				
	Hd=	10.06	kN	sliding force				
	Rd=	32.852	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	3.265	>1.50	Factor of safety against sliding				

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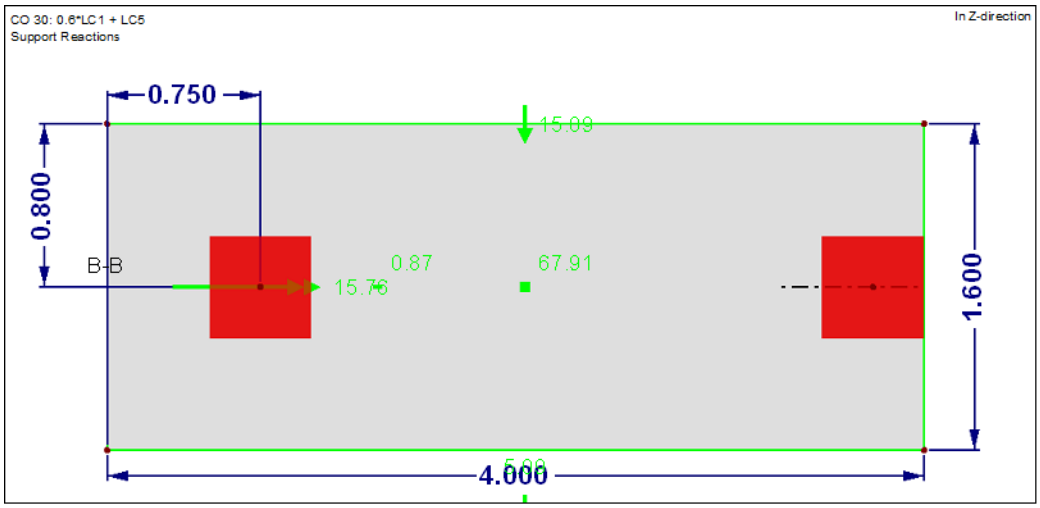


OVERTURN:	My=	30.87	kNm	overturn moment			
	Fz=	79.87	kN	total vertical load			
	e=	0.387	m	eccentricity			
	B/2=	1.954	m				
	F.O.S=	5.056	>1.50	Factor of safety against overturning			
SLIDING:	Fx=	10.02	kN	lateral force in x			
	Fy=	0.81	kN	lateral force in y			
	Hd=	10.05	kN	sliding force			
	Rd=	31.948	kN	resistance to sliding			
	F.O.S=	3.178	>1.50	Factor of safety against sliding			

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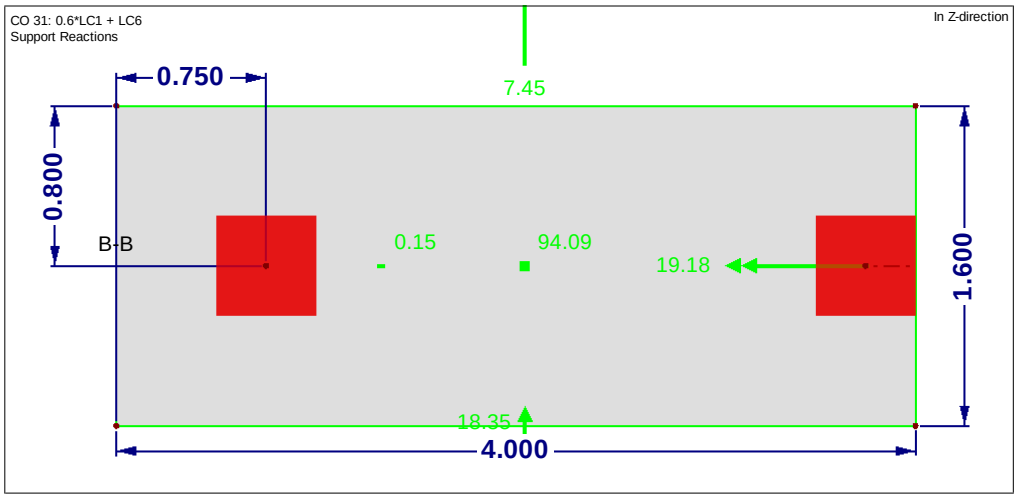


OVERTURN:	Mx=	15.76	kNm	overturn moment			
	Fz=	67.91	kN	total vertical load			
	e=	0.232	m	eccentricity			
	B/2=	0.8	m				
	F.O.S=	3.447	>1.50	Factor of safety against overturning			
SLIDING:	Fx=	0.87	kN	lateral force in x			
	Fy=	15.09	kN	lateral force in y			
	Hd=	15.12	kN	sliding force			
	Rd=	27.164	kN	resistance to sliding			
	F.O.S=	1.797	>1.50	Factor of safety against sliding			

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OVERTURN:	Mx=	19.18	kNm	overturn moment			
	Fz=	94.09	kN	total vertical load			
	e=	0.204	m	eccentricity			
	B/2=	0.8	m				
	F.O.S=	3.925	>1.50	Factor of safety against overturning			
SLIDING:	Fx=	0.15	kN	lateral force in x			
	Fy=	18.35	kN	lateral force in y			
	Hd=	18.35	kN	sliding force			
	Rd=	37.636	kN	resistance to sliding			
	F.O.S=	2.051	>1.50	Factor of safety against sliding			

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8.4. Design of pedestal

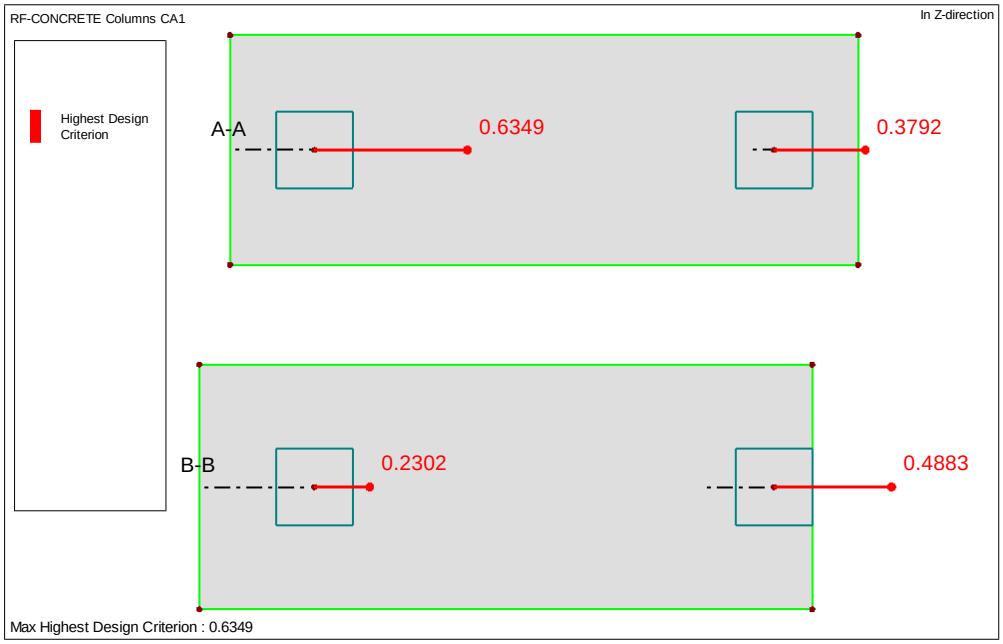


Figure 30 Highest design criterion

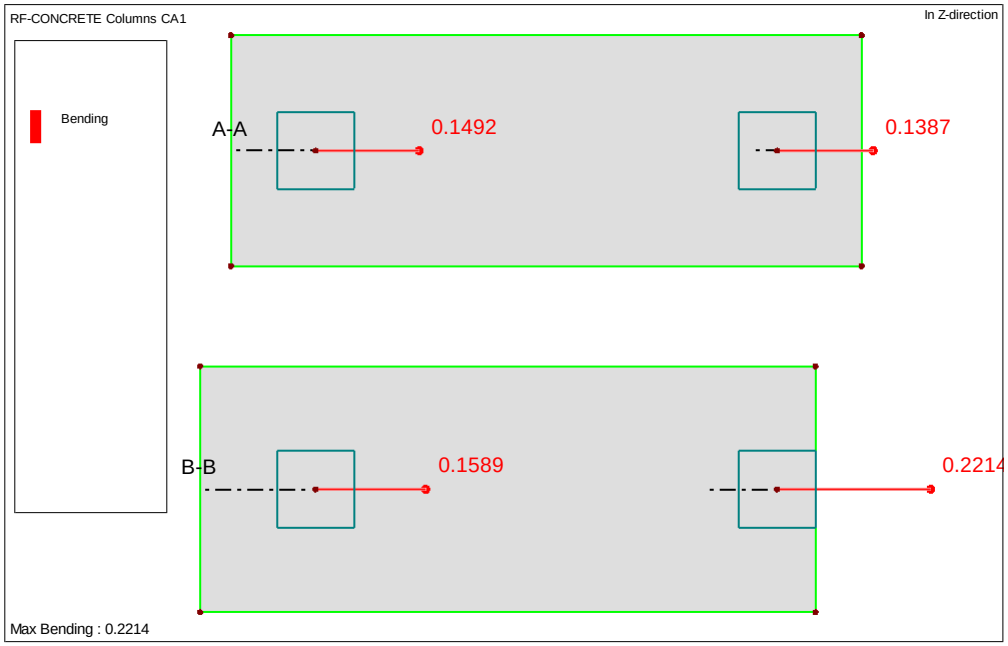


Figure 31 Utility ratio for bending

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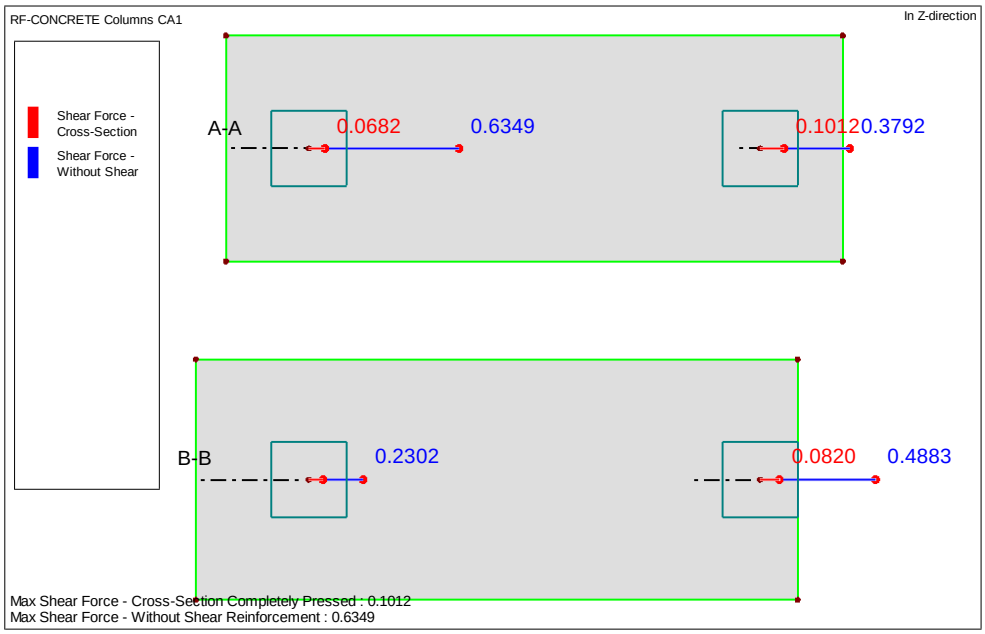


Figure 32 Utility ratio for shear

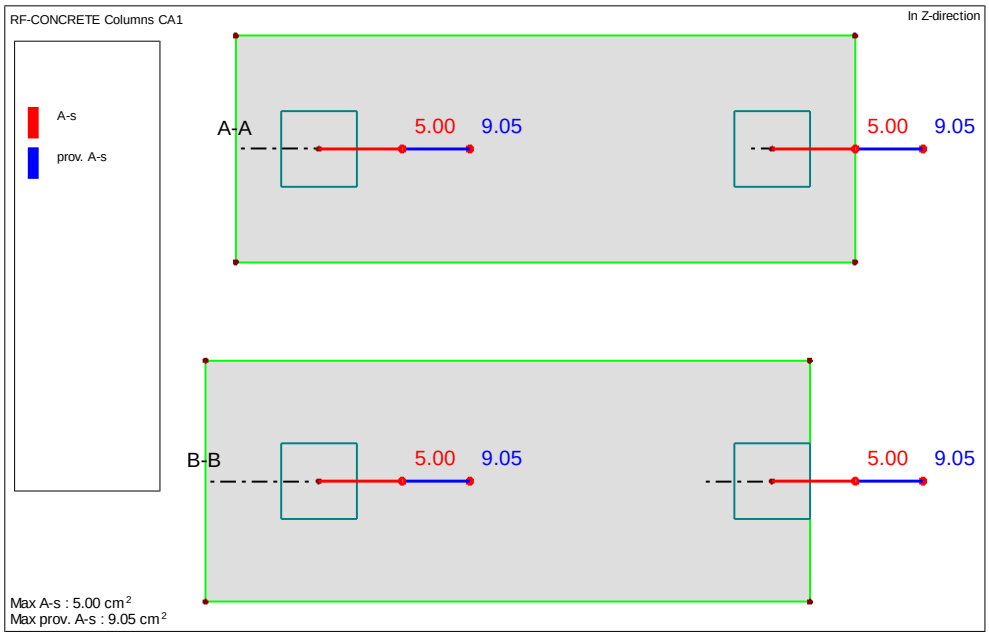


Figure 33 Required and provided longitudinal reinforcement

Provide main bars 8T12, As=9.05cm^2

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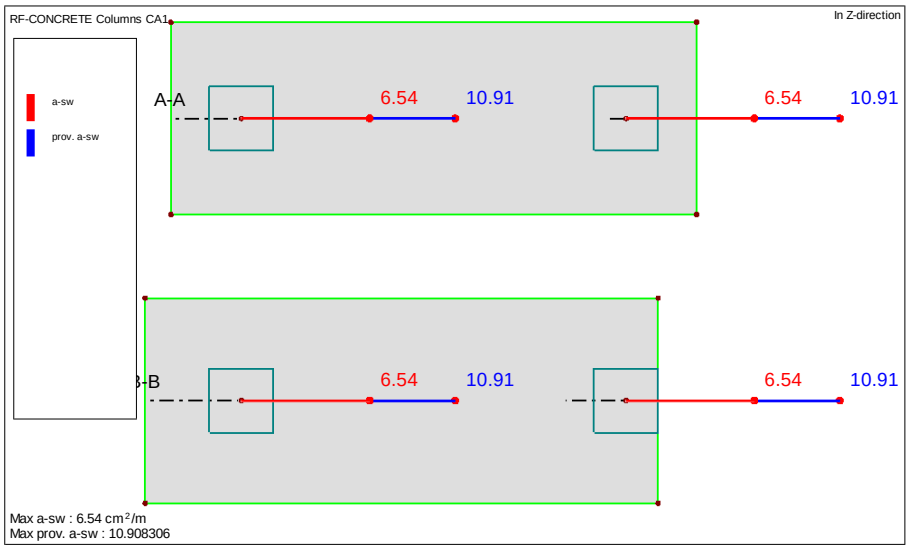


Figure 34 Required and provided shear reinforcement

Provide shear reinforcement T10/150mm, As=10.9cm^2

8.5. Design of raft

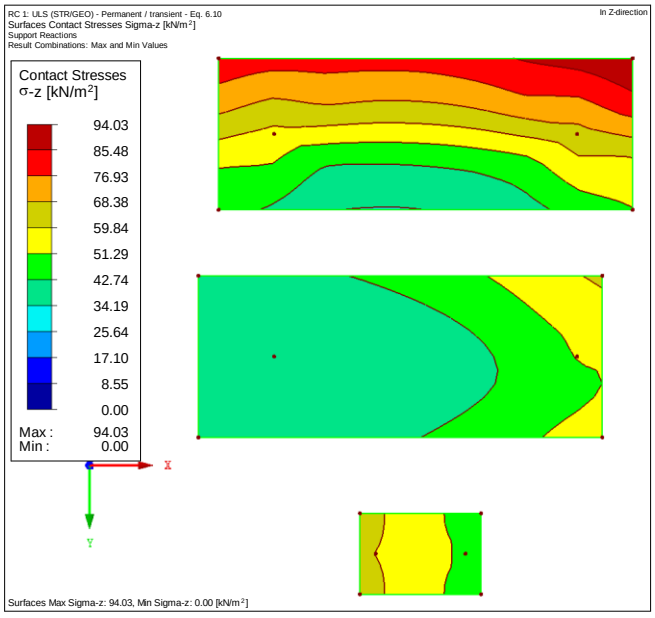


Figure 35 Soil contact stress at ULS

$\sigma_s < \sigma_{s,max} = 375kN/m^2$, ok!

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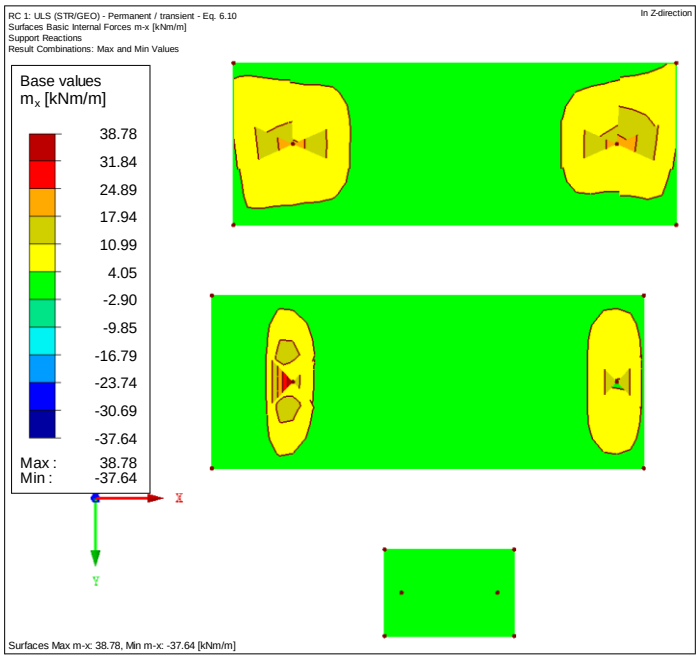


Figure 36 Bending moment Mx

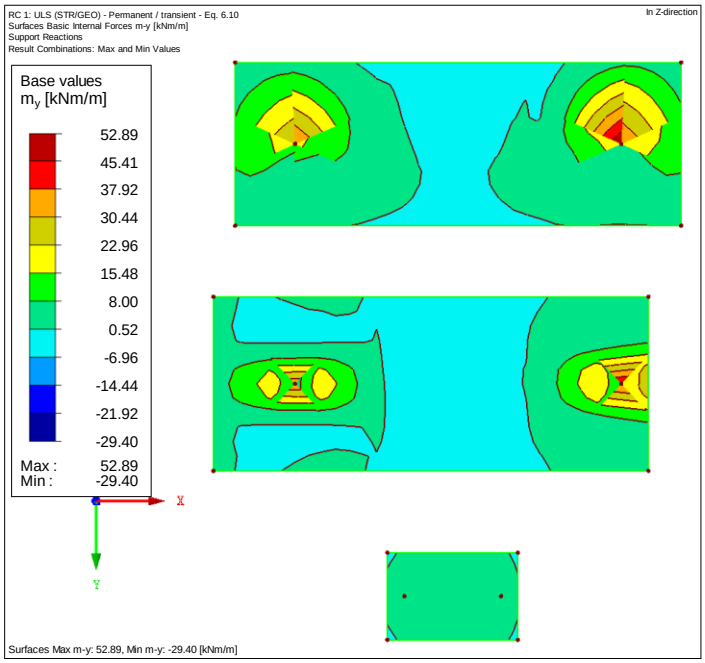


Figure 37 Bending moment My

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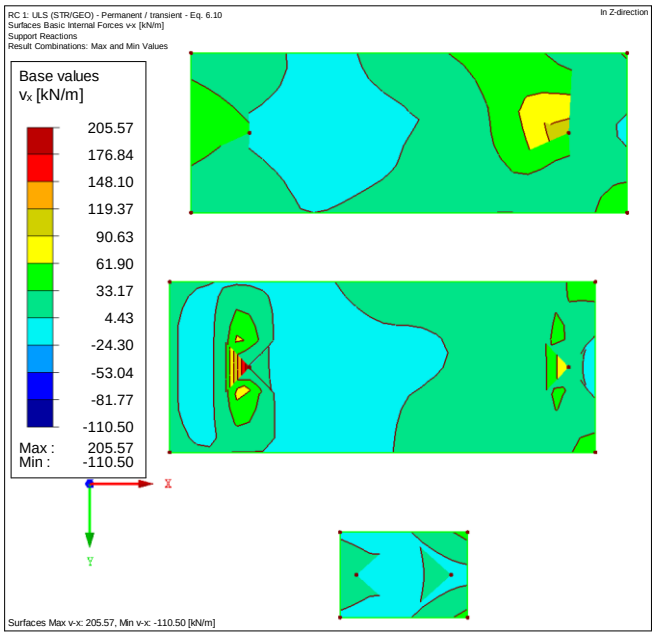


Figure 38 Shear force Vx

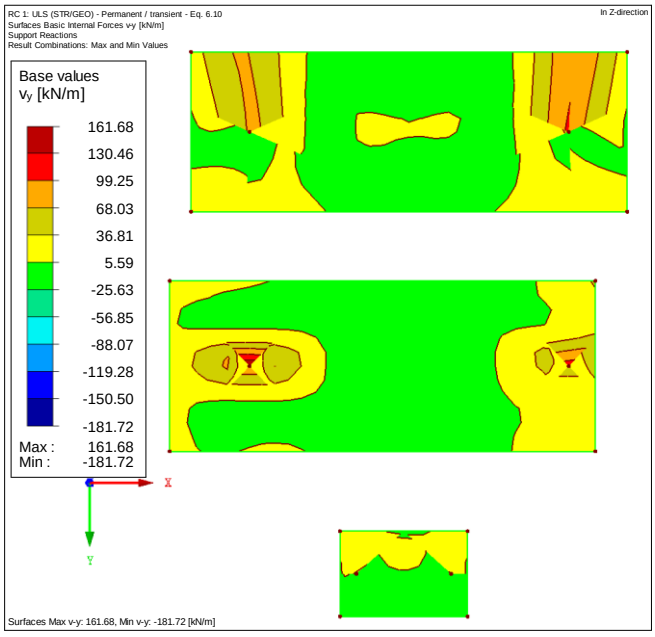


Figure 39 Shear force Vy

Shear force result is concentrated at one point because pedestal dimension is not considered. Applied shear force is much less in reality.

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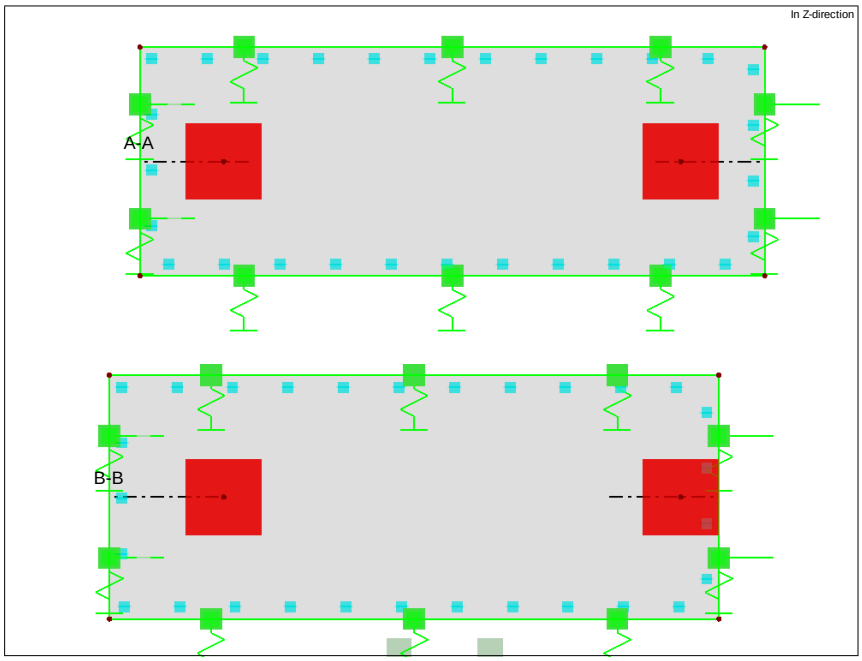


Figure 40 Sections created in RFEM

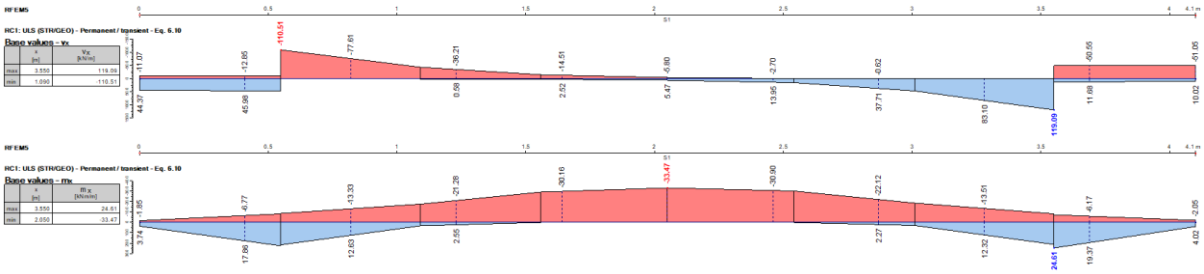


Figure 41 Shear force and bending moment for section A-A

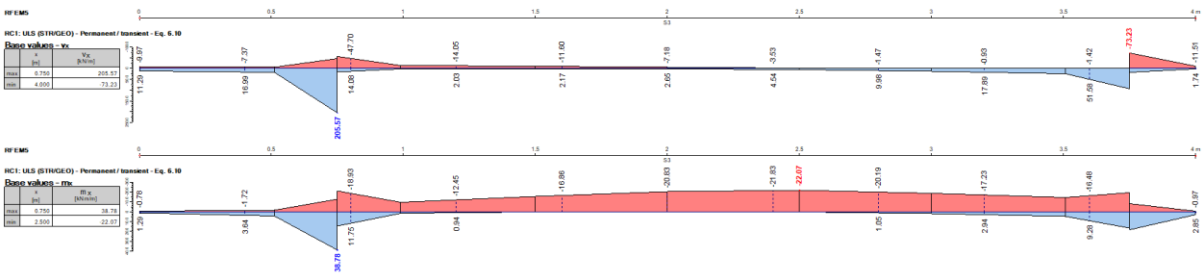


Figure 42 Shear force and bending moment for section B-B

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PUNCHING SHEAR CHECK AT DISTANCE d/2 FROM COLUMN FACE

material property in [MPa]

$f_{ck} := 35 \quad f_{yk} := 420$

effective depth in [m]

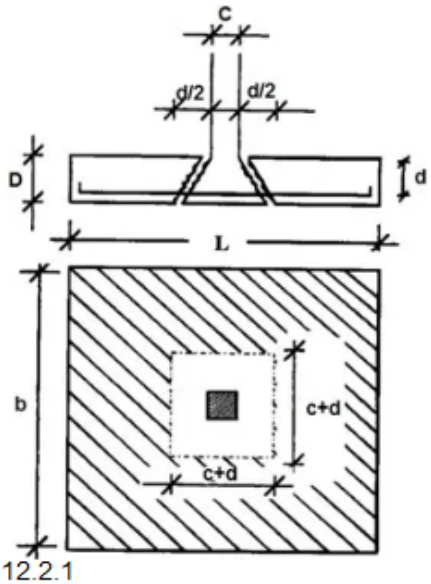
$d := 0.400 - 0.075 - \frac{0.016}{2} = 0.317$

column dimension in [m]

$c_x := 0.5 \quad c_y := 0.5$

perimeter for critical section in [m]

$b_0 := 2 \cdot 2\left(c_x + 2 \cdot \frac{d}{2}\right) = 3.268$



ratio of long side to short side of the column

$\beta_c := 1$

for corner columns

$\alpha_s := 20$

concrete Shear strength in [kN], SBC 304 Ch. 11.12.2.1

$$V_c := \min\left[\left(1 + \frac{2}{\beta_c}\right) \cdot \frac{\sqrt{f_{ck}} \cdot b_0 \cdot d}{6} \cdot 10^3, \left(\frac{\alpha_s \cdot d}{b_0} + 2\right) \cdot \frac{\sqrt{f_{ck}} \cdot b_0 \cdot d}{12} \cdot 10^3, \frac{\sqrt{f_{ck}} \cdot b_0 \cdot d}{3} \cdot 10^3\right] = 2.012 \times 10^3$$

reduction factor for shear

$\phi := 0.75$

Design value of concrete shear strength in [kN]

$\phi \cdot V_c = 1.509 \times 10^3$

max applied shear force for column M6 in [kN]

$V_{Ed} := 205.57$

factored shear force at section in [kN]

$V_u := 1.35 \cdot V_{Ed} = 277.519$

punching shear check

$\frac{V_u}{\phi \cdot V_c} = 0.184$ ok!

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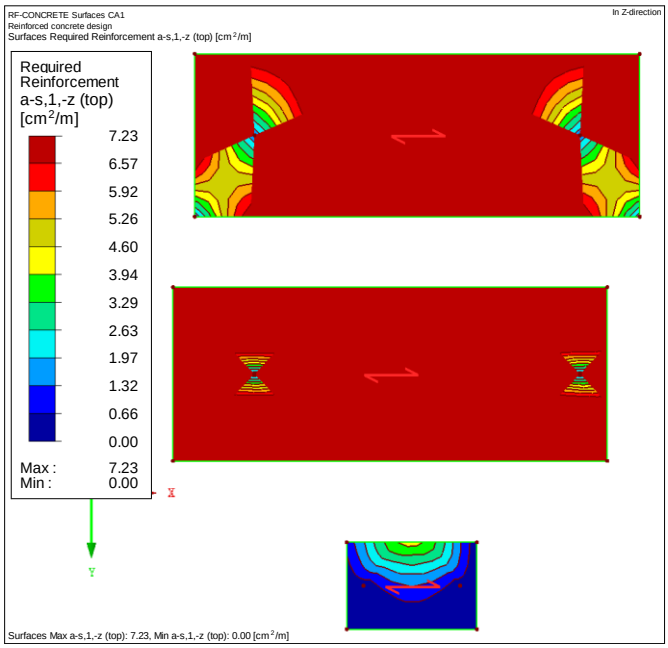


Figure 43 Required top reinforcement $a_{s,1,top}$

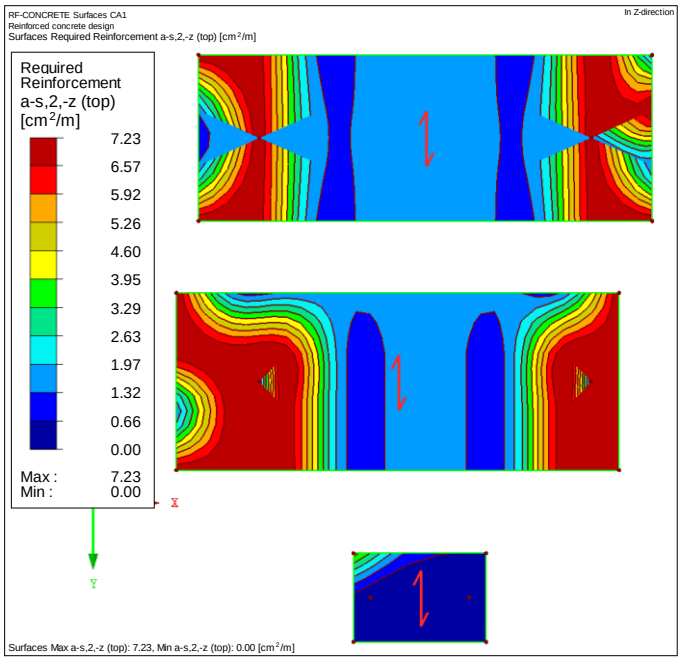


Figure 44 Required top reinforcement $a_{s,2,top}$

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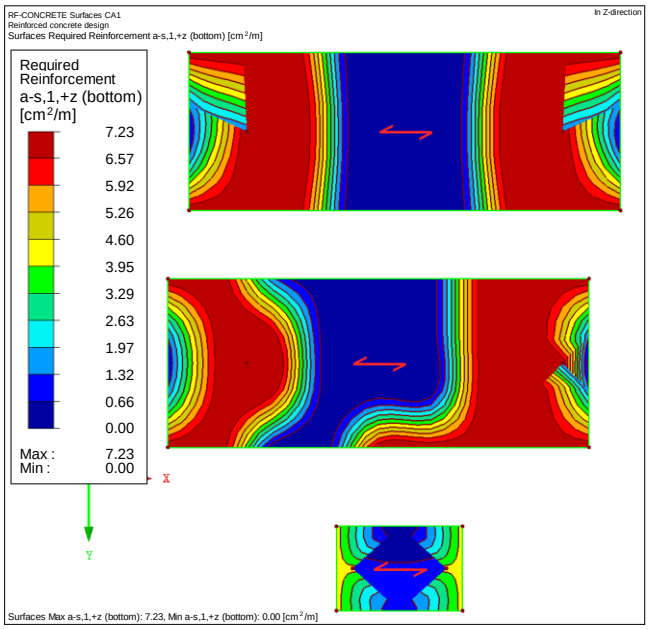


Figure 45 Required top reinforcement $a_{s,1,bottom}$

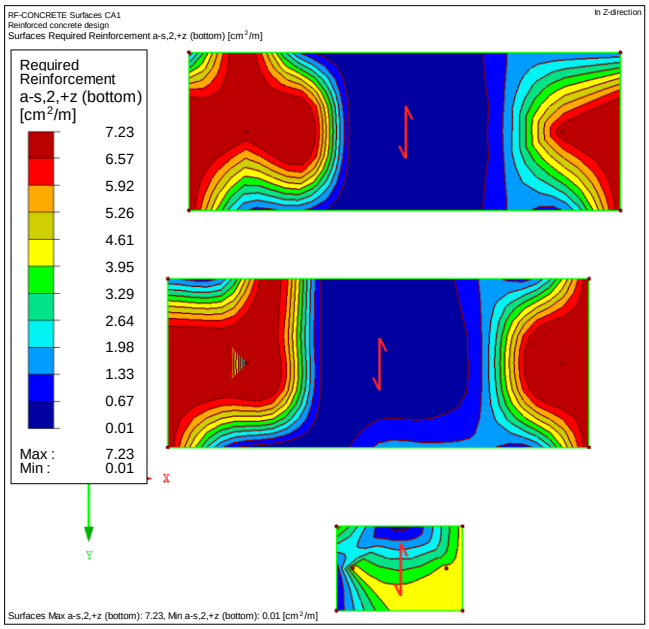


Figure 46 Required top reinforcement $a_{s,2,bottom}$

Provide T16/200mm, A_{sp}= 10,05cm^2/m for combined footing and T12/150mm for stair landing.

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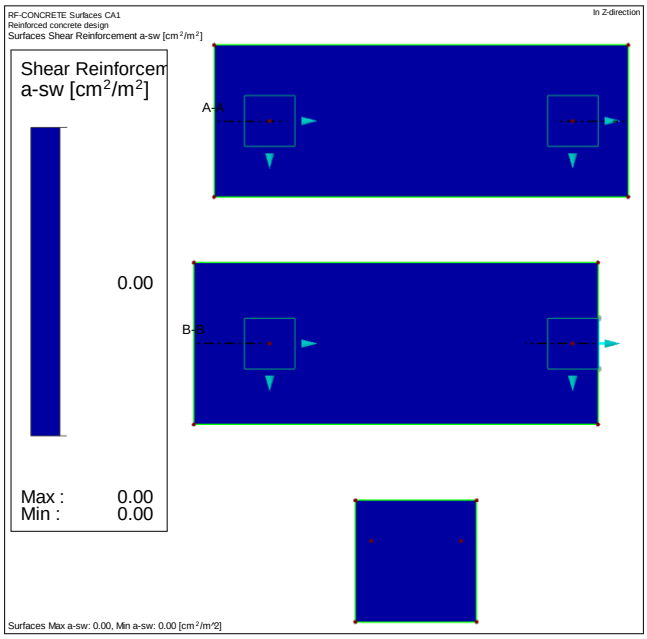


Figure 47 Required shear reinforcement

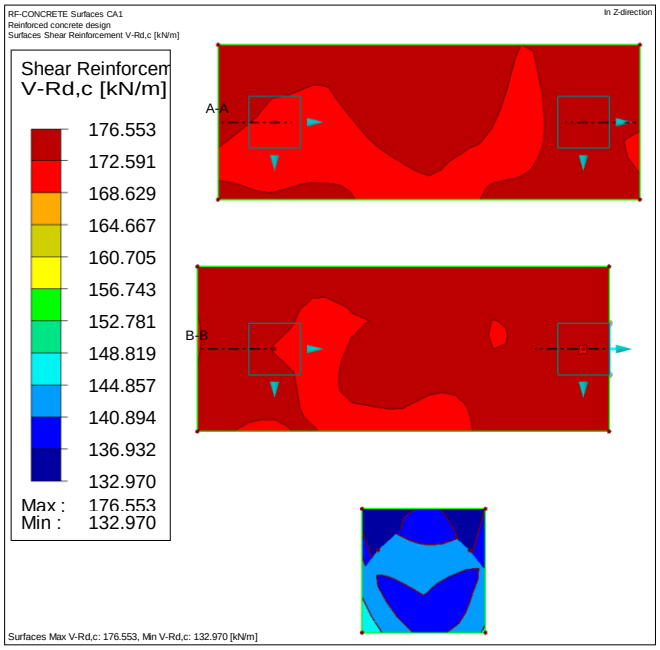


Figure 48 Design shear resistance without shear reinforcement

Provide T12/200mm as shear reinforcement.

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9. SLUDGE BASIN

9.1. Geometry of the structures

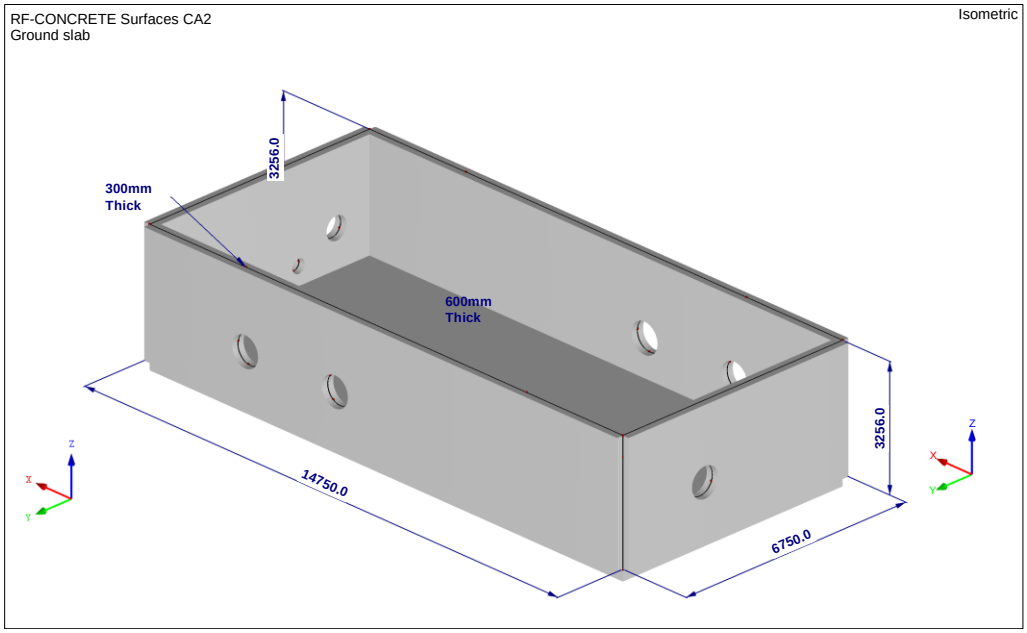


Figure 49. Geometry of the structure

9.2. Design loads

The structure is loaded with dead load, live load in form of fluid load and wind load. In addition to that there is point load on the top the structures. Thus the basic loads can be classified according to SBC-301 as shown in the table below.

1	Dead load (D)
2	Operational load (L)
3	Wind load (W)

The height of the wall is 3.256m with a freeboard of 0.5m for the fluid, thus the maximum height of the fluid is 2.756m. The maximum characteristic fluid pressure can be calculated as product of the Density and height 10 kN/m^3 * 2.756m = 27.56 kN/m^2

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Wind load is calculated according to SBC 301 ch. 6. according to geographical location of the project basic wind speed is 43 m/s. Wind direction factor 0.85, Topographical factor k_{zt} =1, Ground elevation factor k_e =1

9.3. Load combinations

Ultimate Strength Design - Ultimate Load Combinations: SBC_301, 2.3.2)

Load Combination	Expression
101	1,4 D
102	1,2 D + 1,6 L
103	1,2 D + 1,0 L +1,6 W
104	0,9 D + 1,6 W

Allowable Stress Design - Serviceability Load Combinations: SBC_301, 2.4.1)

Load Combination	Expression
201	D
202	D + L
203	D + 0,75 L
204	D + W
205	D + 0,75 W + 0,75 L
206	0,6 D + W

9.4. Result

Displacement

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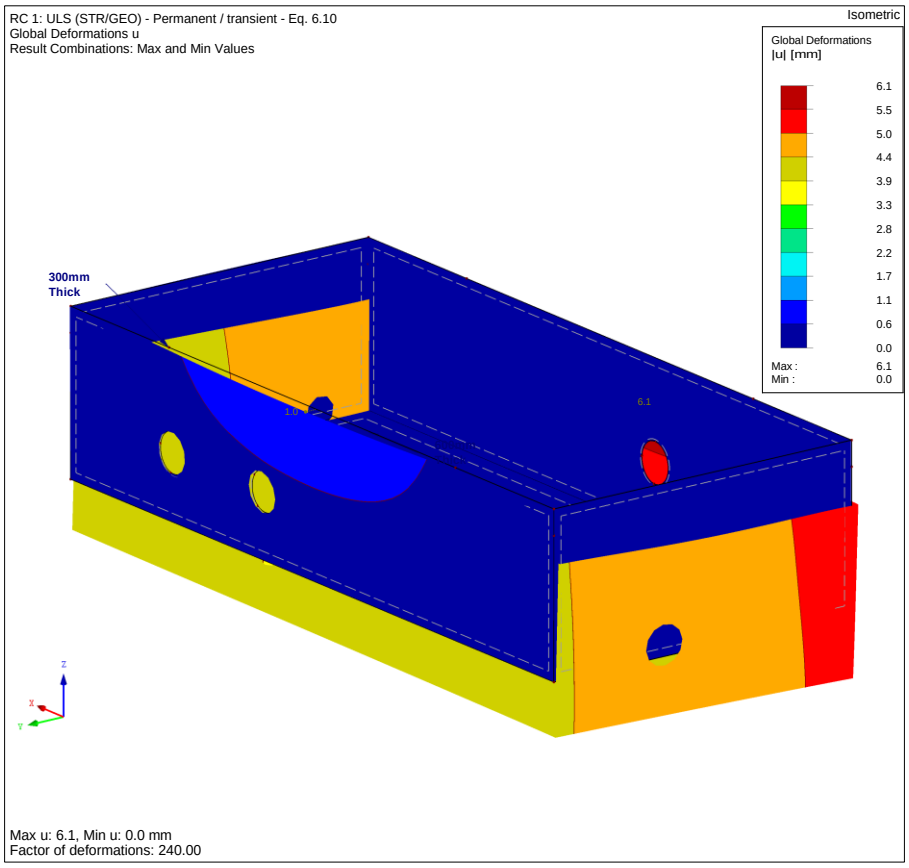


Figure 50. Critical displacement (max 6mm)

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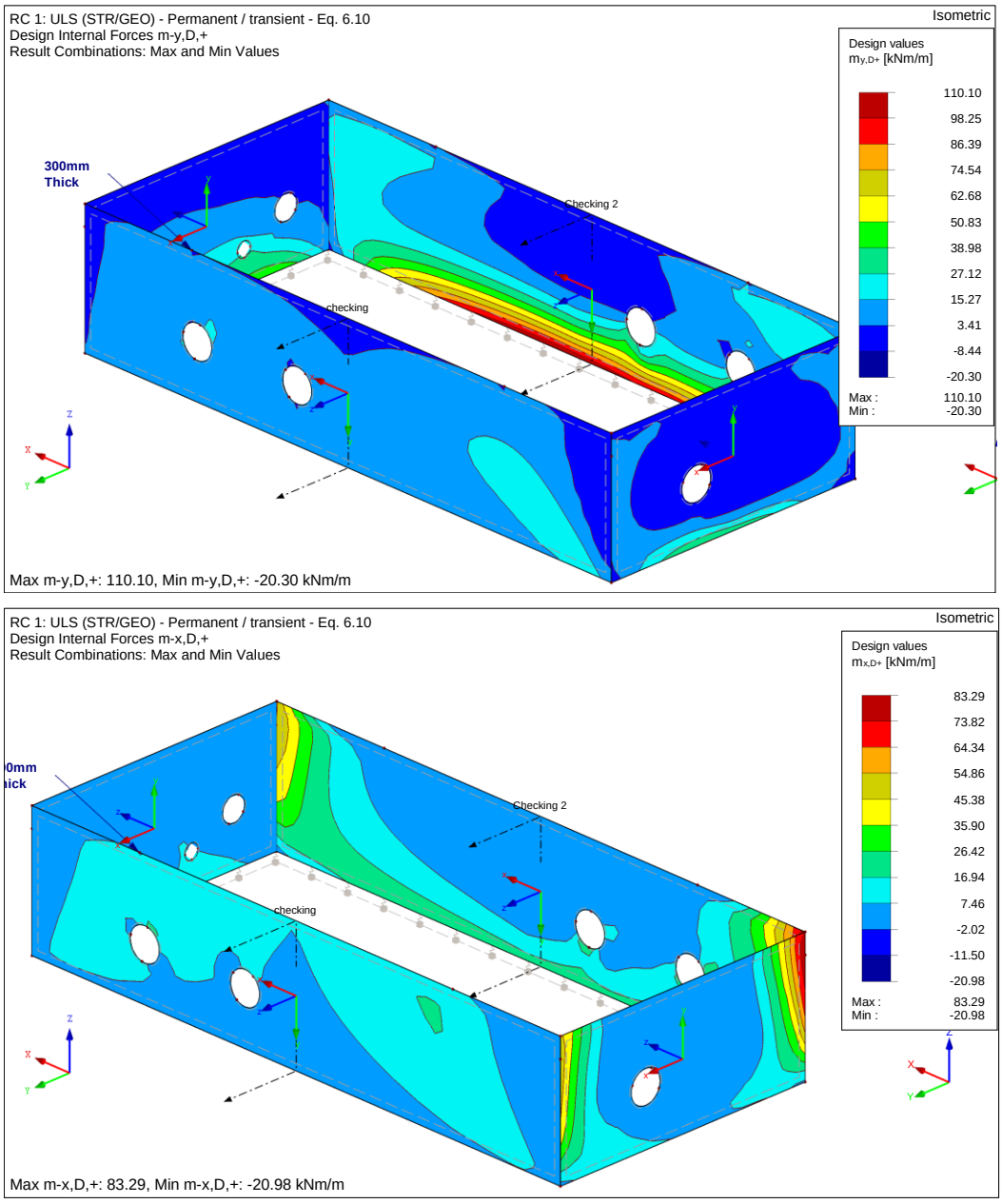


Figure 51. Critical moments in X Y direction

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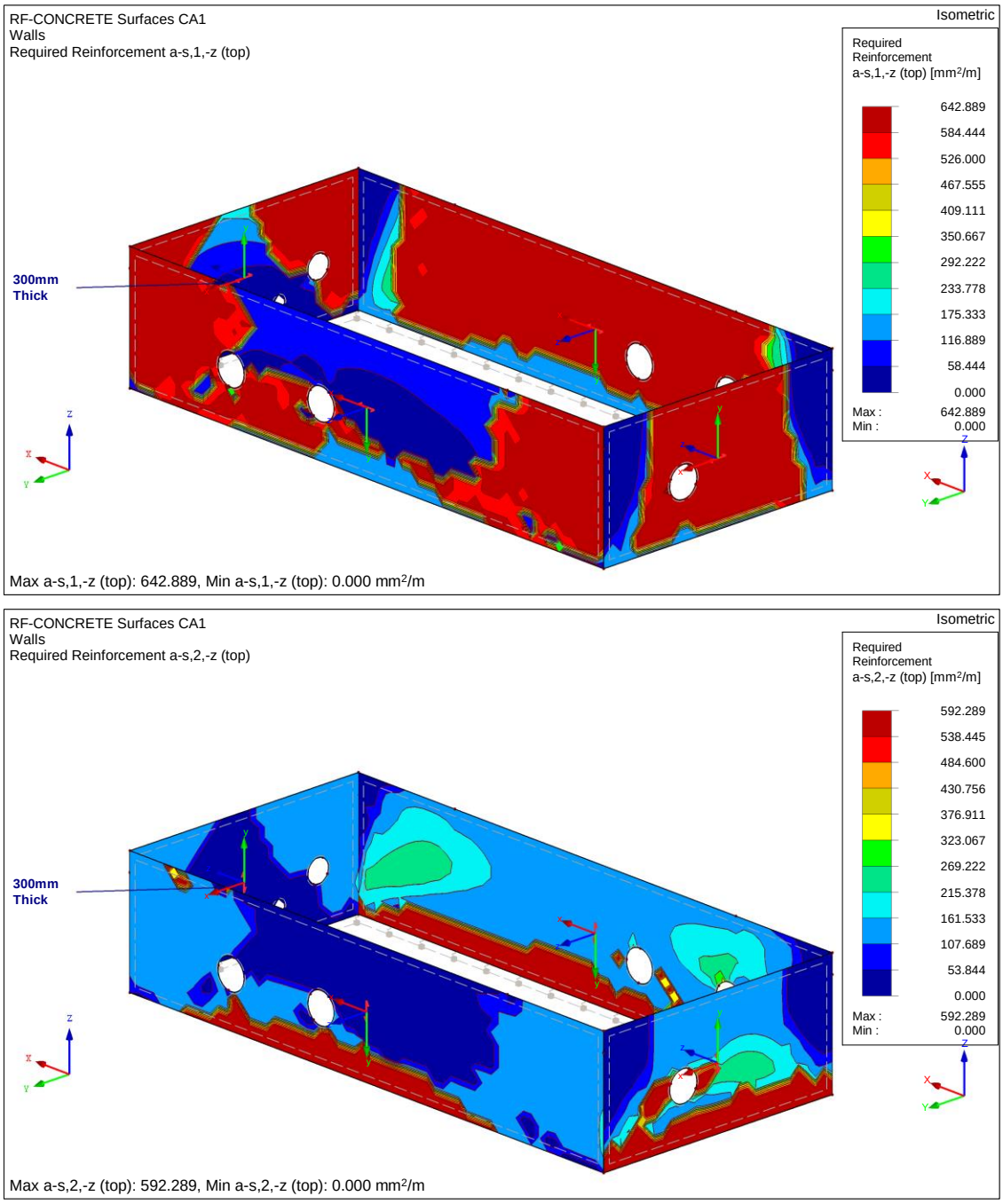
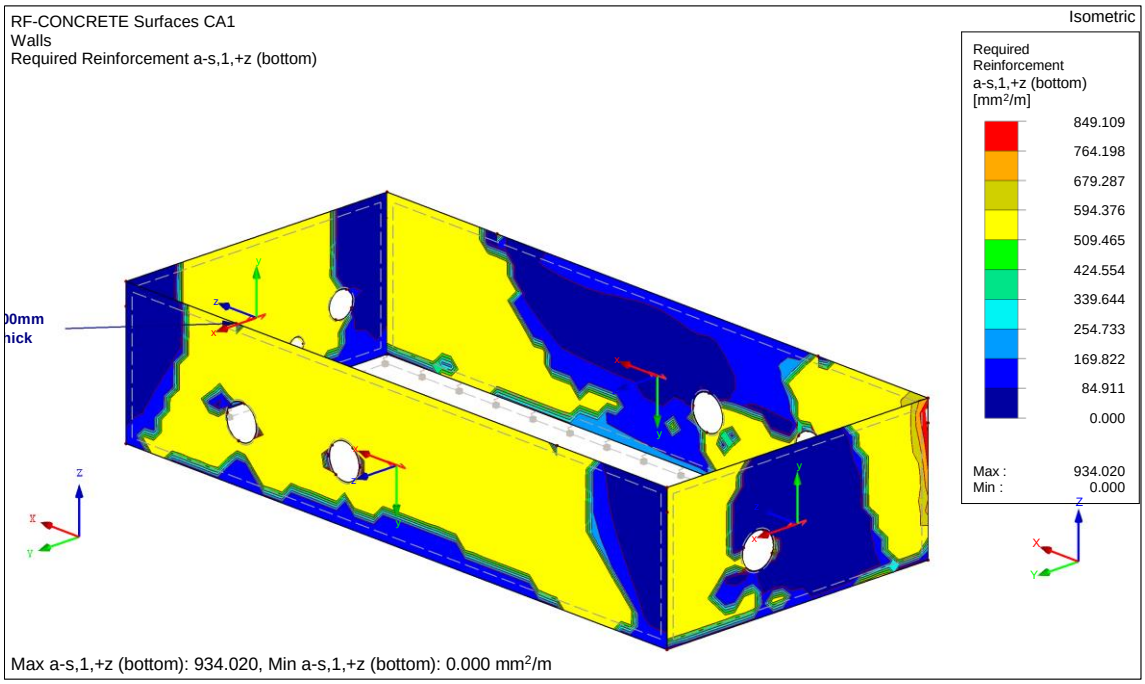


Figure 52. Required inner surface reinforcement (Provide T12K150)

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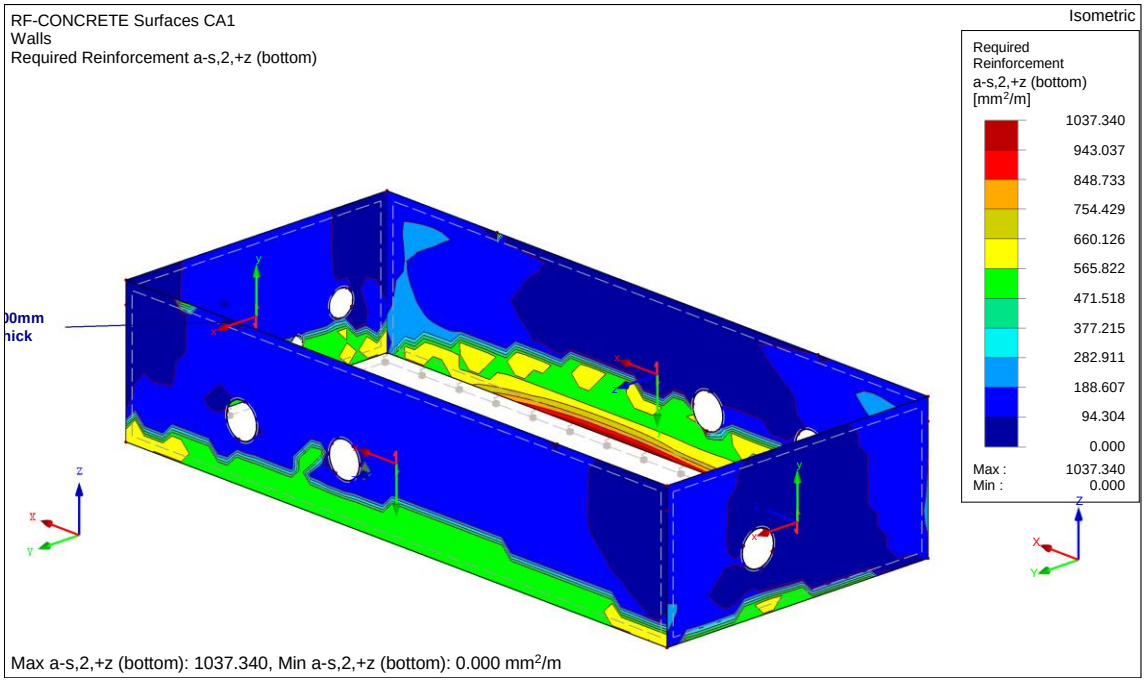


Figure 53. Required outer reinforcement (Provide T12K150)

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Note: Wall are 300mm thick and they are reinforcement with 12mm diameter bars spaced by 150mm center to center. In the outer surface shows slightly higher required reinforcement, Since the starter bars are T16k150 the provided reinforcement will meet the requirements.

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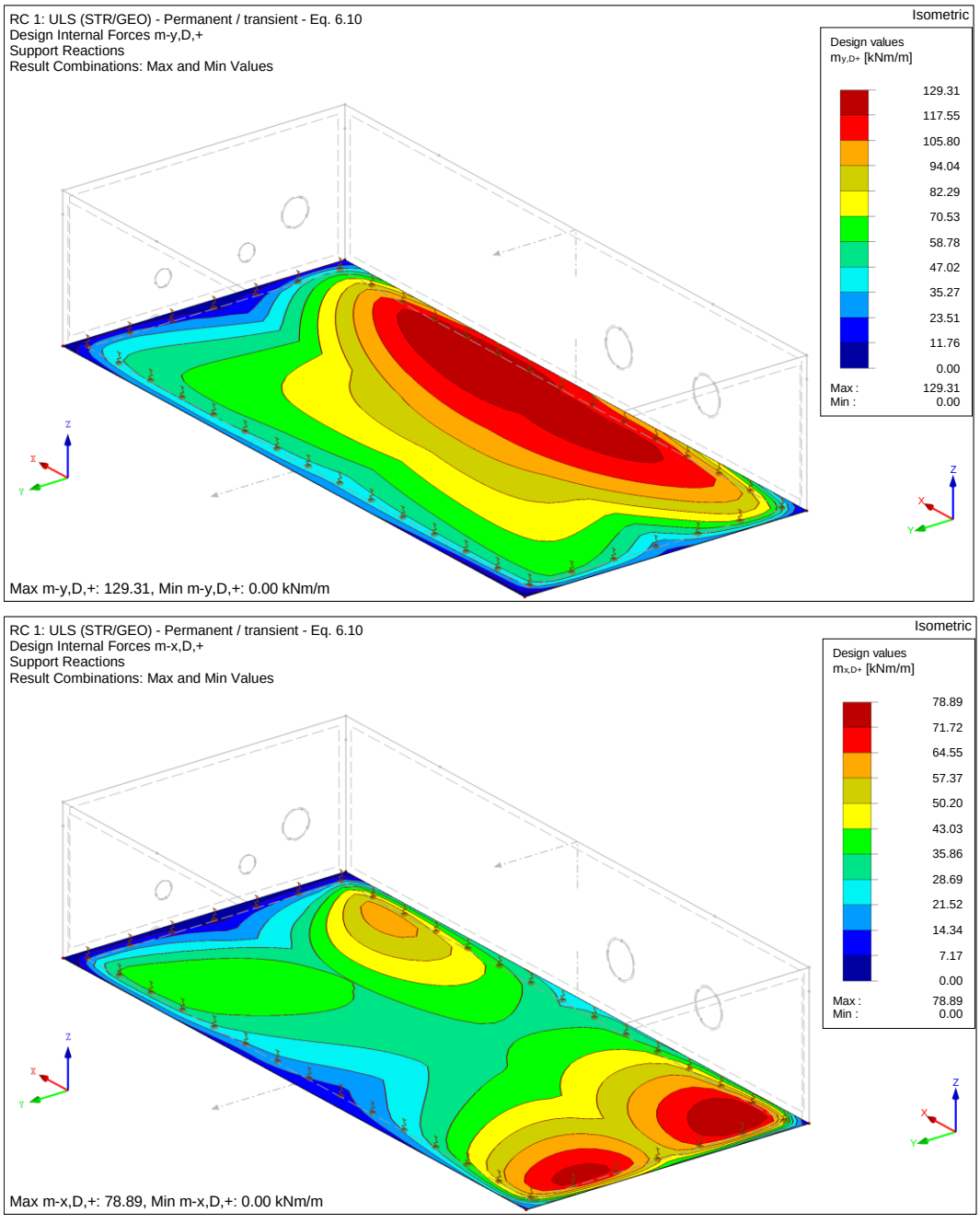
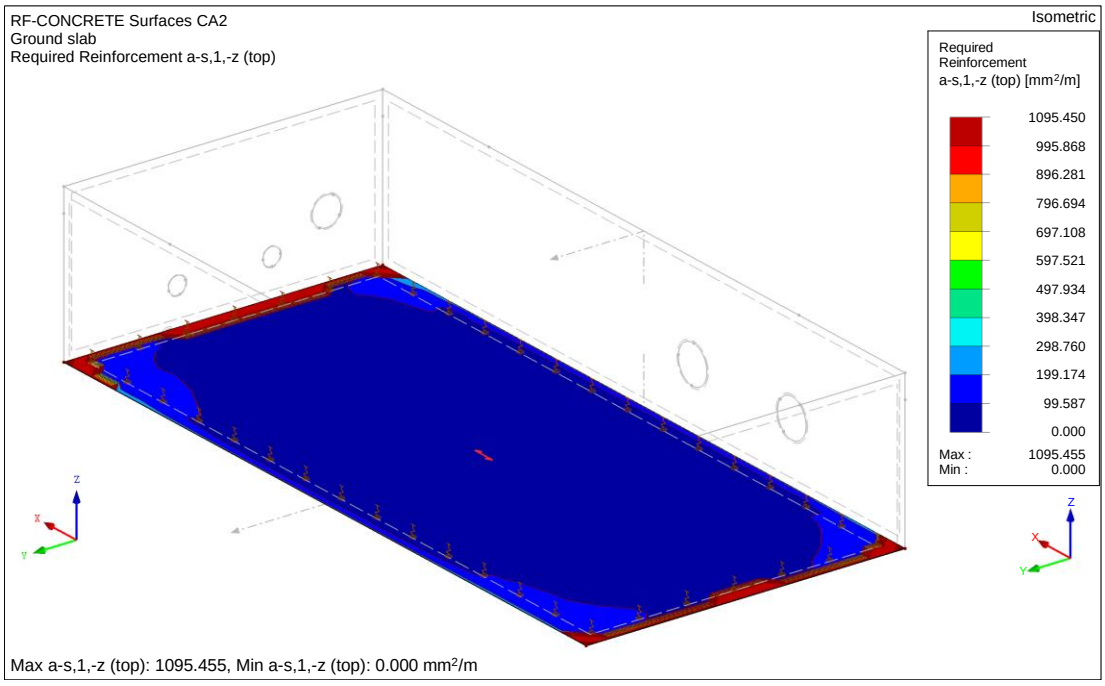


Figure 54. Critical bending moments

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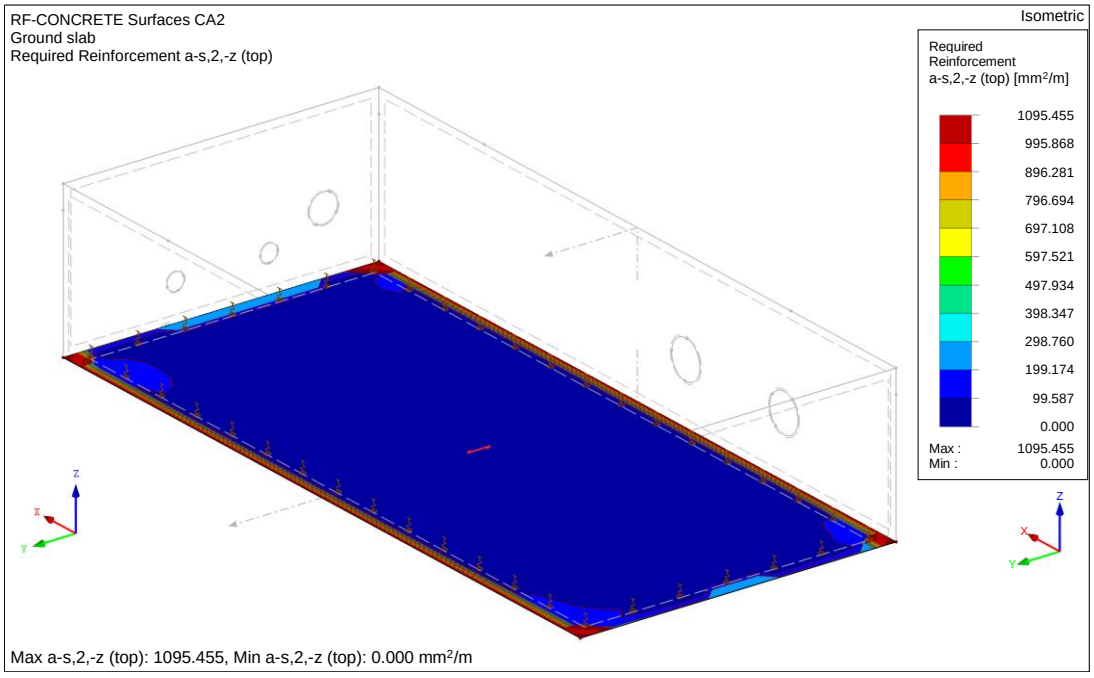
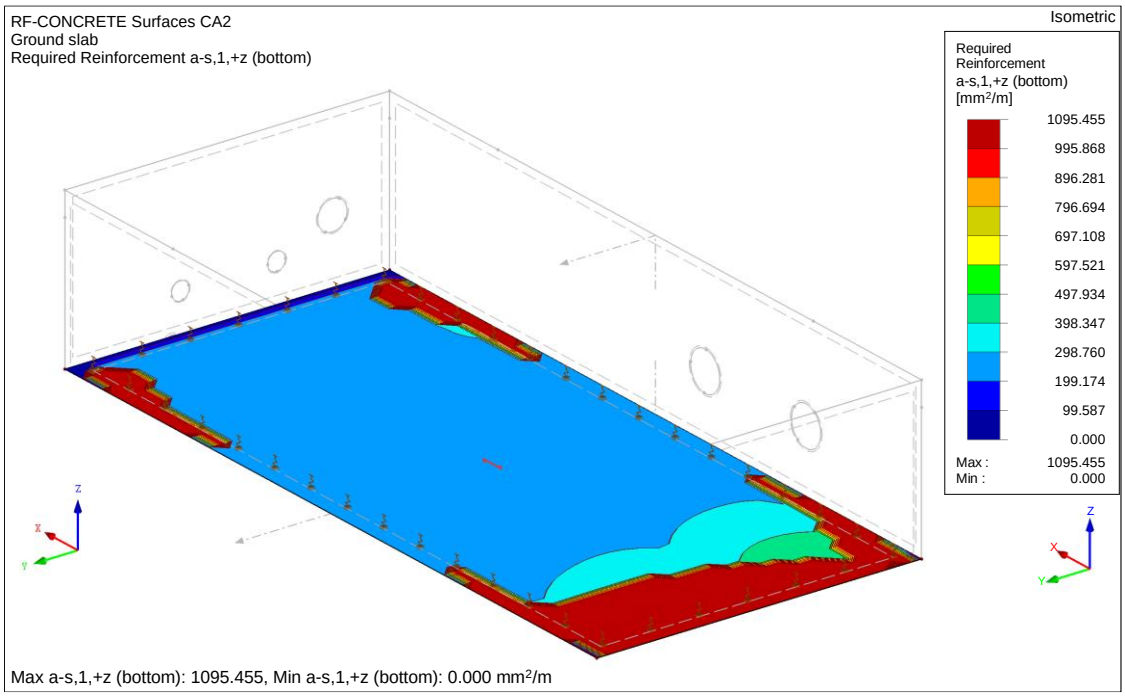


Figure 55. Required top reinforcement (Provide T12K150)

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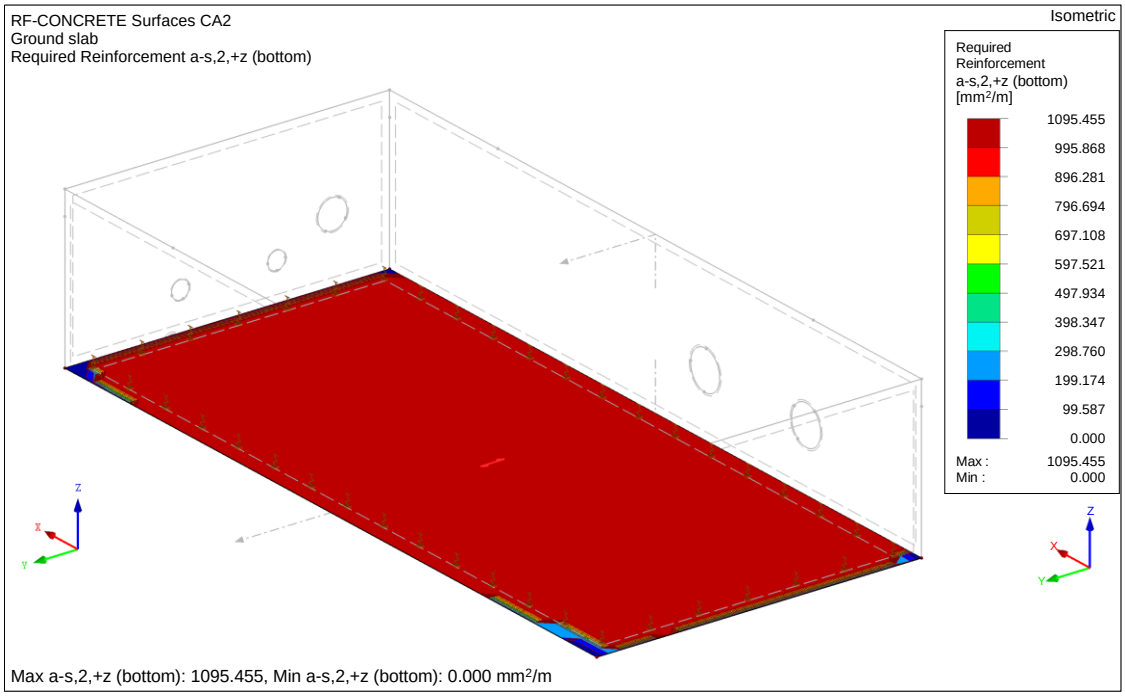


Figure 56. Required bottom reinforcement (Provide T16K150)

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Note: The base slab is 600mm thick, it is reinforced with T12 on the top and T16 on the bottom. All bars are space by 150mm center to center.

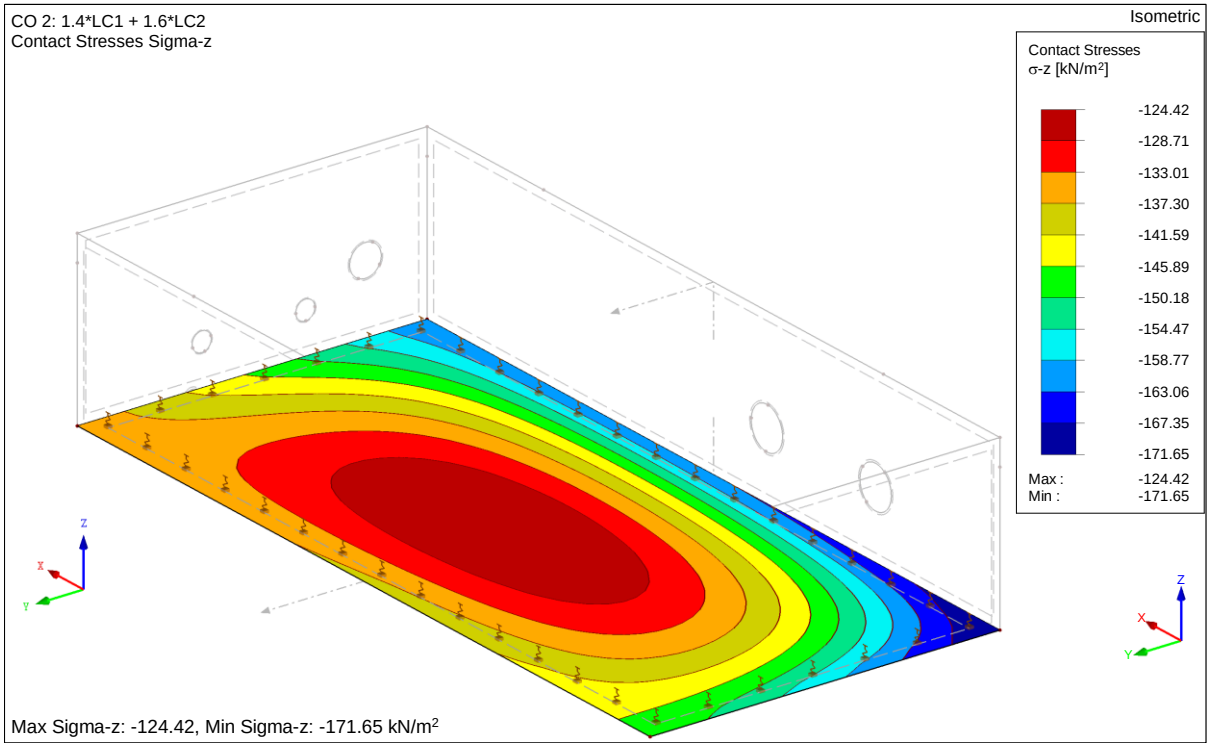


Figure 57. Critical contact stress

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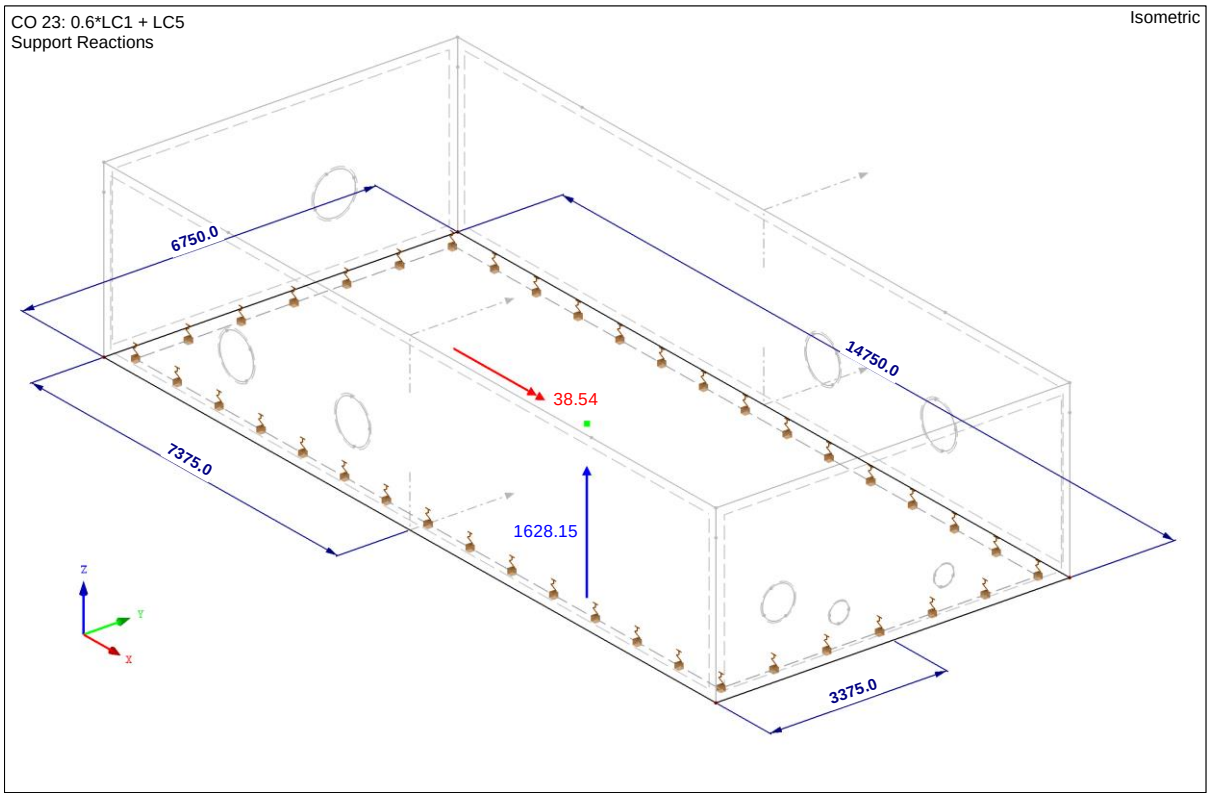
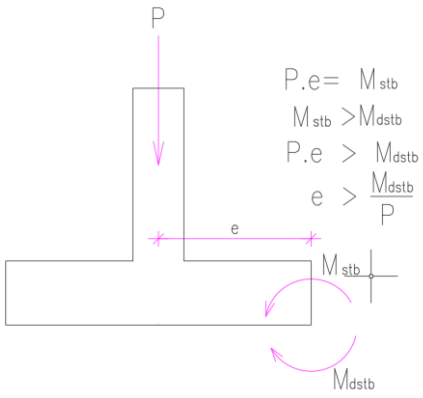


Figure 58. Overturn resistance

The overturn resistance is verified by comparing eccentricities. the eccentricity resulted by overturn moment and total reaction force in z direction is compared to the eccentricity of the structure in the same direction.



The maximum destabilizing moment: $M_{dstb} = 34.54$ kNm

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Reaction force downward P = 1628 kN

Eccentricity $e = \frac{M_{dstb}}{P}$ =0.021 m

Verification $\frac{e}{3.375m} = 0.006$ Safe

Cracking

The maximum crack width is 0.17 mm and it is localized, generally the applied moment is bellow the cracking moment. Since it is a water retaining structure the inner surface should be coated according to Ma'aden design specifications.

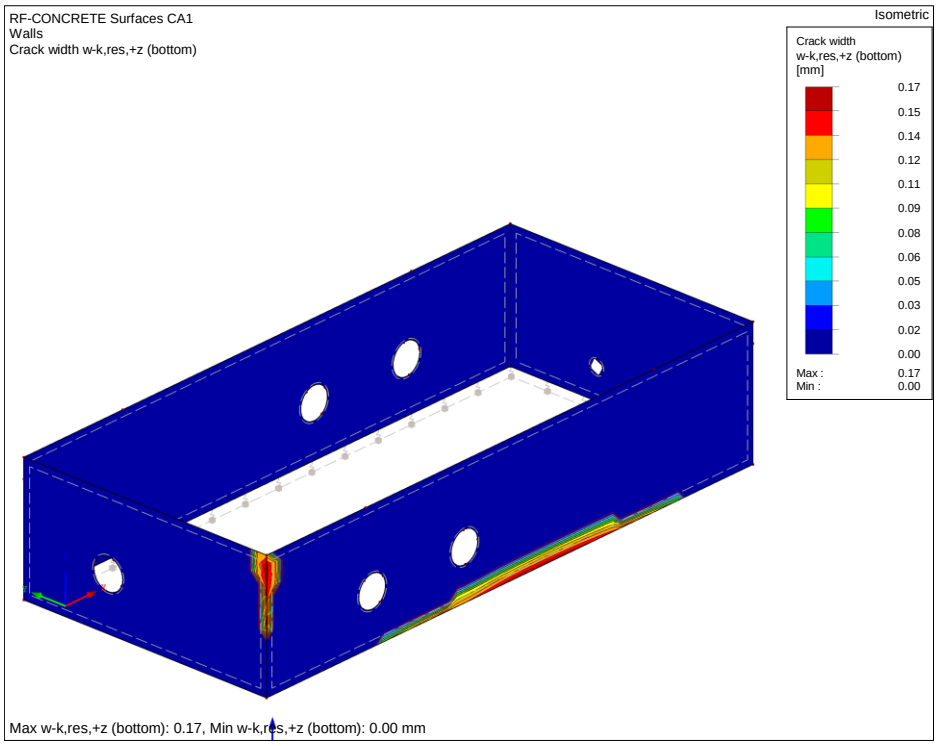


Figure 59. Crack width

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10. DISPERSION VESSEL FOUNDATION

10.1. Geometry and loading

The dispersion vessel is supported by three columns which transfer the weight to the ground slab.

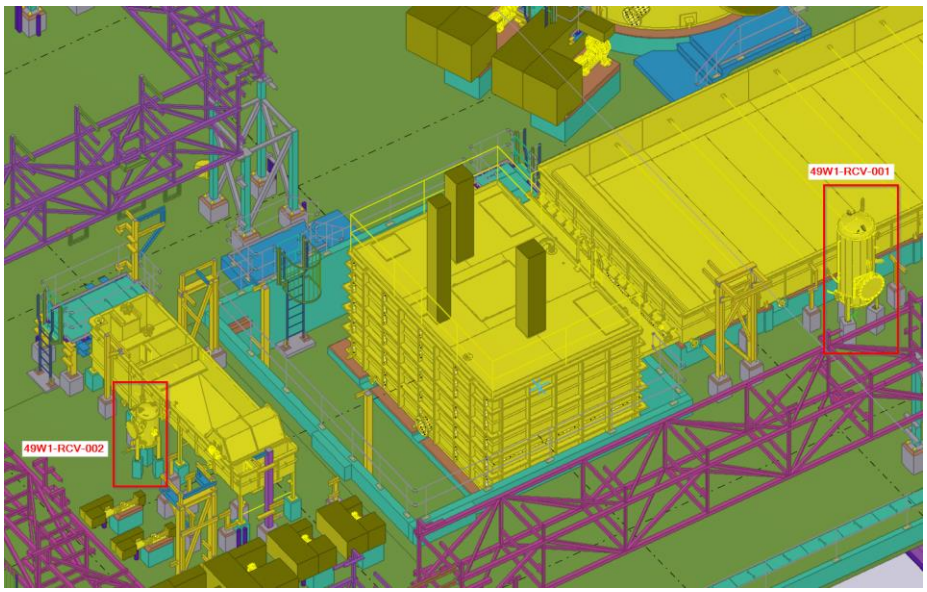


Figure 60 location of the dispersion vessels

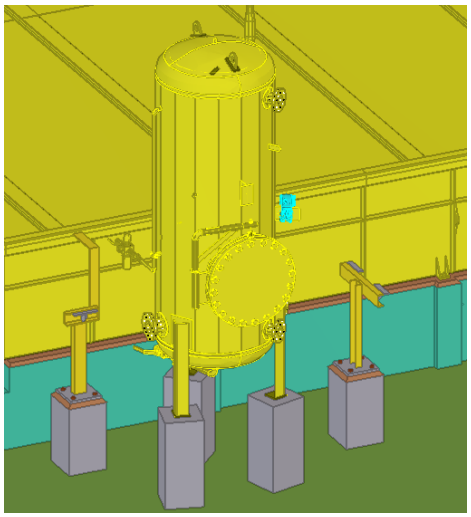


Figure 61 Geometry of the dispersion vessel

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For dispersion vessel 49W1-RCV-001, total weight is 2000 kg when the vessel is full. This will result in **8kN** on each column foundation.

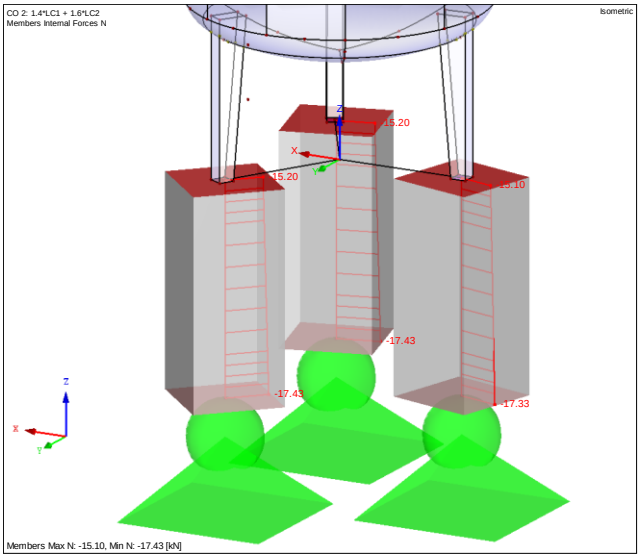


Figure 62. Maximum design normal forces, dimension 300x300

Since the load are low compared to the geometry of the column provide minimum reinforcement.

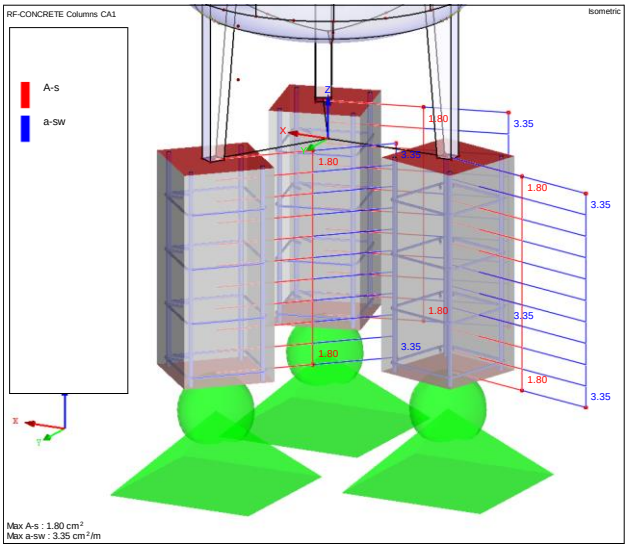


Figure 63 Required main bars A_s and shear reinforcement a_{sw}

Provide 4T16, $A_{s,p}$=805mm^2 and Stirrups T12k150, a_{sw}= 1131mm^2/m.

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Stability check for vessel 001 foundation under load combination 0.9D+ 1.0W. The resultant eccentricity e= M_y/ F_z = 0.77kNm / 8.87kN = 0.09m, r = 0.45m, S.O.F against overturning =5.18 > 1.50, ok

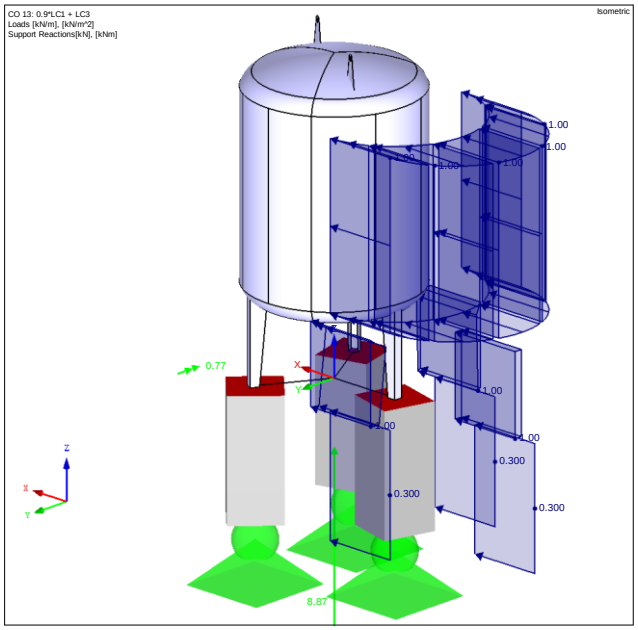


Figure 64 Resultant force and moment for dispersion vessel 001

For dispersion vessel 49W1-RCV-002, total weight is 500 kg when the vessel is full. This will result in **2kN** on each column foundation.

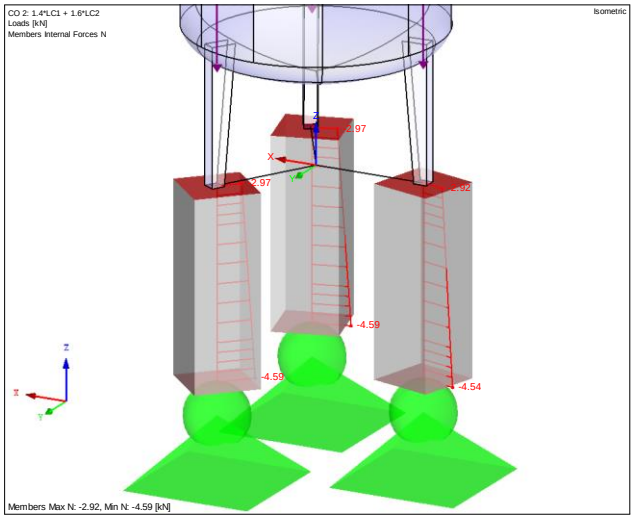


Figure 65 Maximum design normal forces, dimension 250x250

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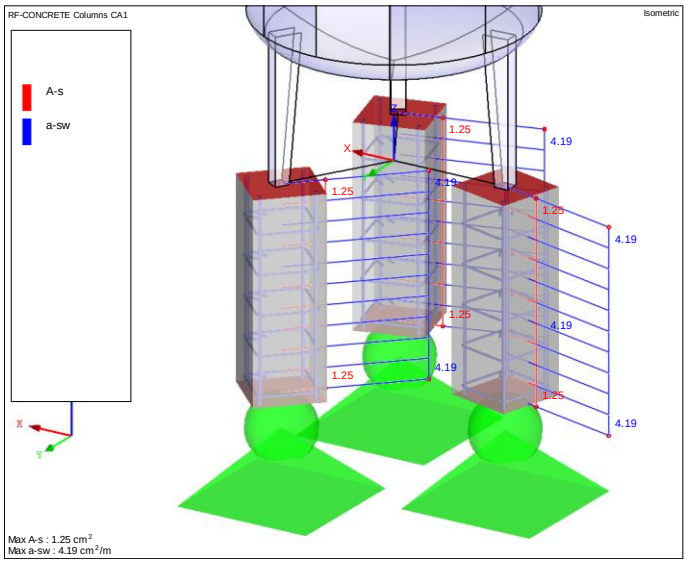


Figure 66 Required main bars As and shear reinforcement asw

Provide 4T16, $A_{s,p}$=805mm^2 and Stirrups T12k150, a_{sw}= 1131mm^2/m.
Stability check for vessel 002 foundation under load combination 0.9D+ 1.0W.
The resultant eccentricity e= M_y/ F_z = 0.93kNm / 5.73kN = 0.18m, r = 0.29m,
S.O.F against overturning =1.58> 1.50, ok

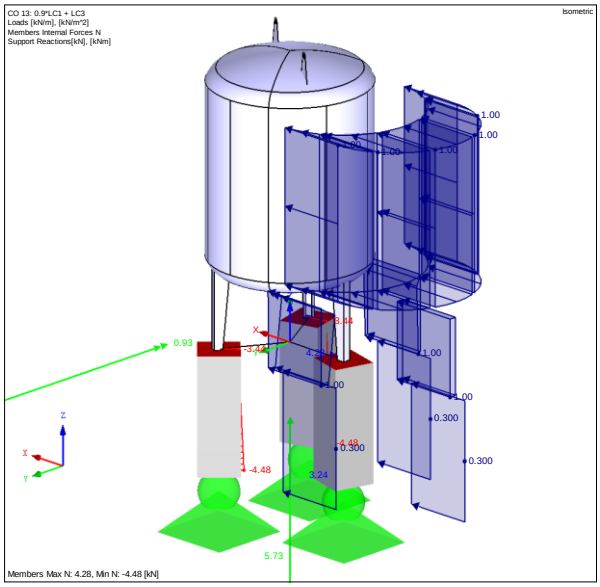


Figure 67 Resultant force and moment for dispersion vessel 002

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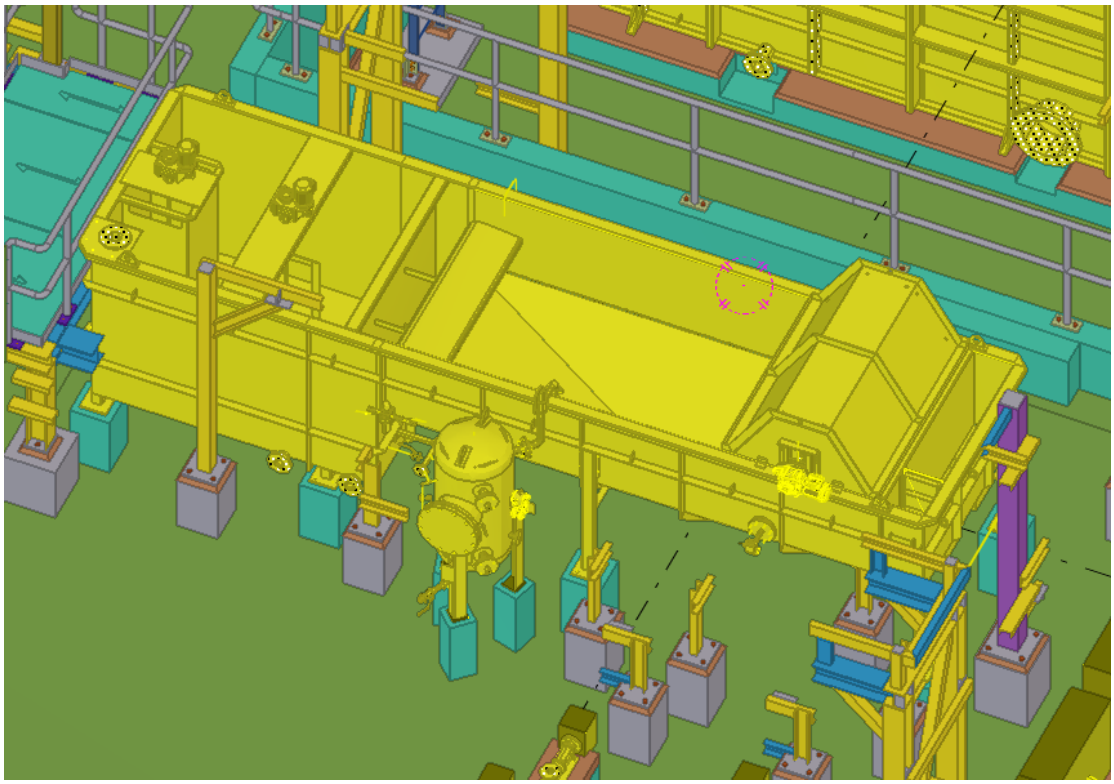


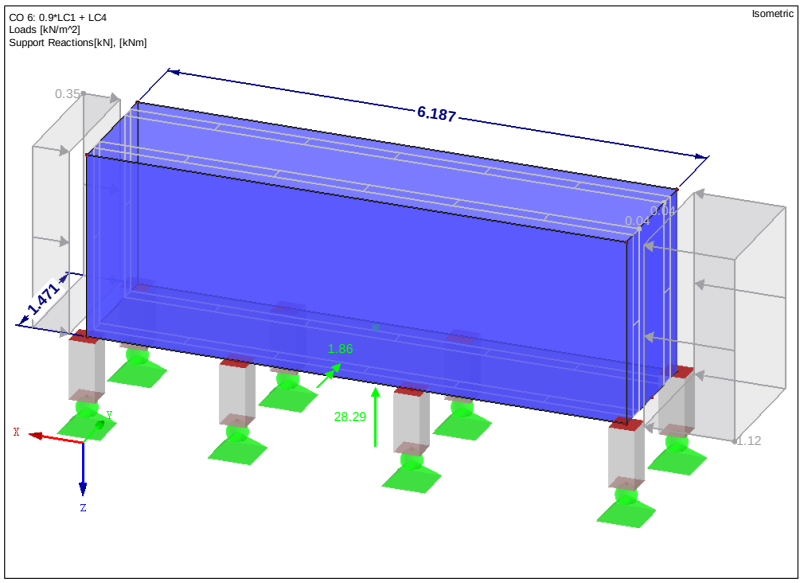
Figure 68 Geometry of Coagulation + Microflotation FlooDAF B-5

The load data is given in drawing P06204M013. The empty load 1900kg and operational load 12000kg.
A simple model is created to investigate the stability. The simple steel structure has a self-weight of ~1900kg included in Load case 1: dead load. A point load of 25kN is added to each column in Load case 2: operational load. Wind load is considered both in X and Y direction in Load case 3 and Load case 4.

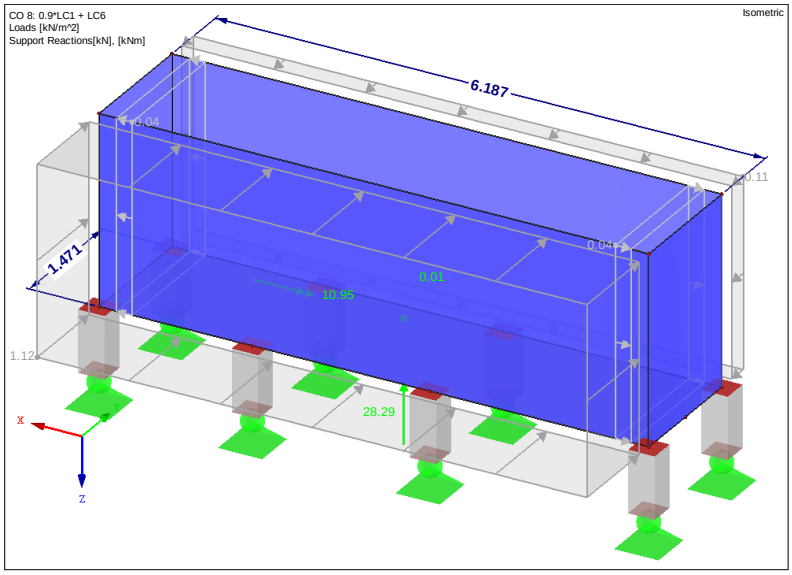
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Stability check under load combination 0.9D+ 1.0W_x. The resultant eccentricity e= M_y/ F_z = 1.86kNm / 28.29kN = 0.07m, B/2 = 3.08m, S.O.F against overturning =44> 1.50, ok



Stability check under load combination 0.9D+ 1.0W_y. The resultant eccentricity e= M_y/ F_z = 10.95kNm / 28.29kN = 0.39m, B/2 = 0.735m, S.O.F against overturning =1.90> 1.50, ok

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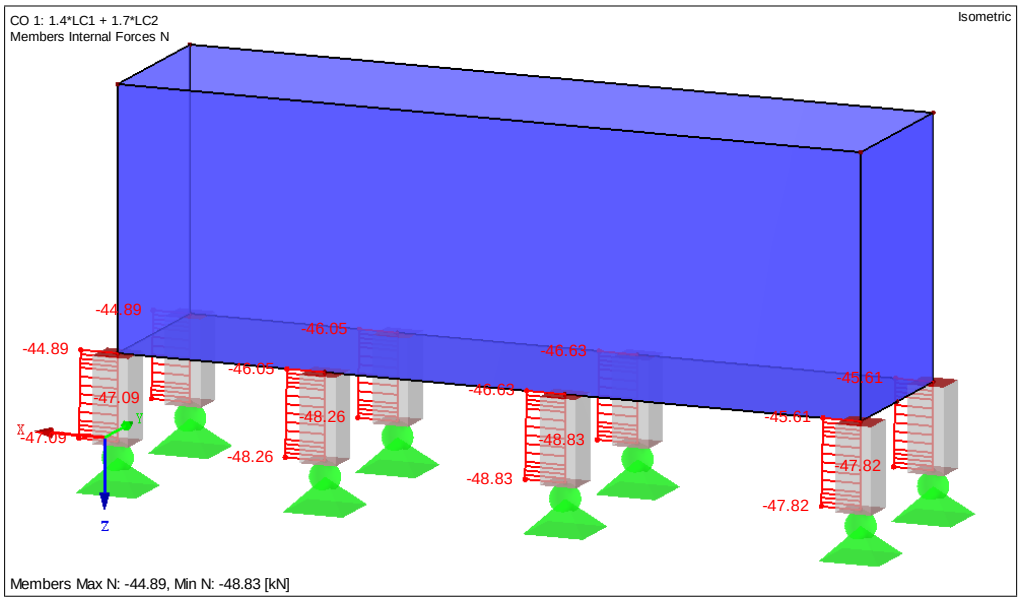


Figure 69 Maximum design normal forces, dimension 300x300

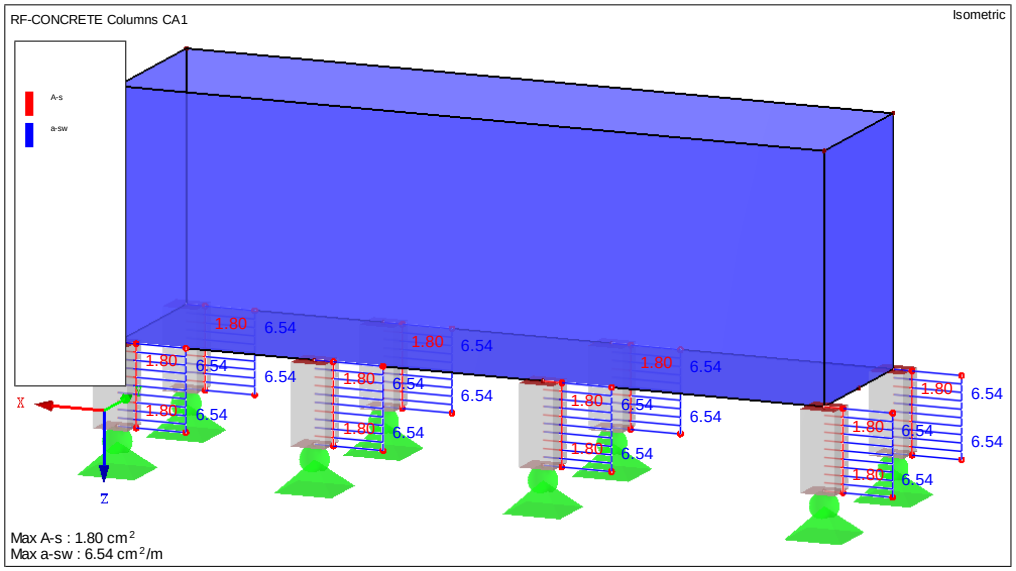


Figure 70 Required main bars As and shear reinforcement asw

Provide 4T16, $A_{s,p}$=805mm^2 and Stirrups T12k150, a_{sw}= 1131mm^2/m.

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11. MAIN GROUND SLAB

11.1. Introduction

Concrete slab on grade is loaded with 10kN/m^2 (live load for ground floor not accessible to trucks) and moving Forklift axle load of 65kN with effective contact area of 0.2m^2. Since it is a moving load different critical point of application should be evaluated alternatively as shown in figure bellow.

Slab thickness is determined using PCA deign chart, suggested by ACI360R-10, Guide to design of slab on grade in Appendix 1: axle load= 65kN= 14.6kips, effective contact area of one wheel = 25in^2, wheel spacing= 1m= 40in, subgrade modulus k=33750kN/m^3=100lb/in^3, concrete compressive strength (C30) = 4350psi, modulus of rupture =9(f_{ck})$^{1/2}$ = 570psi, safety of factor S.O.F =1.7, allowable stress= 335psi, stress/1000lb of axle load = 23, read from chart below that slab thickness t= 5.6in=142mm, take t= 150mm.

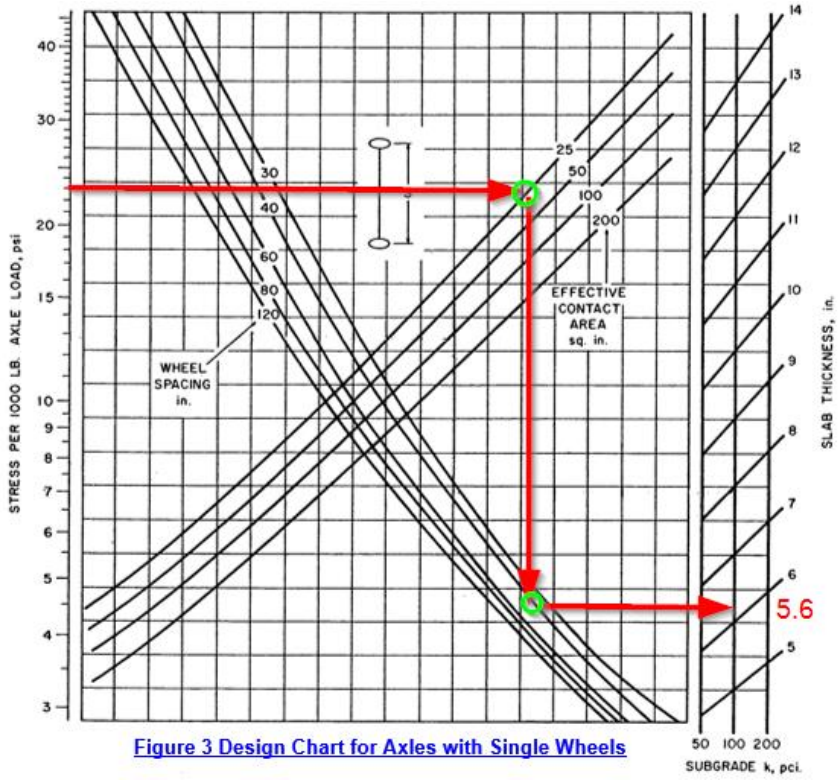


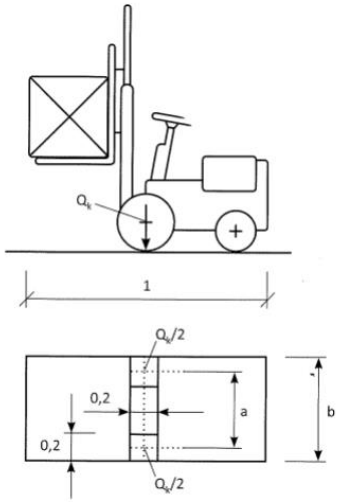
Figure 3 Design Chart for Axles with Single Wheels

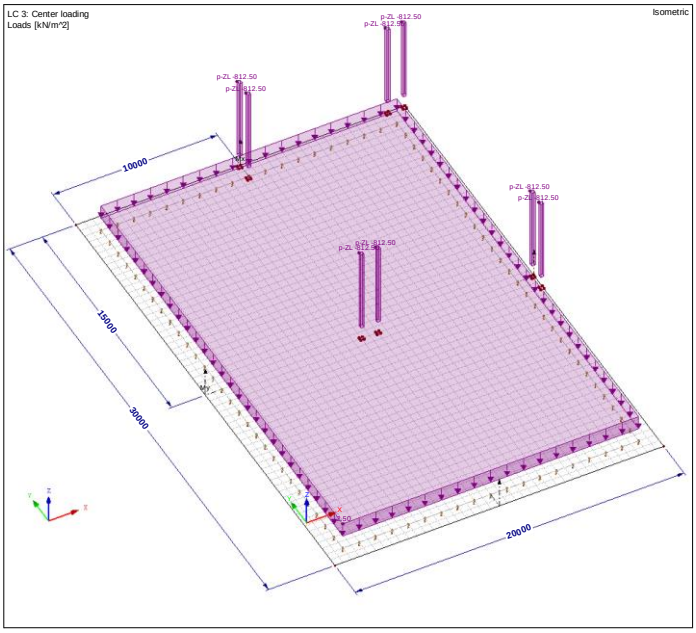
The reinforcement area required to resist flexure and shear is calculated in RFEM. The slab on grade is modeled as a surface supported on soil springs, the vertical spring constant is C_{uz}=33,750 kN/m^3 and the horizontal C_{ux} = Cu_y=155,000 kN/m^2, the maximum soil bearing capacity is 375kN/m^2. Concrete C30 f_{ck}=30MPa, reinforcement f_{yk}=420MPa.

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11.2. Results

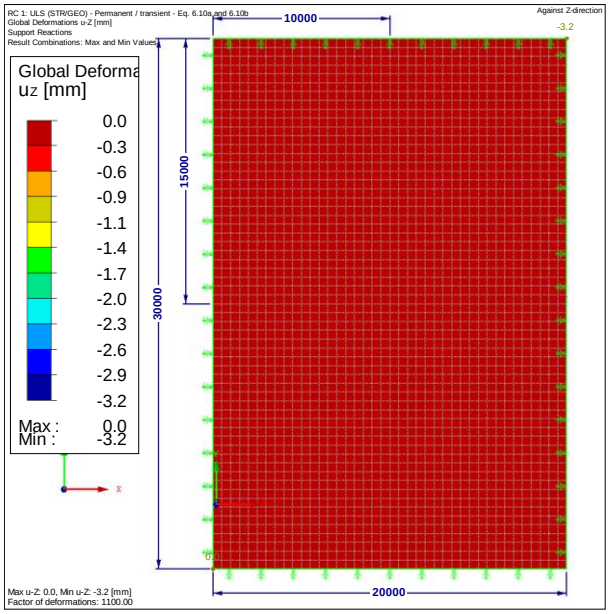


Figure 71 Maximum settlement of the slab $u_z < u_{z,max} = 50mm$,ok

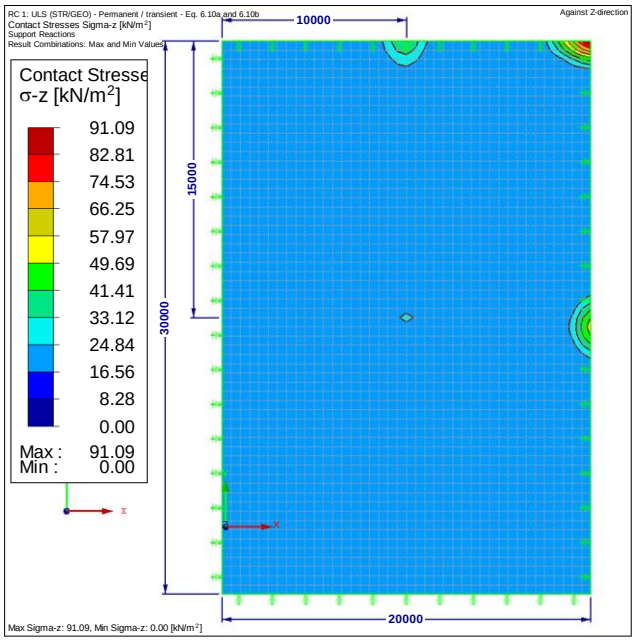


Figure 72 Maximum contact stress

$\sigma < \sigma_{s,max} = 375kN/m^2$, ok

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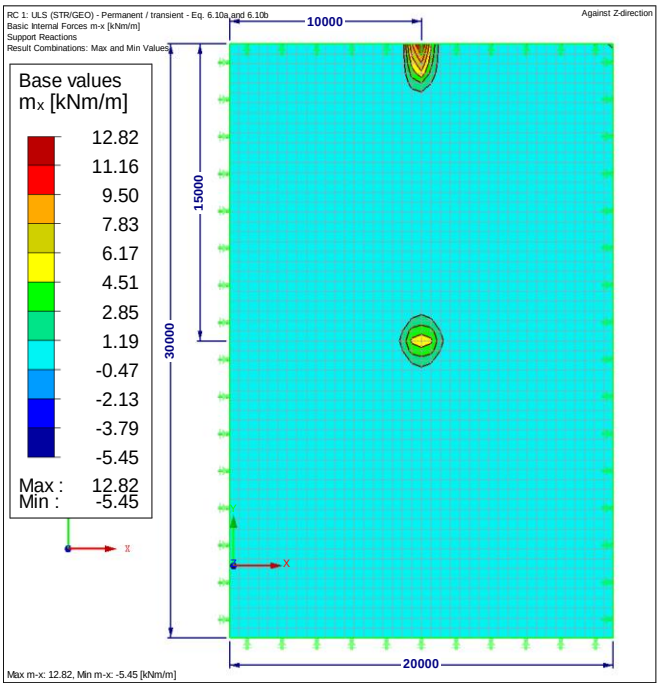


Figure 73 Bending moment Mx

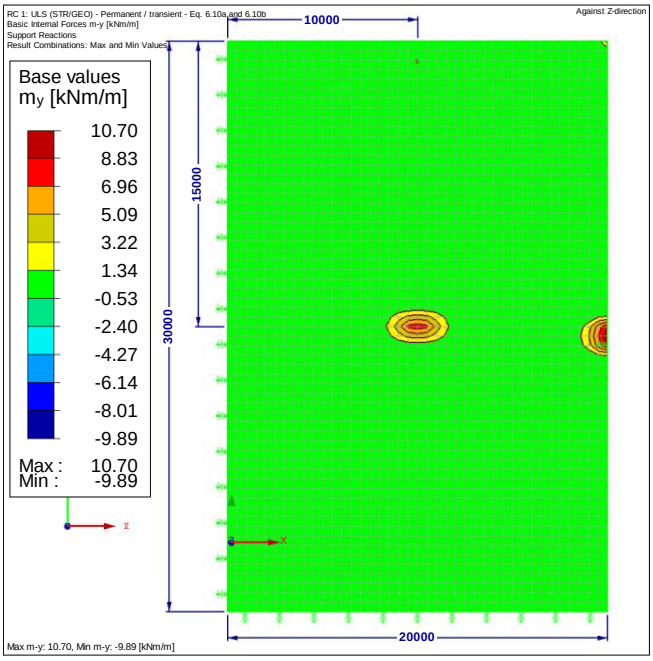


Figure 74 Bending moment My

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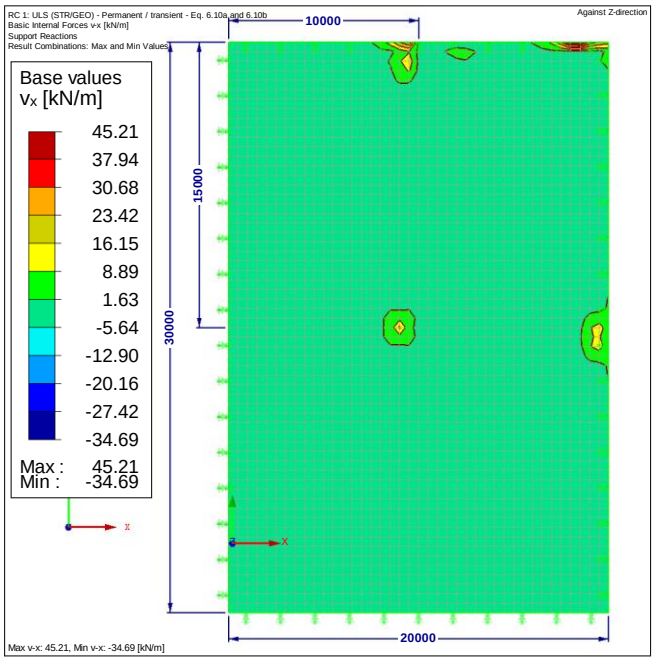


Figure 75 Shear force Vx

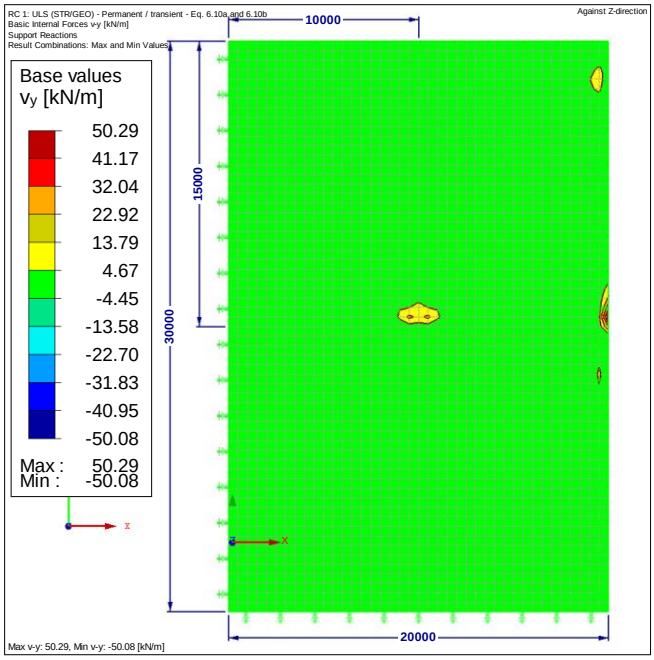


Figure 76 Shear force Vy

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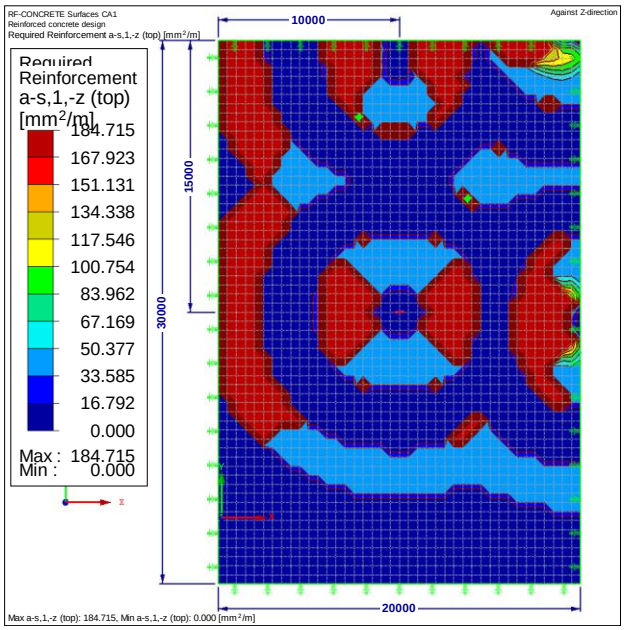


Figure 77 Required top reinforcement $A_{s,top,1}$

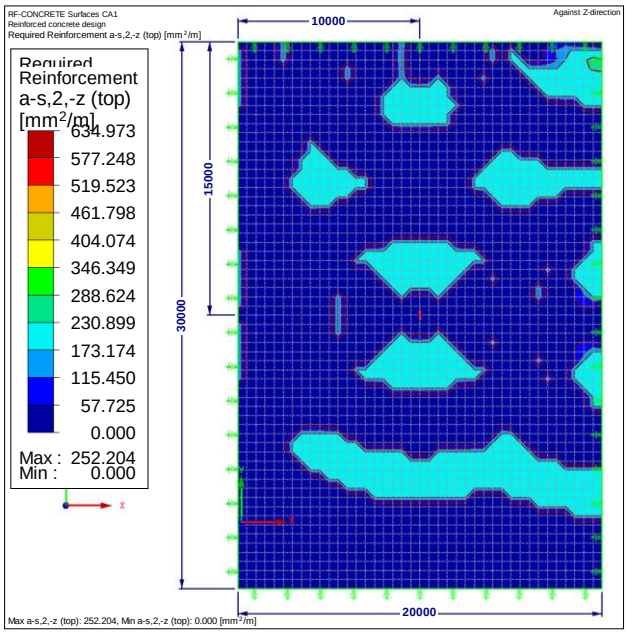


Figure 78 Required top reinforcement $A_{s,top,2}$

Provide T12/200mm, $A_{s,p}$ = 565mm²/m

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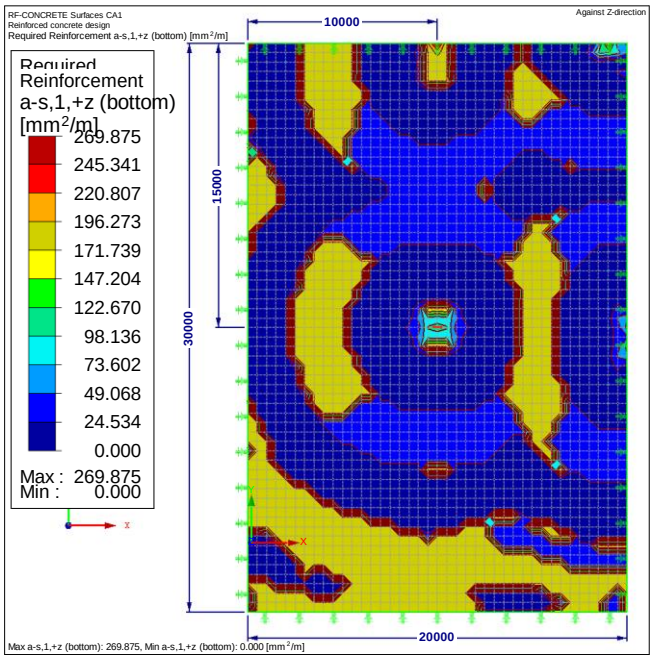


Figure 79 Required bottom reinforcement $A_{s,bottom,1}$

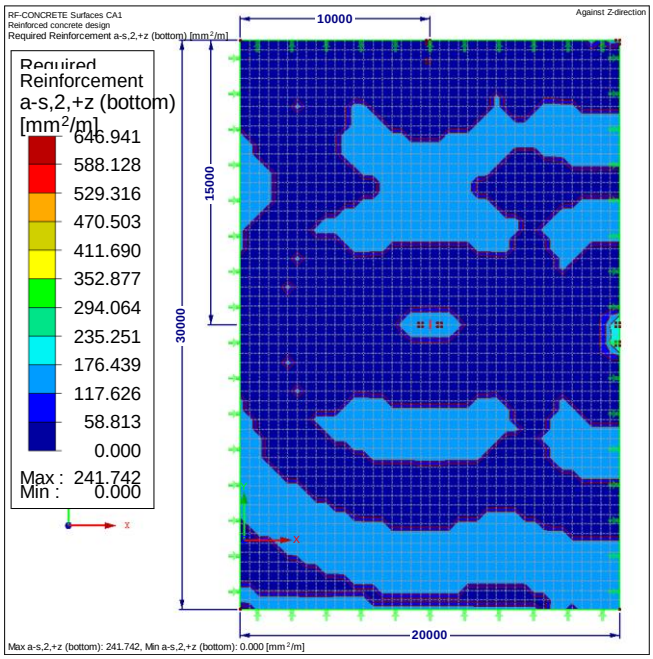


Figure 80 Required bottom reinforcement $A_{s,bottom,2}$

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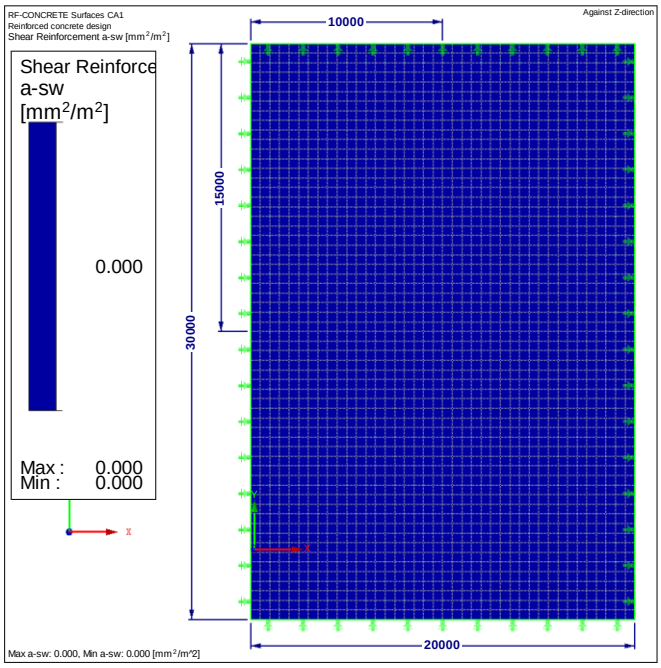


Figure 81 Required shear reinforcement asw

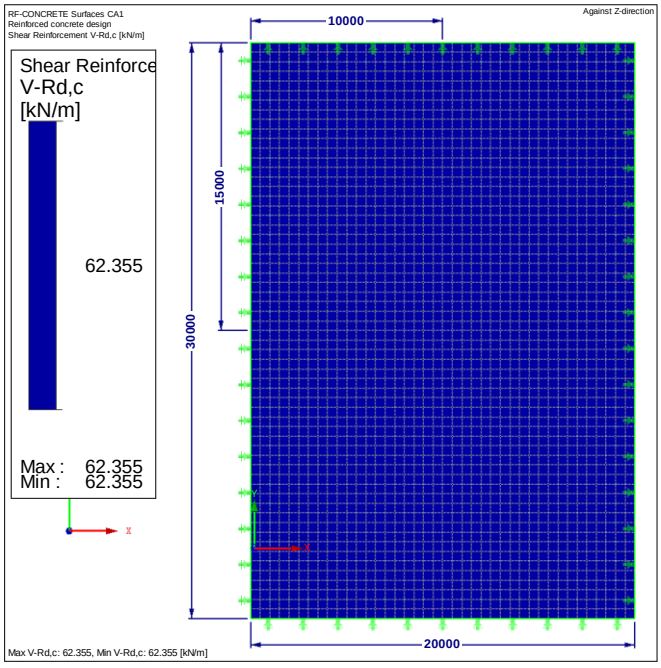


Figure 82 Shear resistance without shear reinforcement $V_{Rd,c}$

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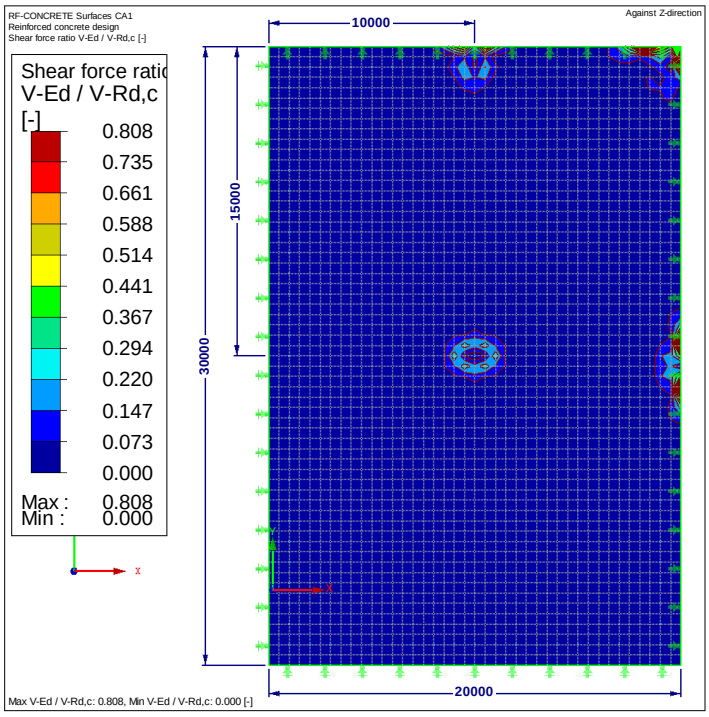


Figure 83 $V_{Ed}/V_{Rd,c}$

utility ratio = applied shear force/ Shear resistance without shear reinforcement = $V_{Ed}/V_{Rd,c} < 1.0$, shear reinforcement is not needed

Design shear resistance without shear reinforcement for ground slab can also be calculated as below:

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Axial load in [N]

$N_{Ed} := 65000$

Shear strength of ground slab in [N]

$$V_{Rd.c} = \left[C_{Rd.c} \cdot k \cdot \left(100 \cdot \rho_1 \cdot f_{ck}\right)^{\frac{1}{3}} + k_1 \cdot \sigma_{cp}\right] \cdot b_w \cdot d$$

where

$C_{Rd.c} := \frac{0.18}{\gamma_c} = 0.12$

$k := \min\left(1 + \sqrt{\frac{200}{d}}, 2.0\right) = 2$

$b_w := 1000$

$\rho_1 := \frac{A_s}{b_w \cdot d} = 5.188 \times 10^{-3}$

$k_1 := 0.15$

$A_c := 200000$ concrete cross-section area in [mm^2]

$\sigma_{cp} := \frac{N_{Ed}}{A_c}$

$V_{Rd.c} := \left[C_{Rd.c} \cdot k \cdot \left(100 \cdot \rho_1 \cdot f_{ck}\right)^{\frac{1}{3}} + k_1 \cdot \sigma_{cp}\right] \cdot b_w \cdot d = 7.063 \times 10^4$

$v_{min} := 0.035 \cdot k^{\frac{3}{2}} \cdot f_{ck}^{\frac{1}{2}} = 0.542$

$V_{Rd.c.min} := \left(v_{min} + k_1 \cdot \sigma_{cp}\right) \cdot \left(b_w \cdot d\right) = 6.442 \times 10^4$

$V_{Rd,c} > V_u - \sigma * A_c$, ok!

11.3. Summary:

The base slab is 150mm thick, it is reinforced with T12k200 in both direction top and bottom. The maximum contact stress is 156 kN/m^2 which is safe compared to maximum allowable value of 375kN/m^2.

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12. CONTAINER FOUNDATION

12.1. Geometry of the structure

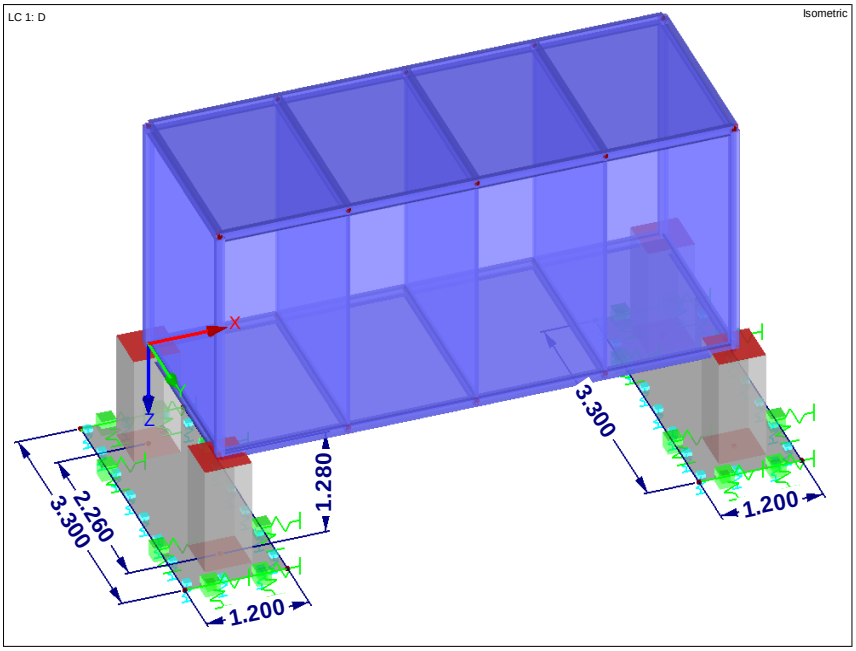


Figure 84 Container foundation geometry

The pedestal dimension 550mmx550mm, max. height 1280mm. The raft dimension 3260mmx1200mm, thickness 400mm.

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12.2. Load calculations

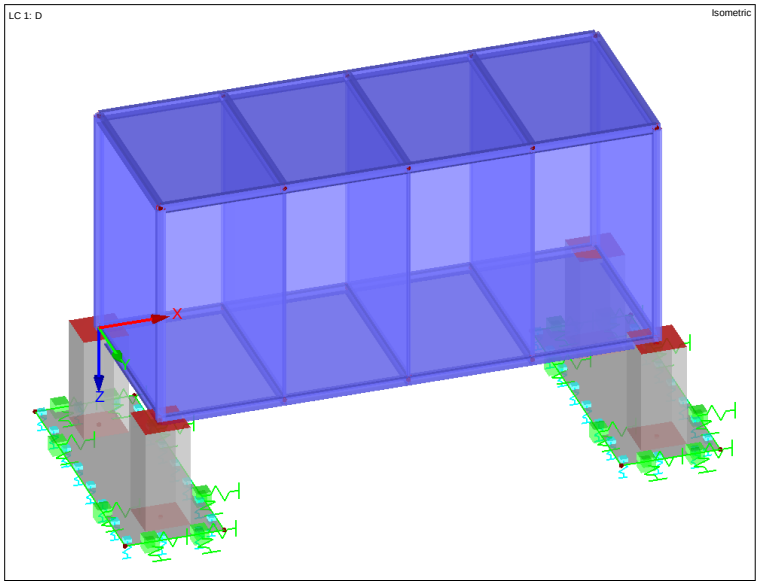


Figure 85 Dead load

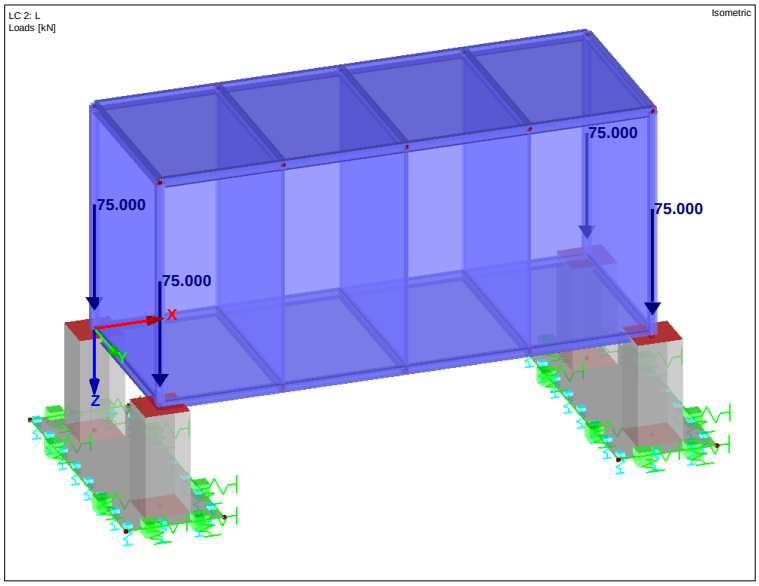


Figure 86 Live load

The 20 ft container has empty weight 30kN and full weight 30 tons = 300kN. Therefore, max. axial load on each pedestal is 75kN.

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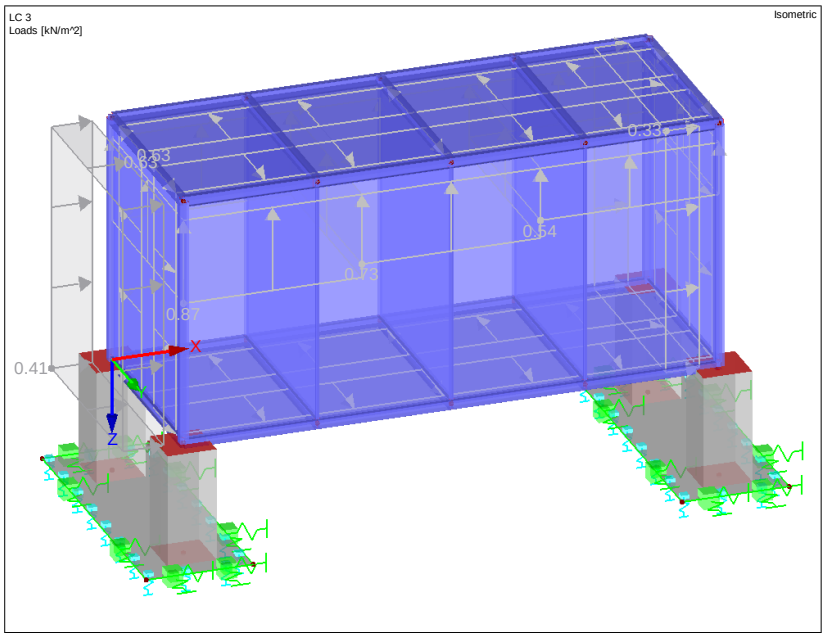


Figure 87 Wind X, +Cpi

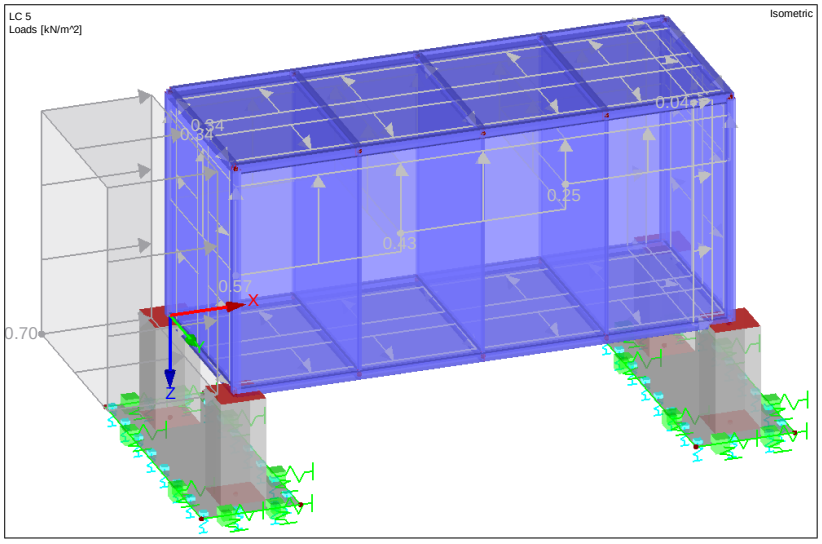


Figure 88 Wind X, -Cpi

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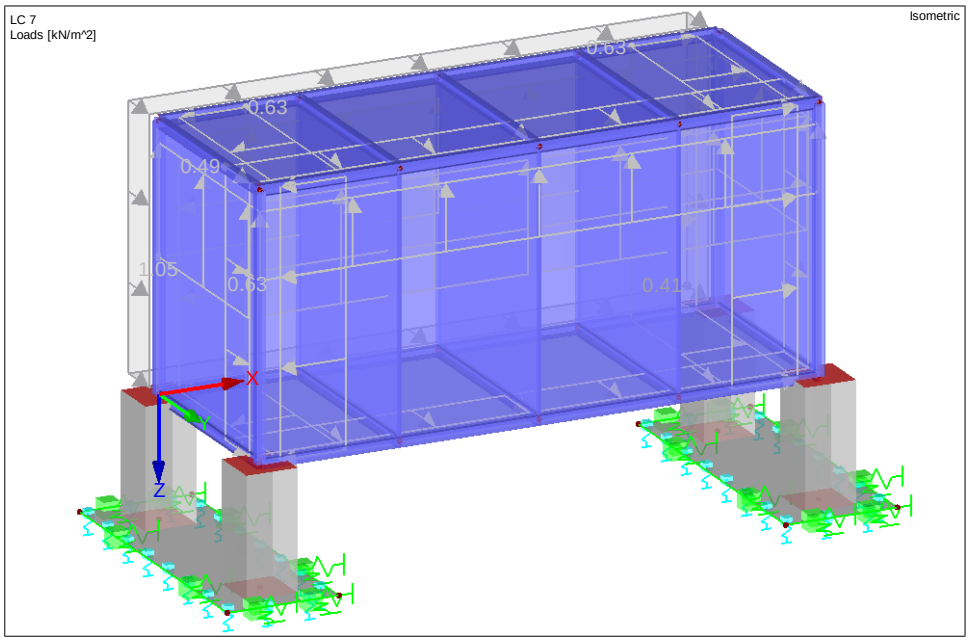


Figure 89 Wind Y, +Cpi

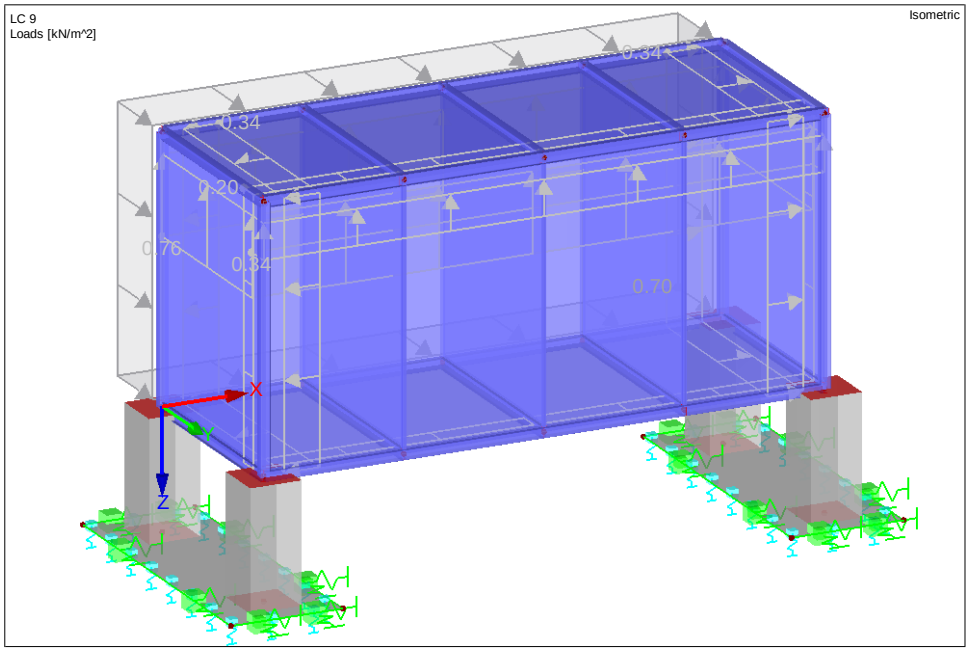


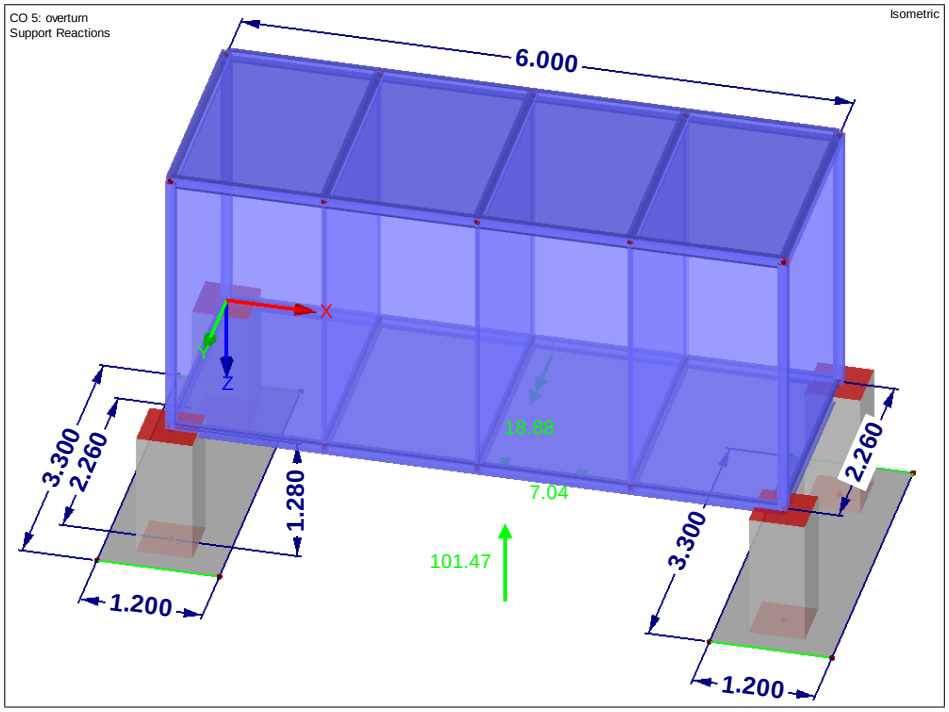
Figure 90 Wind X, -Cpi

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12.3. Overturning check

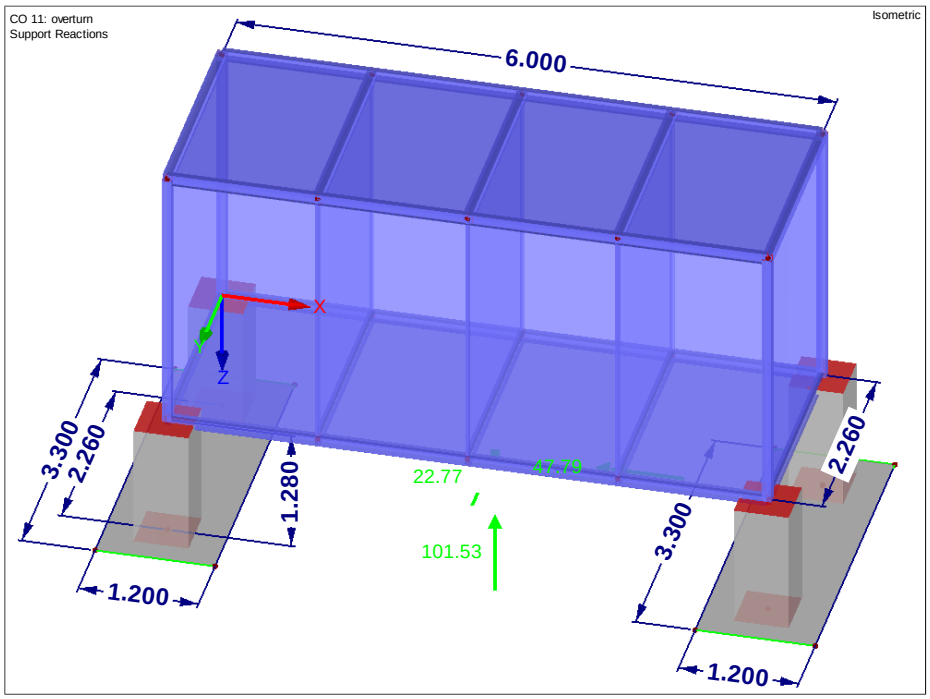


OVERTURN:	My=	18.88	kNm	overturn moment				
	Fz=	101.47	kN	total vertical load				
	e=	0.186	m	eccentricity				
	B/2=	3	m					
	F.O.S=	16.123	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	7.04	kN	lateral force in x				
	Fy=	0	kN	lateral force in y				
	Hd=	7.04	kN	sliding force				
	Rd=	40.588	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	5.765	>1.50	Factor of safety against sliding				

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OVERTURN:	Mx=	47.79	kNm	overturn moment			
	Fz=	101.53	kN	total vertical load			
	e=	0.471	m	eccentricity			
	B/2=	1.13	m				
	F.O.S=	2.401	>1.50	Factor of safety against overturning			
SLIDING:	Fx=	0	kN	lateral force in x			
	Fy=	22.77	kN	lateral force in y			
	Hd=	22.77	kN	sliding force			
	Rd=	40.612	kN	resistance to sliding			
	F.O.S=	1.784	>1.50	Factor of safety against sliding			

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12.4. Design of pedestal

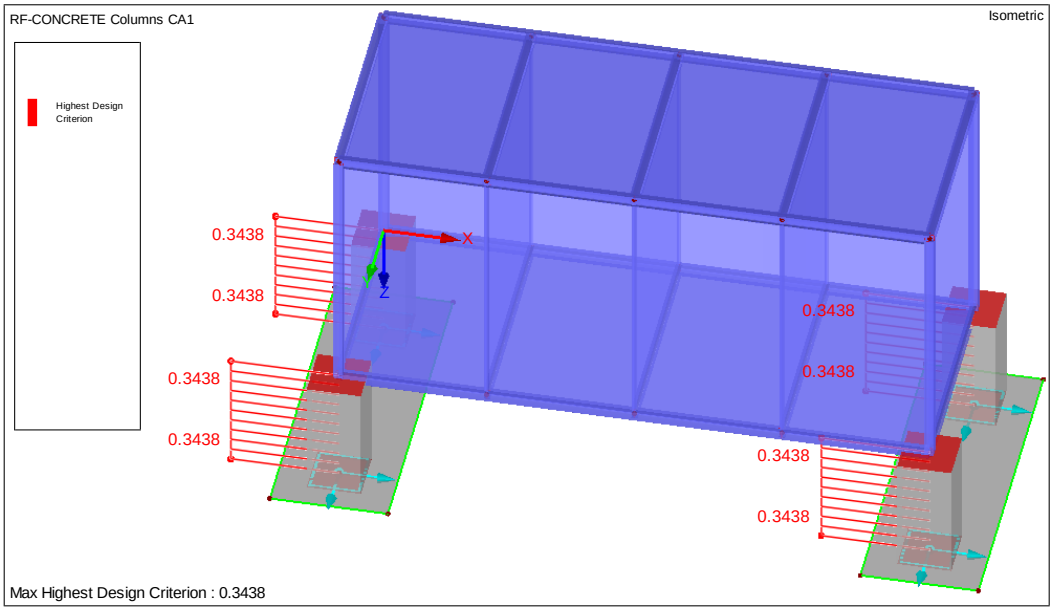


Figure 91 Highest utility ratio for pedestal

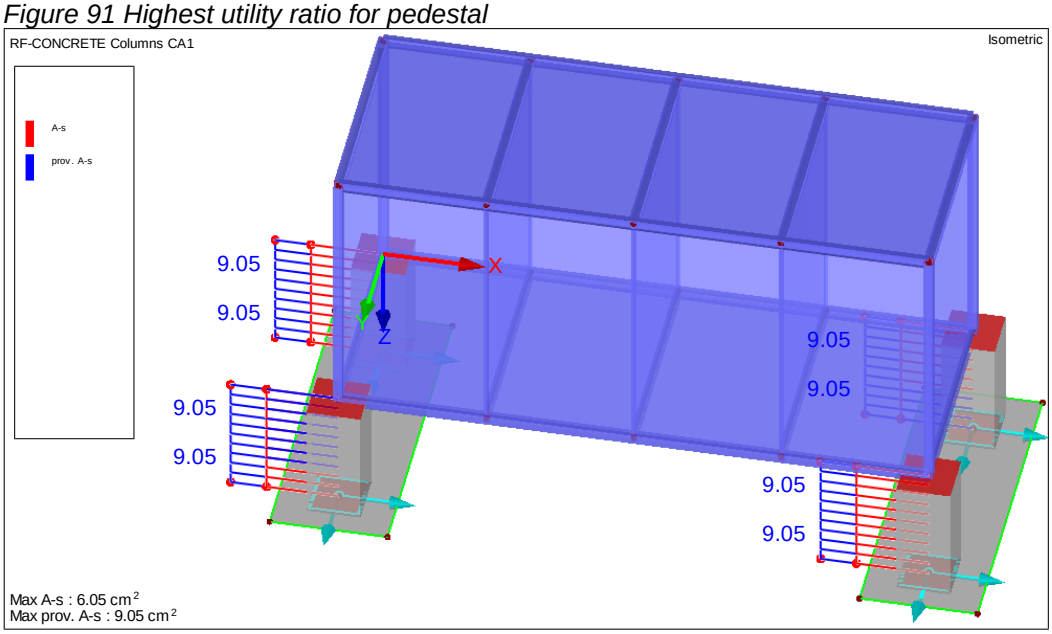


Figure 92 Required (in red) and provided (in blue) longitudinal reinforcement

Provide 8T12 for longitudinal rebars, Asp = 9.05cm²

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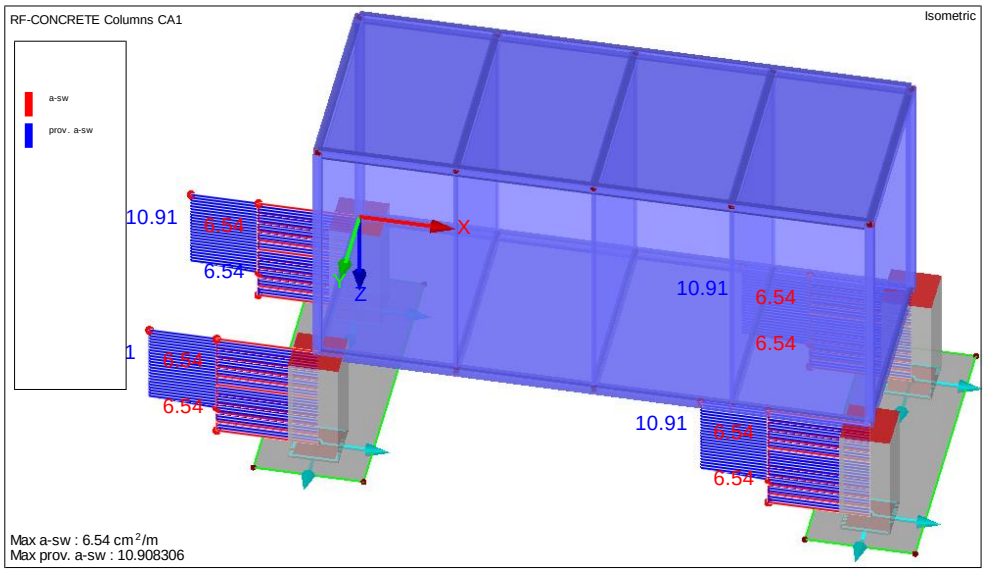


Figure 93 Required (in red) and provided (in blue) shear reinforcement

Provide T10/150mm as shear reinforcement.

12.5. pad footing design

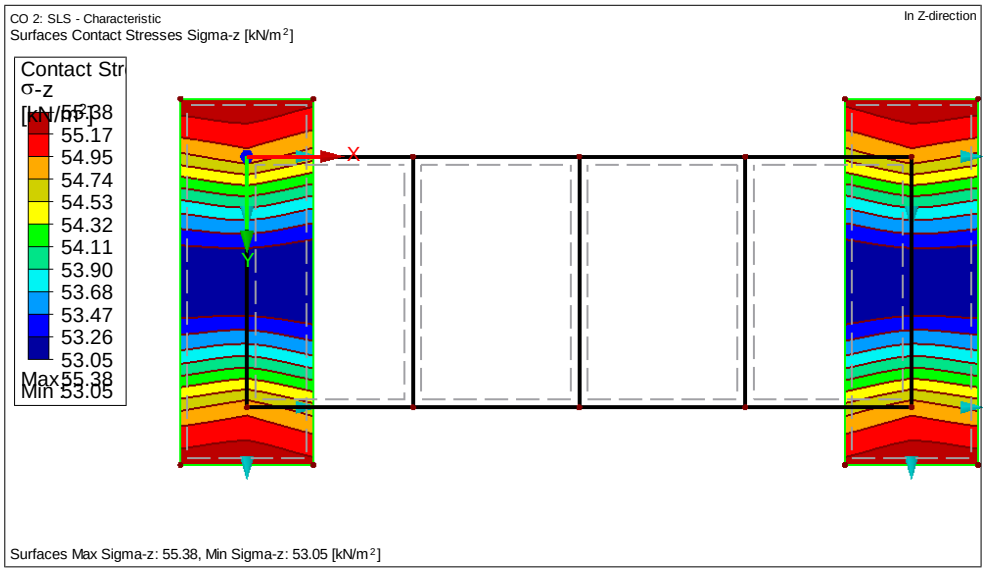


Figure 94 Soil contact stress at SLS

$\sigma_s < \sigma_{s.max}$, ok

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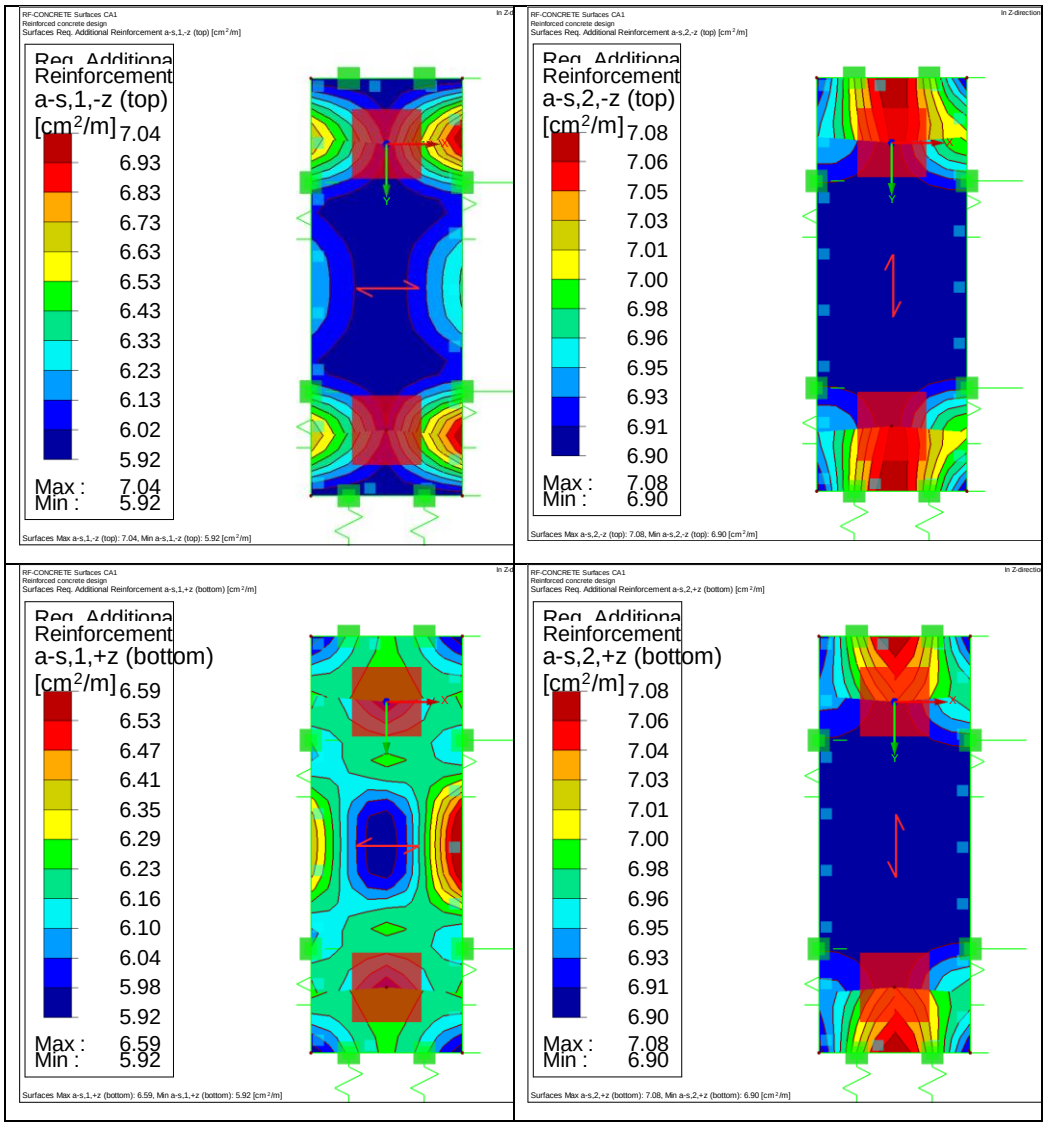


Figure 95 Required reinforcement for pad footing

Provide T12/150mm, As = 7.54cm^2/m

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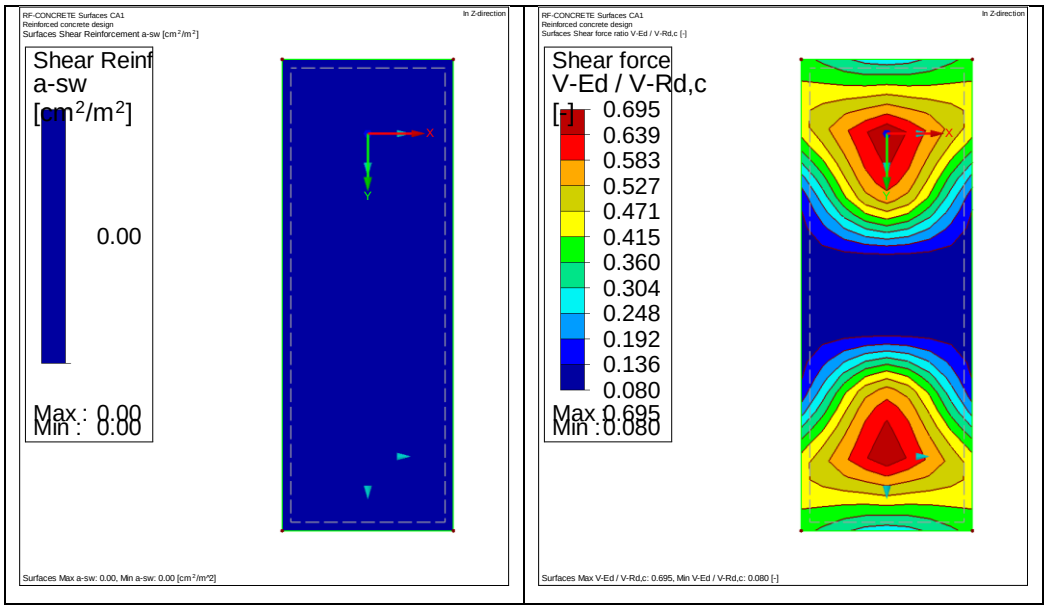


Figure 96 Required shear reinforcement (left) and $V_{Ed}/V_{Rd,c}$ (right)

Utility ratio = applied shear force/ shear resistance without shear reinforcement= $V_{Ed}/V_{Rd,c} < 1,0$. no shear reinforcement needed. Apply stirrups T12/150mm

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13. CONCRETE PLATFORM

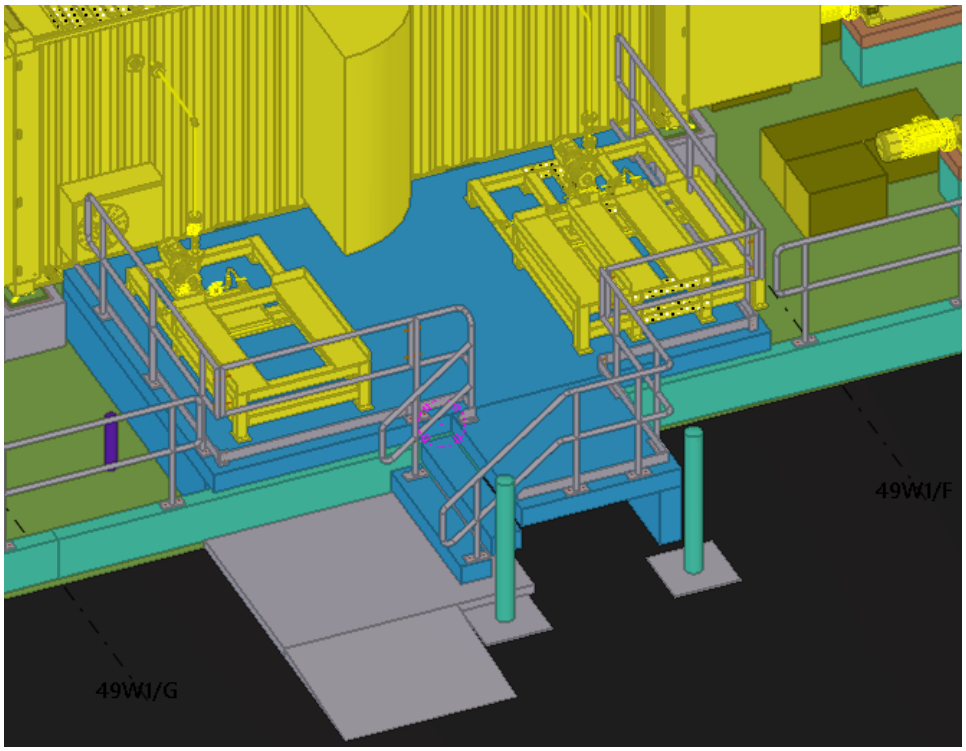


Figure 97 location of the structure

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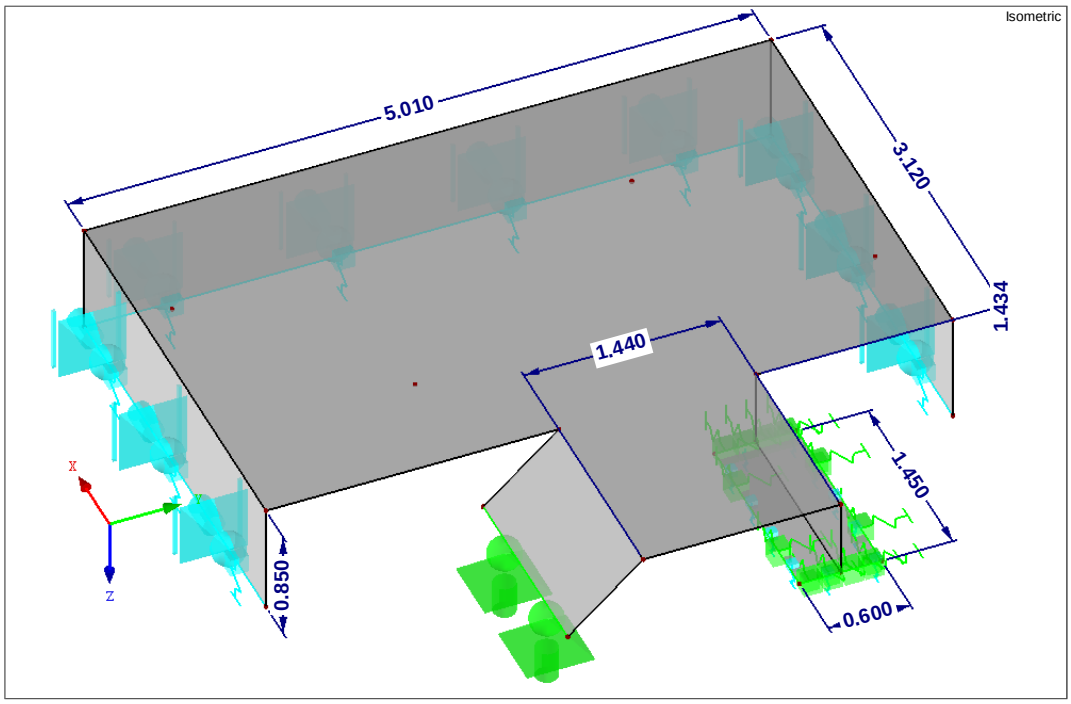


Figure 98 Geometry of the structure in calculation model

The slab has length L=5010mm, width B=3120mm and thickness t=200mm. It is supported on compacted gravel and the side wall has a minimum thickness 300mm. Average thickness for the stairs is 150mm and strip foundation 200mm.

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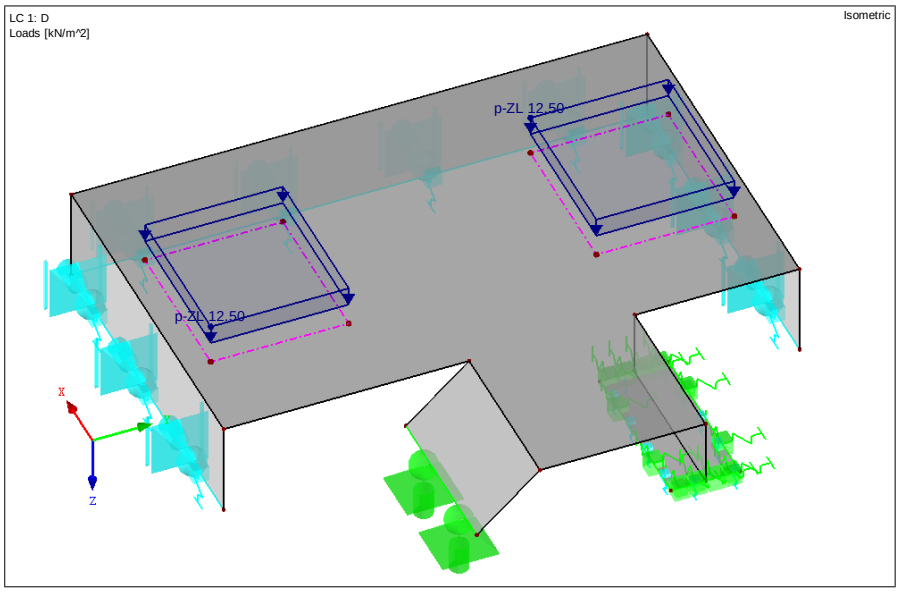


Figure 99 LC1: dead load

dead load including structure self-weight and equipment load 12.5kN/m^2 at certain area.

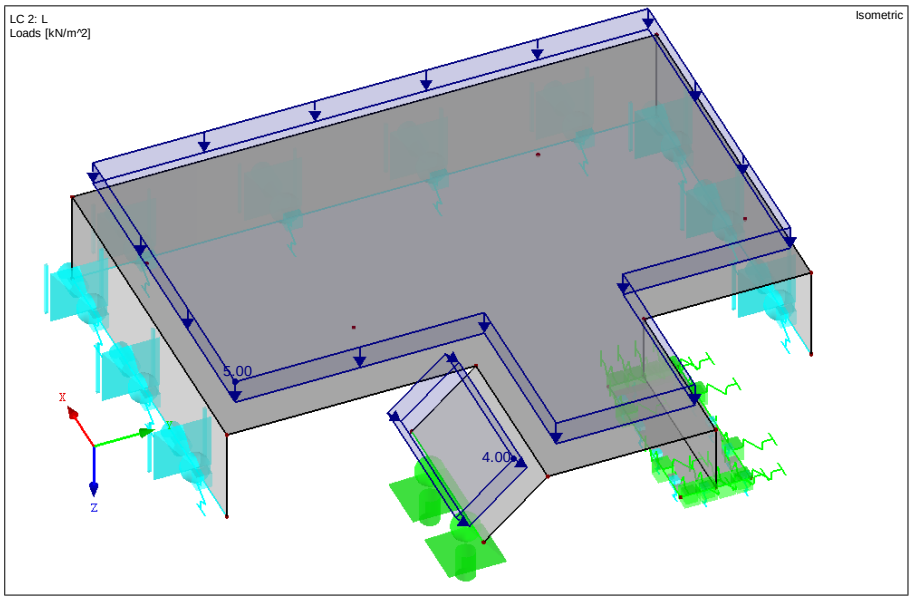


Figure 100 LC2: live load

live load on operating platforms 5kN/m^2 and on stairways 4kN/m^2

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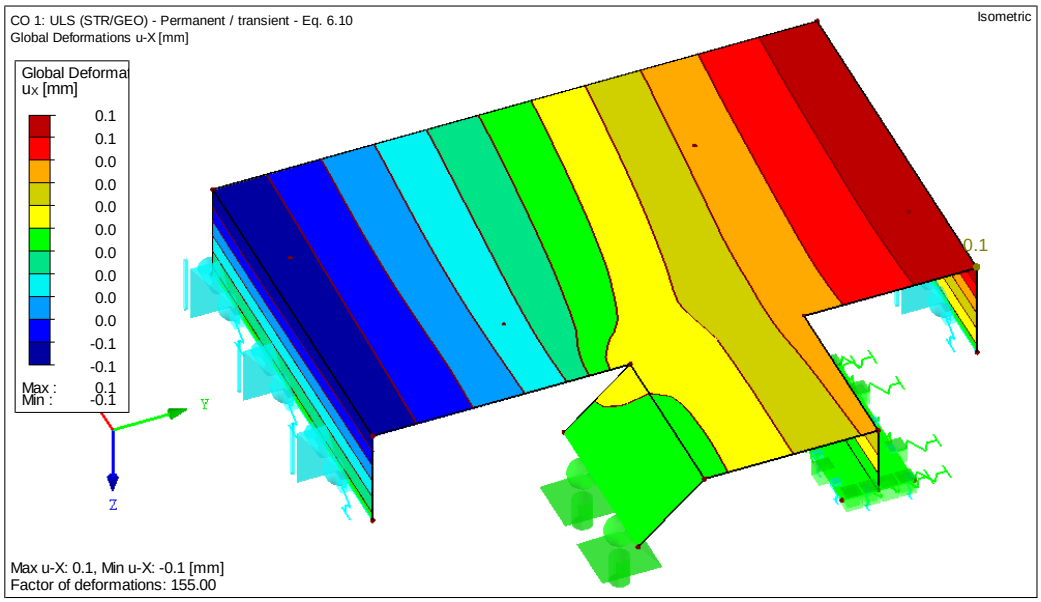


Figure 101 Global deformation u_x

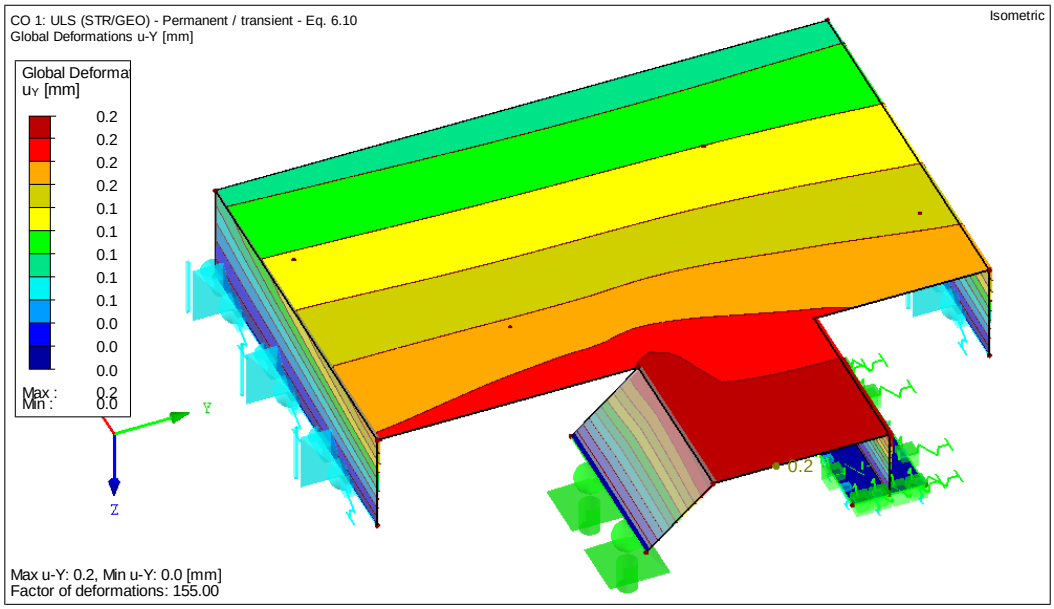


Figure 102 Global deformation u_y

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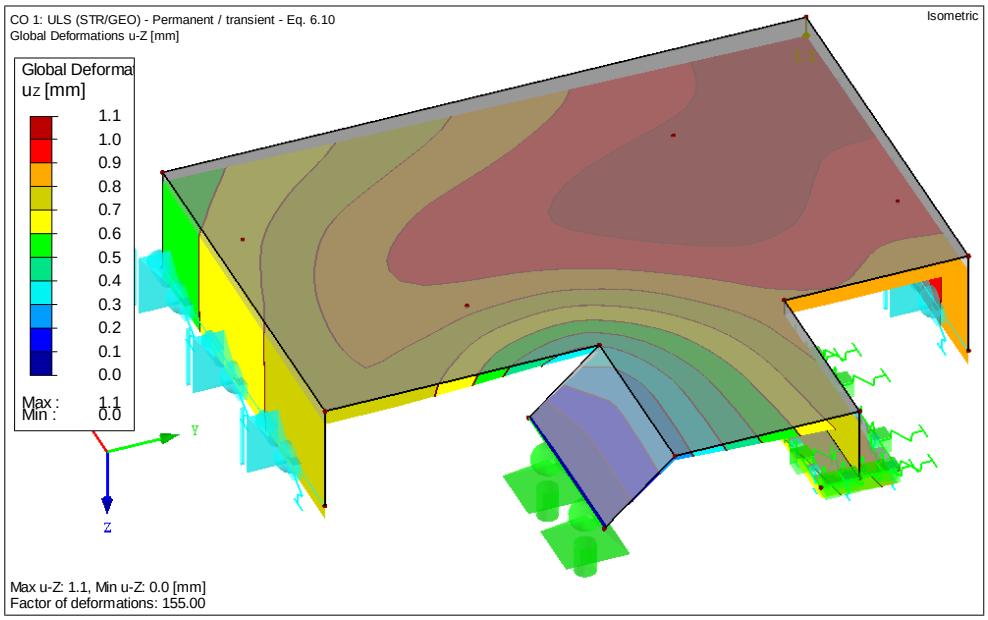


Figure 103 Global deformation uz

deformation u< u_{max} =50mm, ok

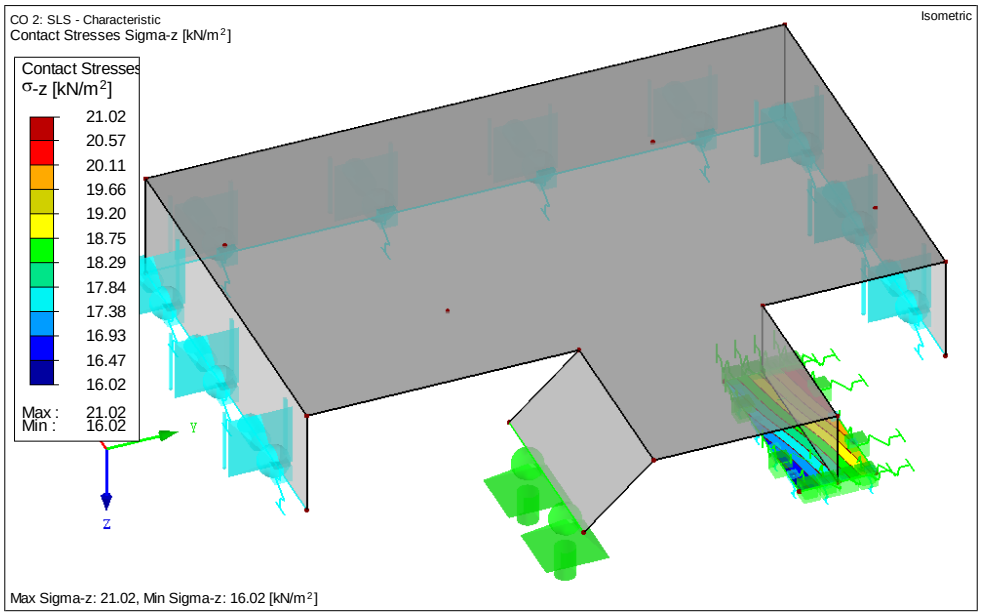


Figure 104 Soil contact stress at SLS

σ <$σ_{s,max}$= 375kPa,ok

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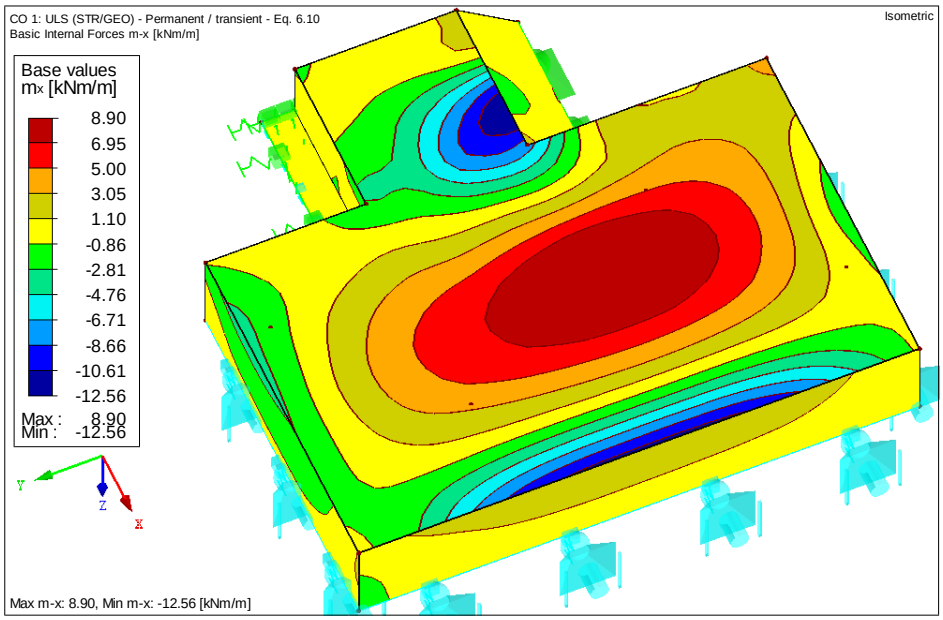


Figure 105 Bending moment Mx

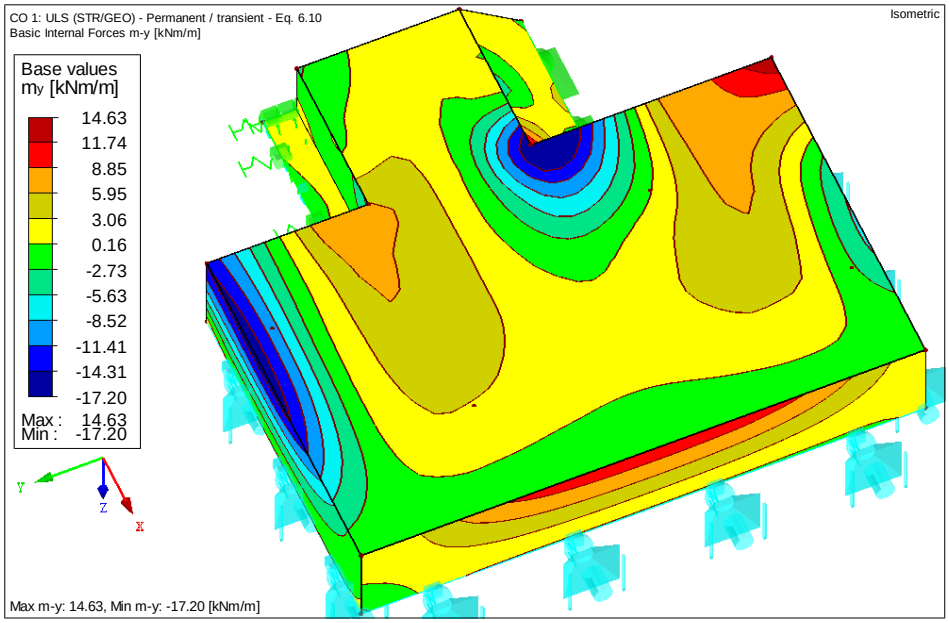


Figure 106 Bending moment My

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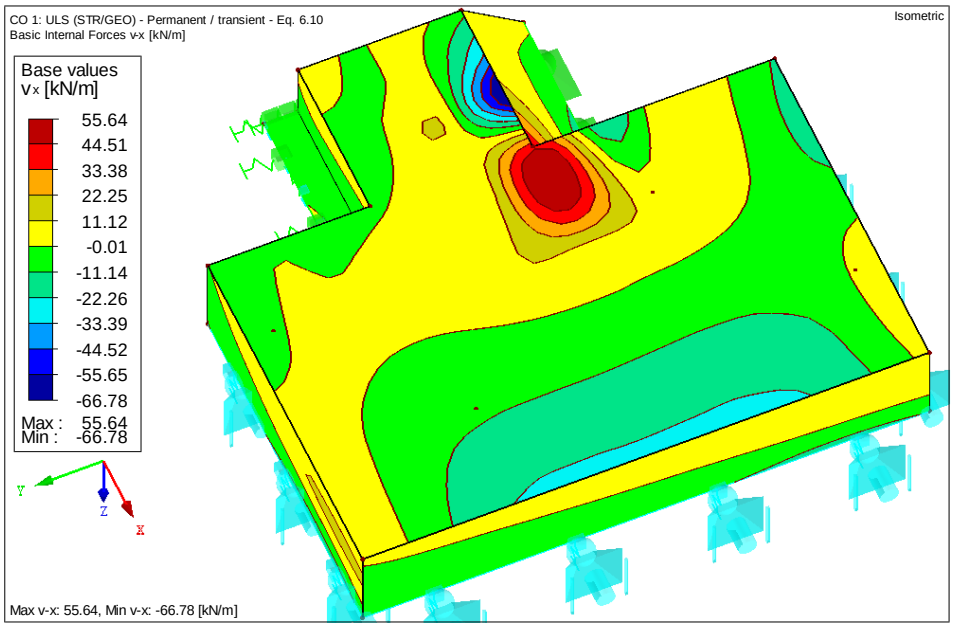


Figure 107 Shear force Vx

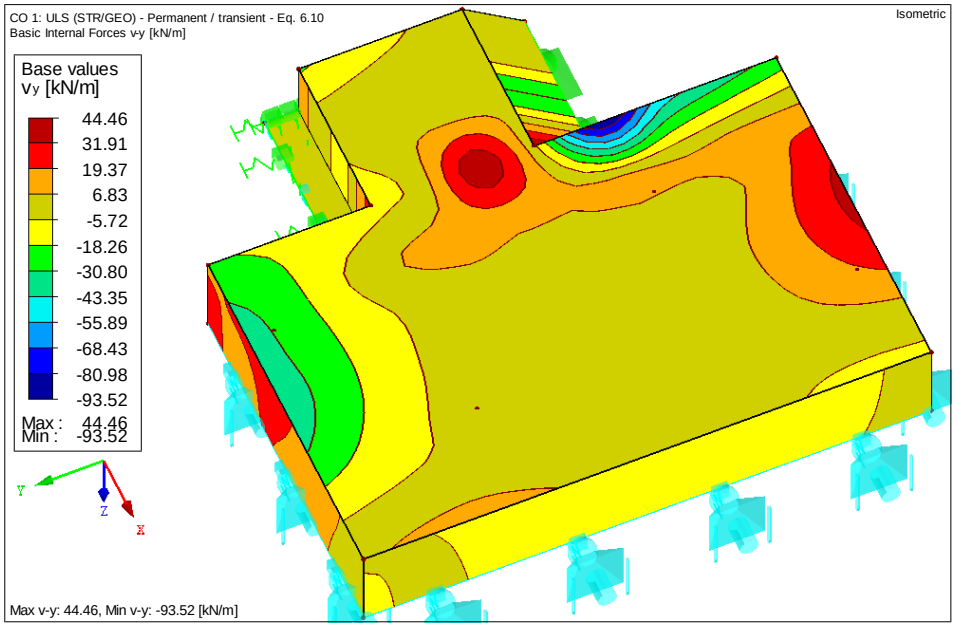


Figure 108 Shear force Vy

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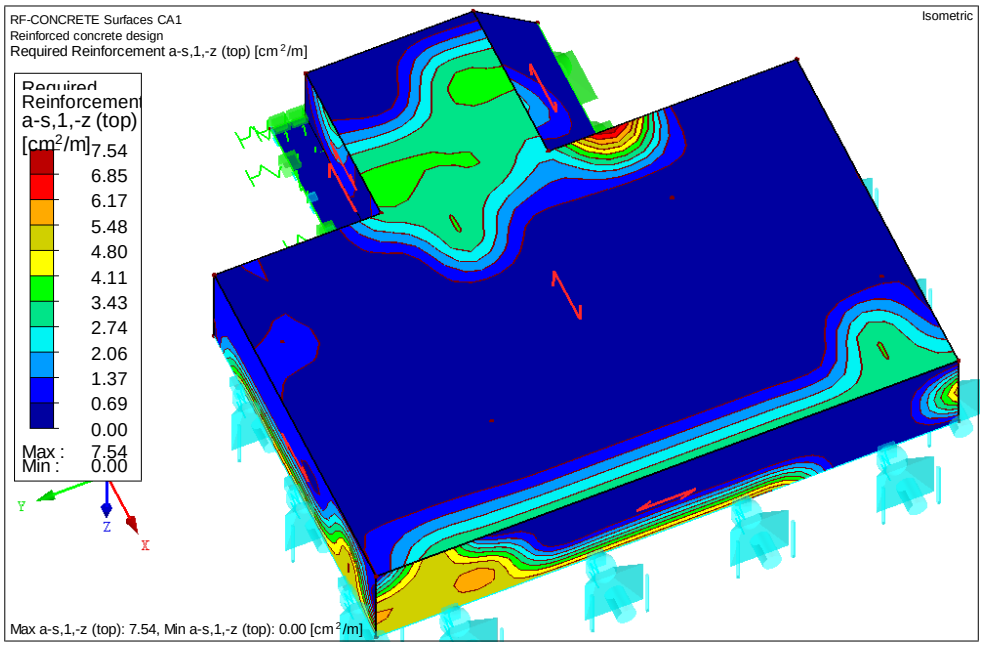


Figure 109 Required top reinforcement $A_{s,top,1}$

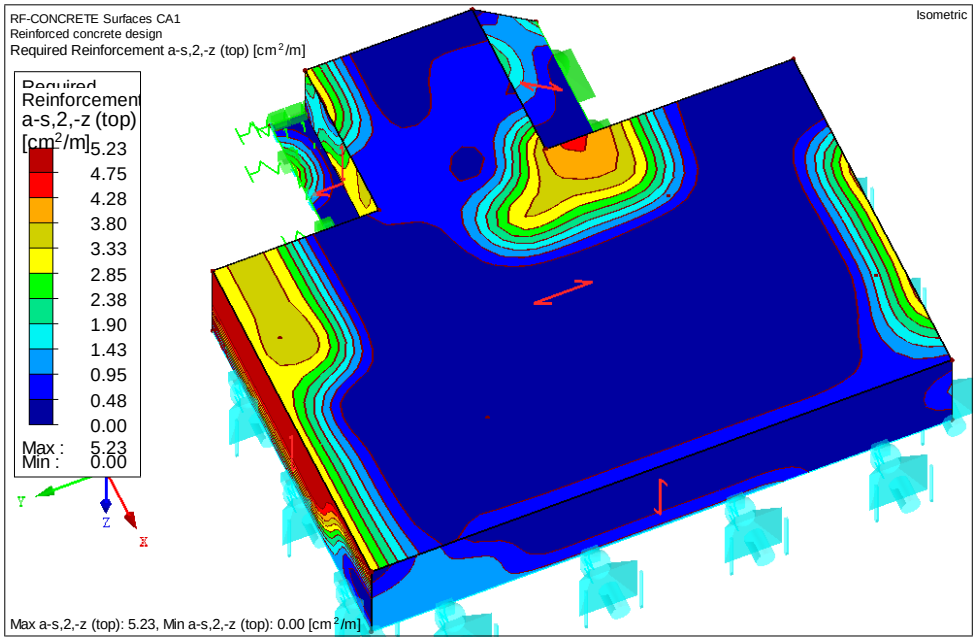


Figure 110 Required top reinforcement $A_{s,top,2}$

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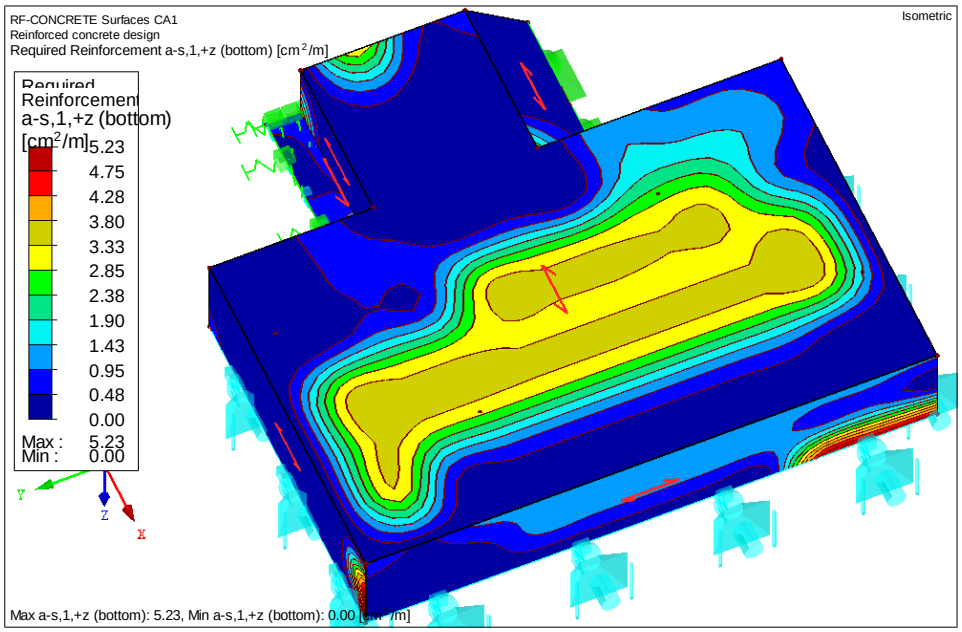


Figure 111 Required bottom reinforcement $A_{s,bottom,1}$

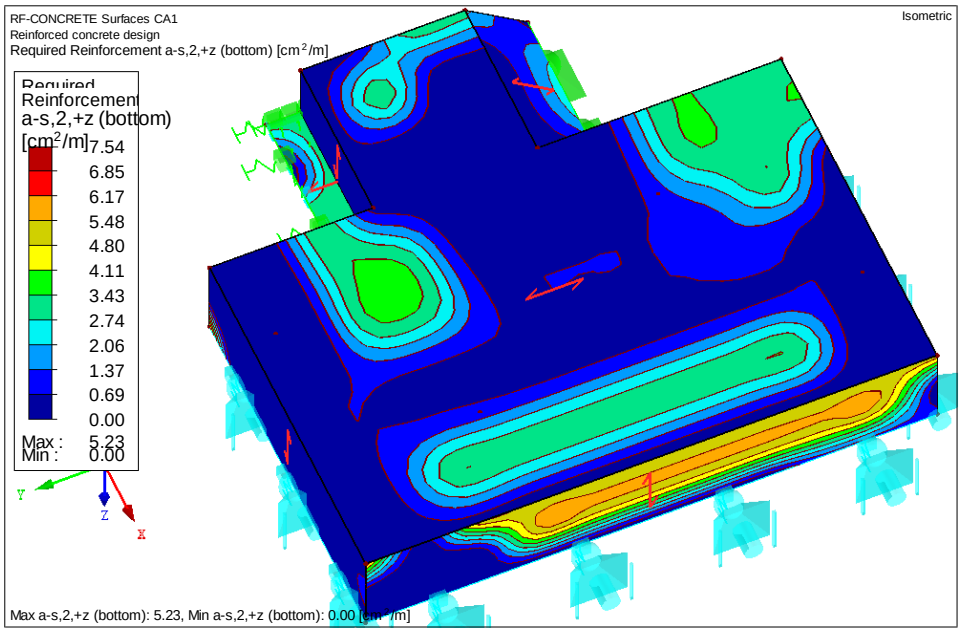


Figure 112 Required bottom reinforcement $A_{s,bottom,2}$

Provide basic reinforcement area T12/150mm, $A_{s,p}$=7,54cm^2/m

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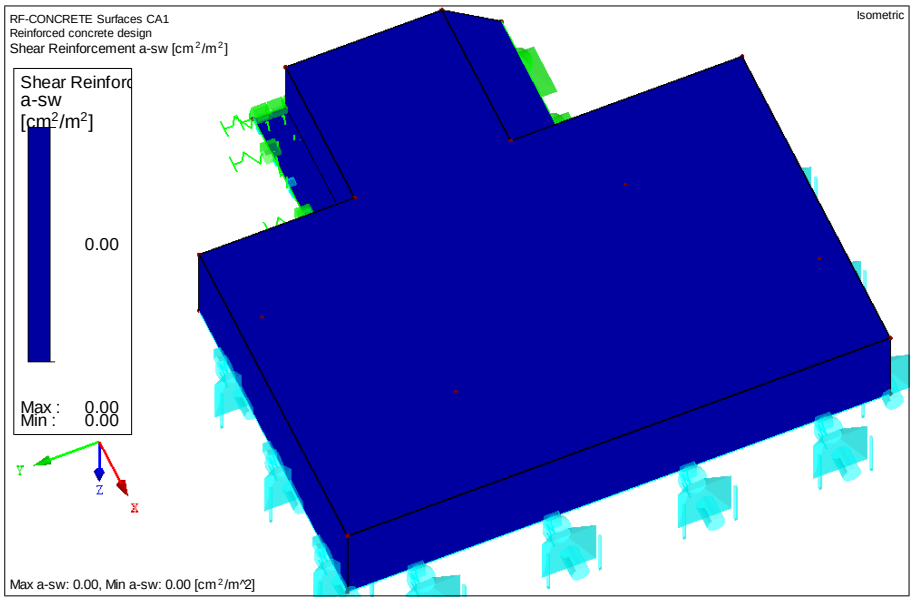


Figure 113 Required shear reinforcement a_{sw}

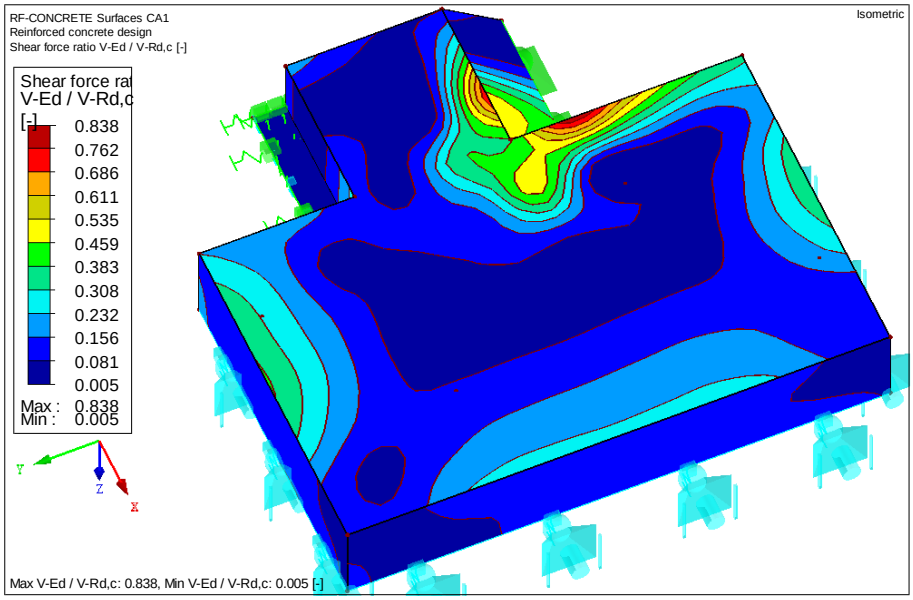


Figure 114 $V_{Ed}/V_{Rd,c}$

Utility ratio=applied shear force/ design shear resistance without shear reinforcement = $V_{Ed}/V_{Rd,c} < 1.0$, no shear reinforcement needed

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REFERENCES

[1] Saudi Building Code (SBC) 301, Structural - Loading and Forces.

[2] Saudi Building Code (SBC) 304, Structural – Concrete Structures.

[3] Project Design Criteria, MD-216-1000-EG-ST-CRT-1001-A03.

[4] Overall geotechnical report, MD-216-3000-EG-CV-RPT-1008-C01.

[5] General arrangement drawing, raw water treatment plan, 1-8/E-J, MD-216-49W1-EG-CV-GAD-1001

[6] General arrangement drawing, raw water treatment plan, 7-12/E-J, MD-216-49W1-EG-CV-GAD-1002

[7] General arrangement drawing, FD074M001, Microflotation basin FlooDAF B-74, 49W1-TNK-002,

[8] General arrangement drawing, P06204M002, Coagulant mixing basin 75m3, 49W1-TNK-001

[9] General arrangement drawing, P06204M003, 49W1-RCV-001, Dispersion water tank 1

[10] General arrangement drawing, P06204M014, 49W1-RCV-002, Dispersion water tank 2

[11] General arrangement drawing, P06204M013, Coagulation + Microflotation FlooDAF B-5, 49W1-TNK-011, 49W1-SCP-003, 49W1-TNK-010

[12] Ma'aden Engineering Structural Design Criteria, MD-101-SMEM-EG-ST-CRT-0001

[13] ACI360R-10, Guide to design of slab on grade

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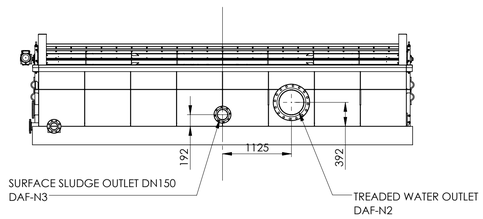
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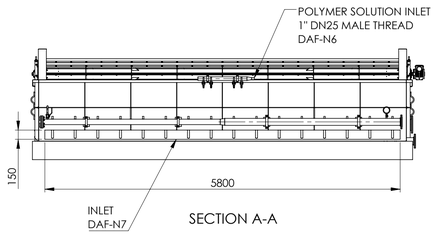
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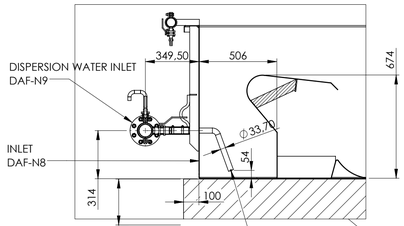
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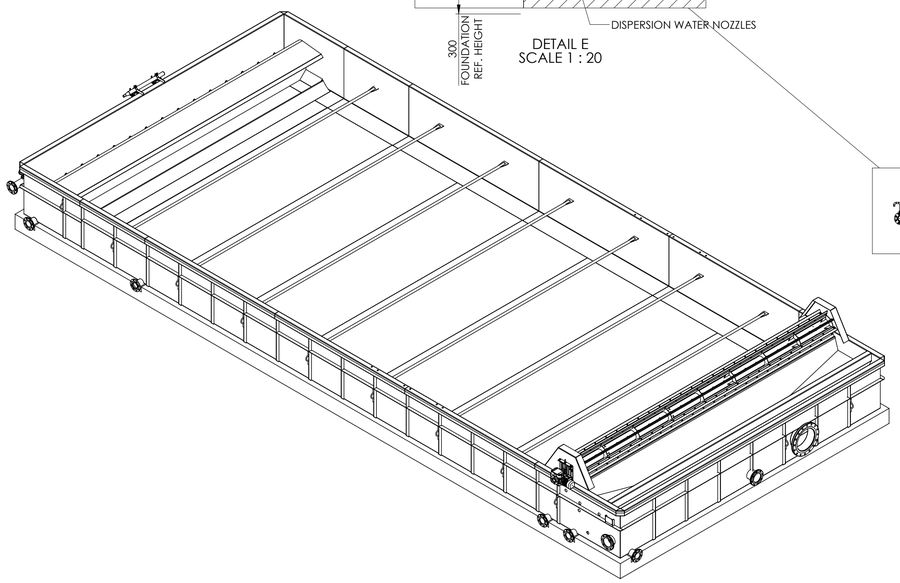


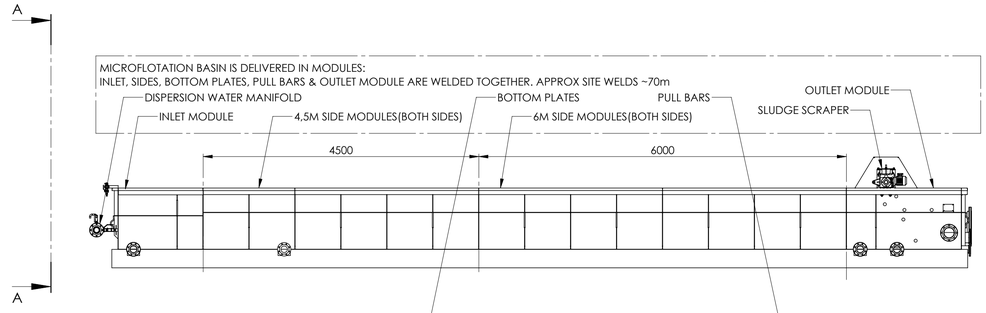


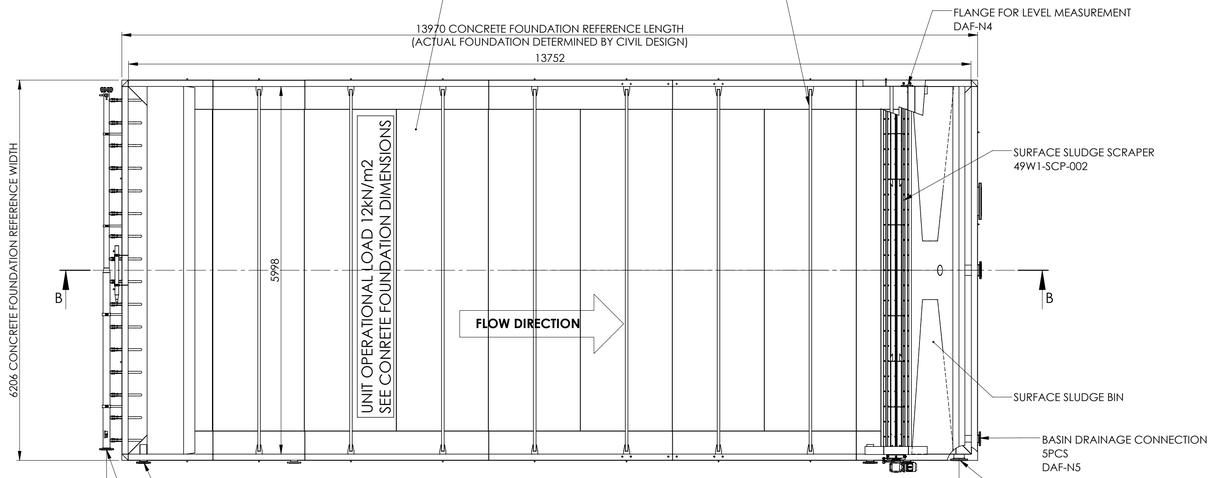
SECTION A-A

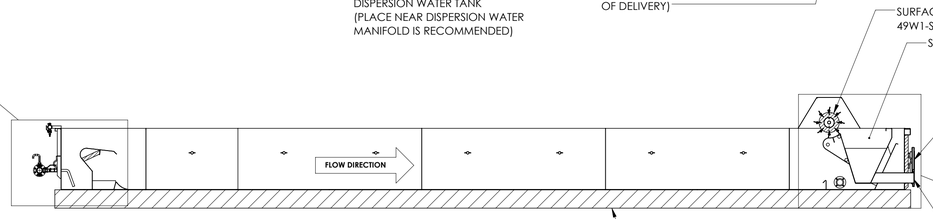


DETAIL E
SCALE 1 : 20



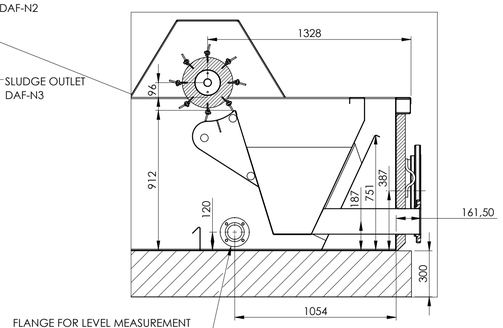






SECTION B-B

		ALL FLANGES ACCORDING ANSI B16.5	
CONNECTION	DESCRIPTION	SIZE	PN CLASS
DAF-N1			
DAF-N2	TREATED WATER OUT	16"(DN400) RF FLANGE	#150
DAF-N3	SLUDGE OUT	6"(DN150) RF FLANGE	#150
DAF-N4	LEVEL MEASUREMENT	3"(DN80) RF FLANGE	#150
DAF-N5	DRAIN	4"(DN100) RF FLANGE	#150
DAF-N6	POLYMER IN	NPT 1"(DN25) MALE THREAD	
DAF-N7	WATER IN FROM COAGULATION	TRAY FEED INLET 150x5800	
DAF-N8	DISPERSION WATER SUCTION	6"(DN150) RF FLANGE	#150
DAF-N9	DISPERSION WATER IN	4"(DN100) RF FLANGE	#150
SEE CONNECTIONS TO DISPERSION WATER TANK(DB) IN DRAWING P06174M003			



DETAIL D
SCALE 1 : 20

BASIN DRY WEIGHT 5200kg

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		01	12-08-2021	REVISED AS PER CLIENT COMMENT	MVa	NL		SL	

Flootech

FLOOTECH DRAWING NUMBER: FD074M001.01 REVISION: 01

LARSEN & TOUBRO DRAWING NUMBER: – REVISION: –

OUTOTEC DRAWING NUMBER: – – – – – DOCUMENT ID: REVISION: –

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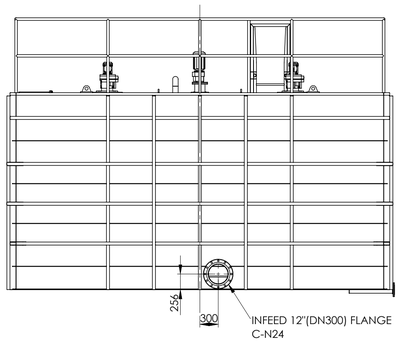
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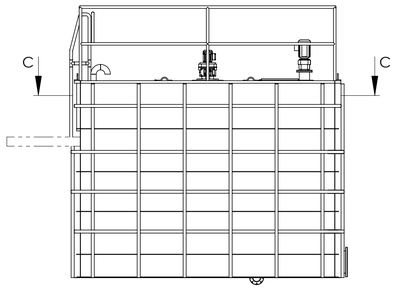
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APPROVED	PPa	2.5.2016
EMPLOYER		
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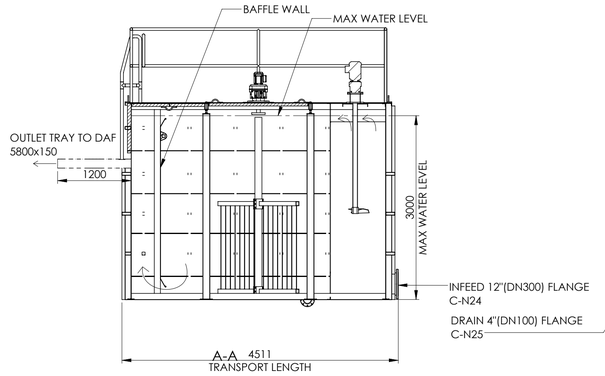
MANSOURAH - MASSARAH GOLD PROJECT
Microflotation basin FlooDAF B-74
General arrangement drawing
49W1-TNK-002

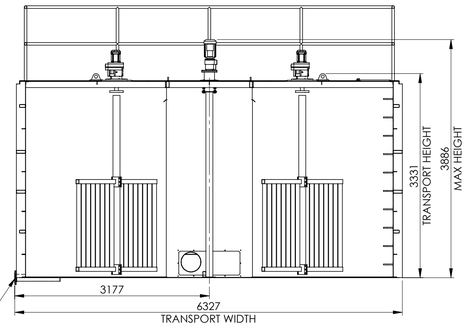
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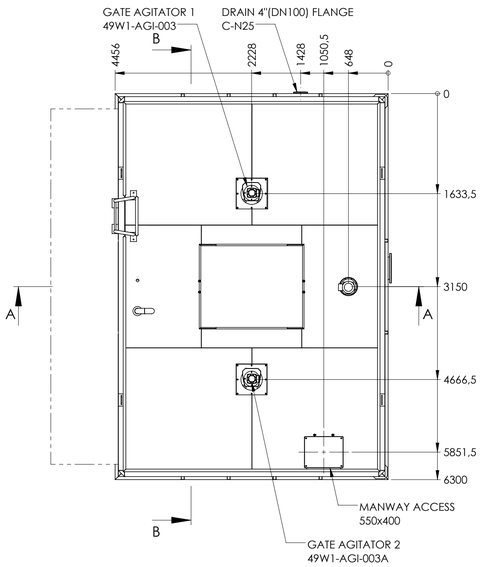


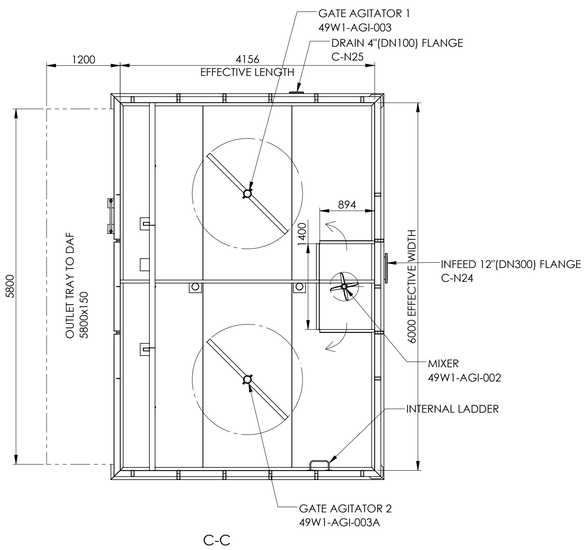




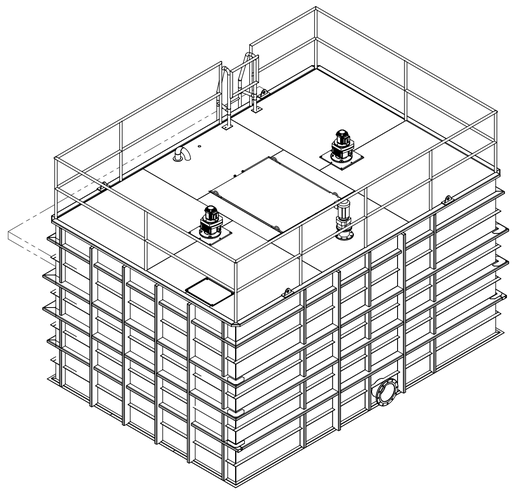


B-B





C-C



PRE-ENGINEERING DRAWINGS
- NOT FINAL -

DRY WEIGHT 6500kg
OPERATIONAL LOAD 33kN/m2

CONNECTION	EQUIPMENT	TYPE	SIZE	NOMINAL PRESSURE
C-N24	49W1-TNK-001	INLET	12" (DN300) RF FLANGE	#150
C-N25	49W1-TNK-001	DRAIN	4" (DN100) RF FLANGE	#150
C-N26	49W1-TNK-001	OUTLET	TRAY 150X5800	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		01	12-08-2021	REVISED AS PER CLIENT COMMENT	MVa	NL		SL	

Flootech

FLOOTECH DRAWING NUMBER P06204M002.00 REVISION –

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OUTOTEC DRAWING NUMBER – – – – – – DOCUMENT ID REVISION –

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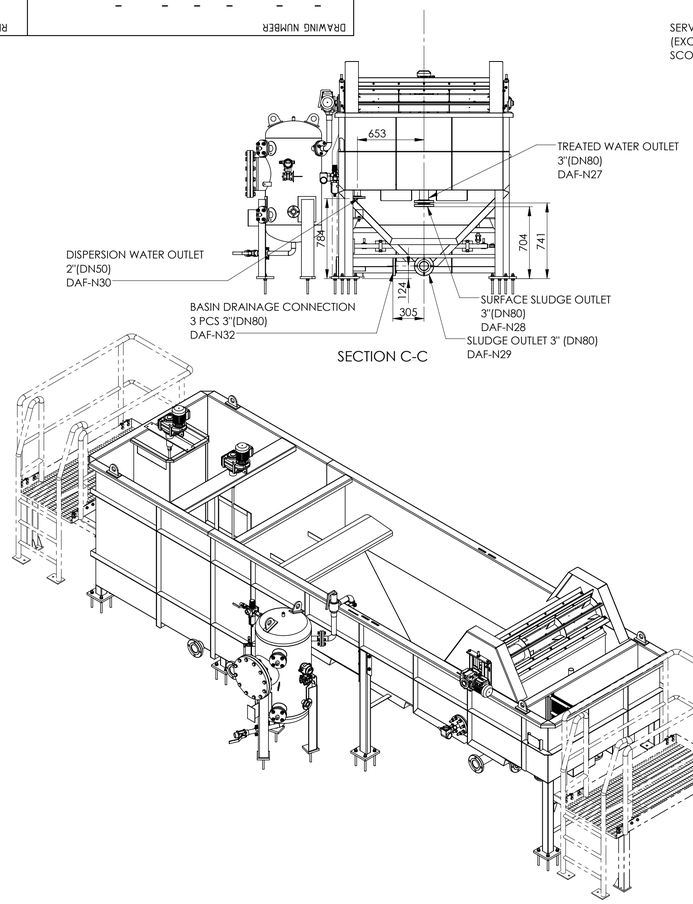
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APPROVED	SL	2.6.2021
EMPLOYER		
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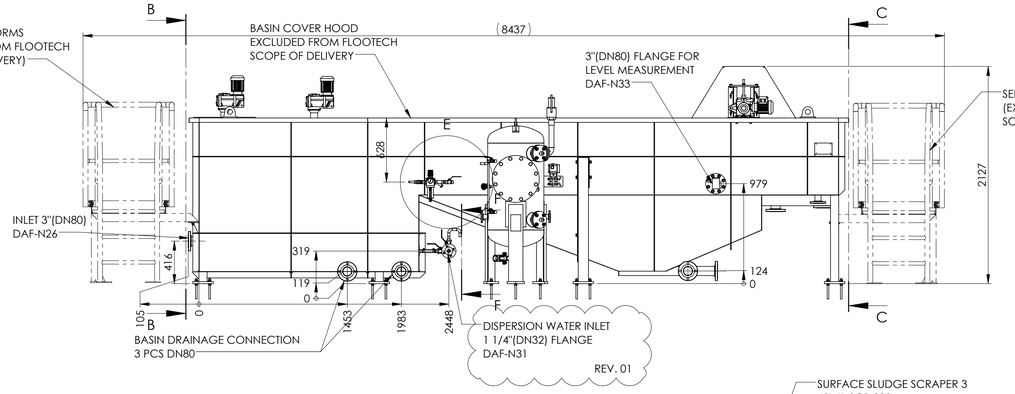
MANSOURAH - MASSARAH GOLD PROJECT
COAGULANT MIXING BASIN 75m3
General arrangement drawing
49W1-TNK-001

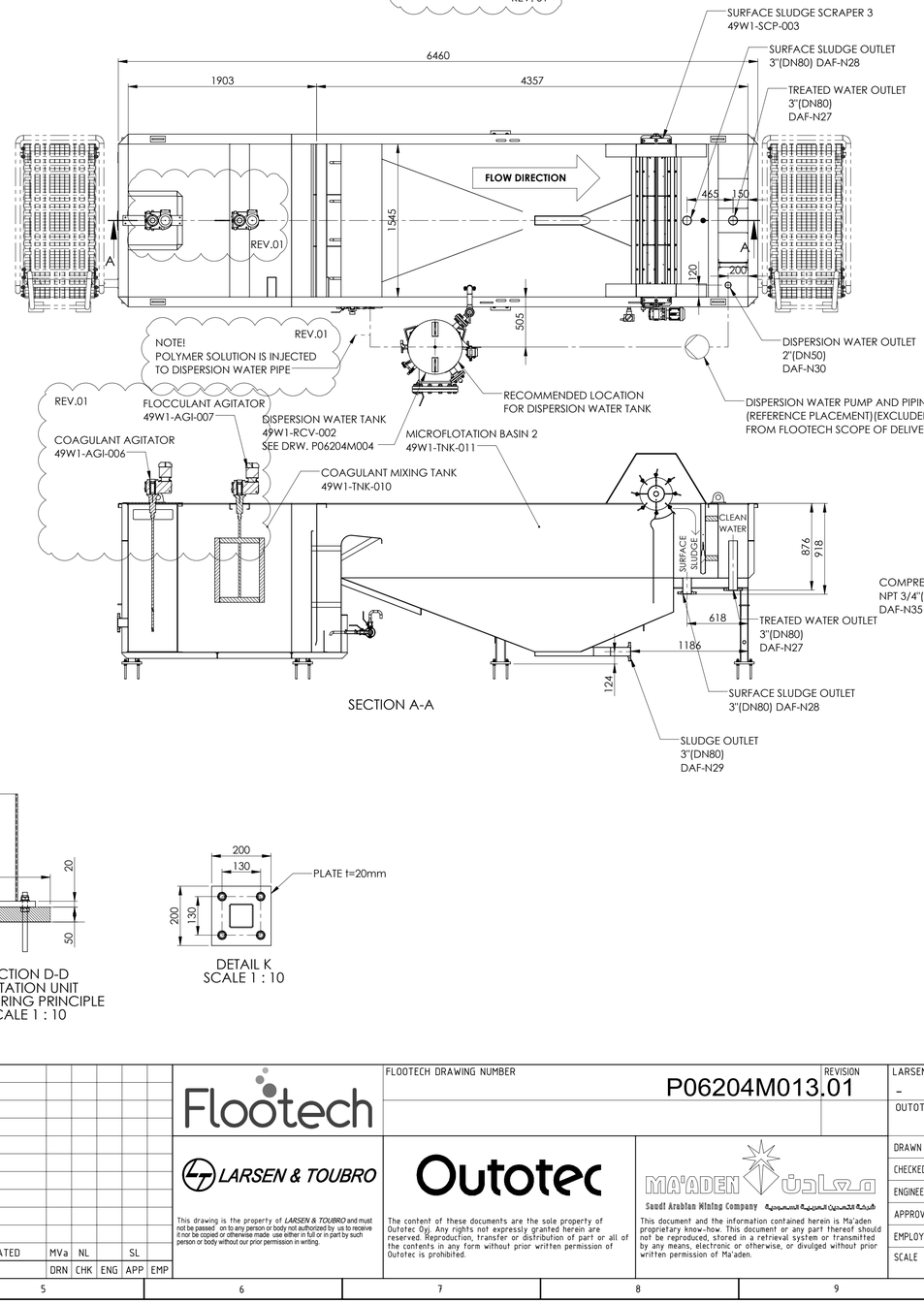
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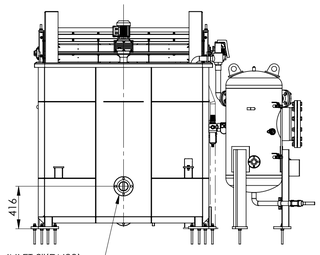


SECTION C-C

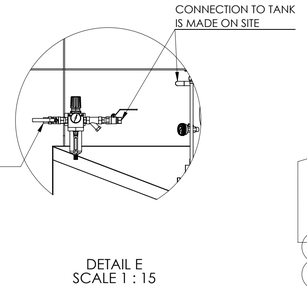




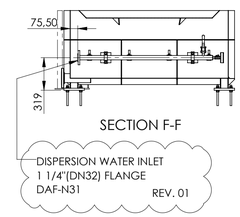
SECTION A-A



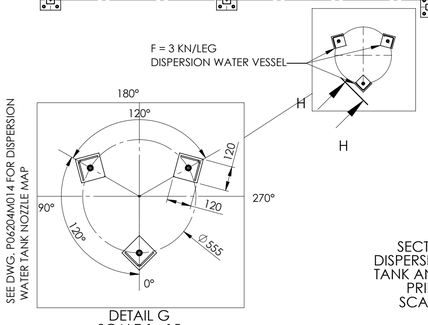
SECTION B-B



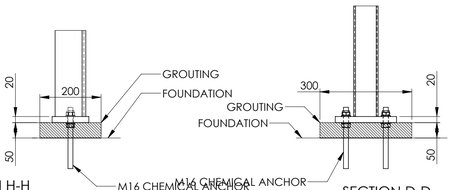
DETAIL E
SCALE 1 : 15



SECTION F-F



DETAIL G
SCALE 1 : 15



SECTION H-H
DISPERSION WATER
TANK ANCHORING
PRINCIPLE
SCALE 1 : 10

SECTION D-D
FLOTATION UNIT
ANCHORING PRINCIPLE
SCALE 1 : 10

DETAIL K
SCALE 1 : 10

ALL FLANGES ACCORDING ANSI B16.5
SEE NOZZLES FOR DISPERSION WATER TANK IN DRW. P06204M004

DAF-N26	INLET	3" (DN80) RF FLANGE	#150
DAF-N27	TREATED WATER OUTLET	3" (DN80) RF FLANGE	#150
DAF-N28	SURFACE SLUDGE OUTLET	3" (DN80) RF FLANGE	#150
DAF-N29	BOTTOM SLUDGE OUTLET	3" (DN80) RF FLANGE	#150
DAF-N30	DISPERSION WATER OUTLET	2" (DN50) RF FLANGE	#150
DAF-N31	DISPERSION WATER MANIFOLD INLET	1 1/4" (DN32) RF FLANGE	#150
DAF-N32	BASIN DRAINAGE	3" (DN80) RF FLANGE	#150
DAF-N33	FLANGE FOR LEVEL MEASUREMENT	3" (DN80) RF FLANGE	#150
DAF-N35	COMPRESSED AIR	NPT3/4" (DN20) FEMALE THREAD	#150

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		01	12-08-2021	REVISED AS PER CLIENT COMMENT, NOZZLE LIST UPDATED	MVa	NL		SL	

Flootech

FLOOTECH DRAWING NUMBER: P06204M013.01 — REVISION:

LARSEN & TOUBRO DRAWING NUMBER: – — REVISION: –

OUTOTEC DRAWING NUMBER: — — — — — DOCUMENT ID: — REVISION: –

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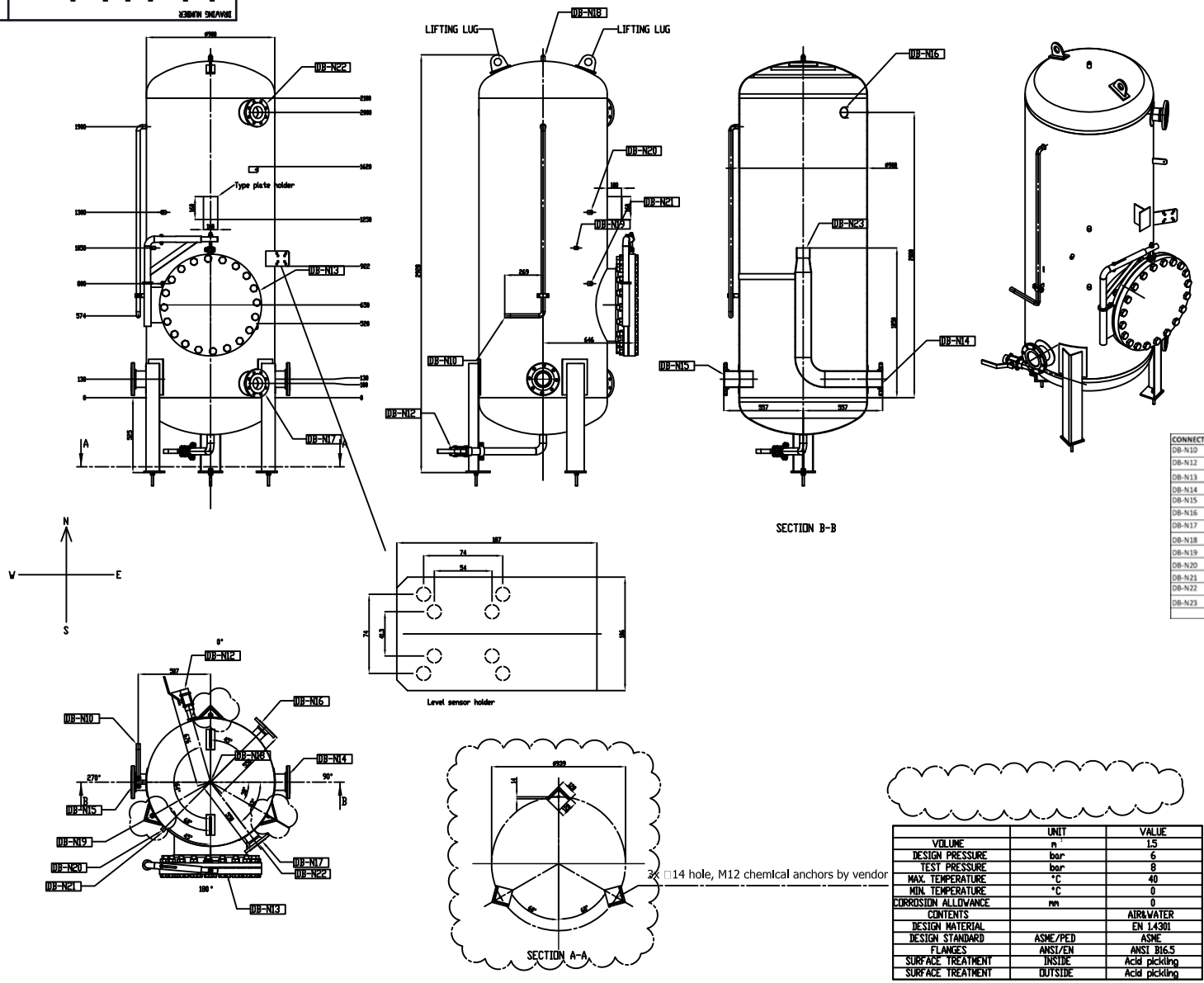
MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

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DRAWN	MVa	2.6.2021
CHECKED	MVa	2.6.2021
ENGINEER	NL	2.6.2021
APPROVED	SL	2.6.2021
EMPLOYER		
SCALE	1:30	A1

MANSOURAH - MASSARAH GOLD PROJECT
Coagulation + Microflotation FlooDAF B-5
49W1-TNK-011, 49W1-SCP-003, 49W1-TNK-010
General arrangement drawing

DRAWING NUMBER: – – – – – – — REV: –



DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	49W1-RCV-001
DESCRIPTION	Dispersio water tank
DESIGN CODE	API 650 / EN14015
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90,97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS (TOTAL)	3 mm
BOTTOM THICKNESS (TOTAL)	3 mm
CORROSION ALLOWANCE: SHELL INSIDE	-
CORROSION ALLOWANCE: BOTTOM	-
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	1114cmx1114mmx2928mm
CAPACITY: FULL VOLUME	1,5m³
CAPACITY: LIVE VOLUME	1,5m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	2000 kg
TOTAL WEIGHT (EMPTY)	500 kg
FINAL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	EN1.4404
STRUCTURALS	EN1.4404
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
PAINTING	SEE MD-216-1000-EG-ME-SPC-1007

CONNECTION	TYPE	LOCATION	SIZE	NOMINAL PRESSURE
DB-N10	COMPRESSED AIR IN	270°	NPT 3/4"(DN20) MALE THREA	#150
DB-N12	DRAIN	344°	NPT 1½"(DN40) MALE THREA	#150
DB-N13	HANDHOLE	180°	20"(DN500) RF FLANGE	#150
DB-N14	INLET	90°	4"(DN100) RF FLANGE	#150
DB-N15	OUTLET	270°	4"(DN100) RF FLANGE	#150
DB-N16	SAFETY VALVE	45°	2,5"(DN65) RF FLANGE	#150
DB-N17	LEVEL MEAS. LOWER	145°	3"(DN80) RF FLANGE	#150
DB-N18	PRESSURE TEST	AT TOP	NPT 1/2"(DN15) SOCKET	#150
DB-N19	PRESSURE GAUGE	240°	NPT 1/2"(DN15) WELDING N	#150
DB-N20	SIGHT GLASS	225°	NPT 1/2"(DN15) SOCKET	#150
DB-N21	SIGHT GLASS	225°	NPT 1/2"(DN15) SOCKET	#150
DB-N22	LEVEL MEAS. UPPER	145°	DN80 RF FLANGE	#150
DB-N23	DISPERSION NOZZLE	CENTER OF VESSEL	BSP R3" FEMALE THREAD	#150

NOTE:
1. SUFFICIENT TEMPORARY ATTACHEMENT AND SUPPORTS FOR TRANSPORTATION TO BE PROVIDED BY TANK FABRICATOR
2. NAME PLATE WITH BRACKET AND EARTHING LUGS TO BE PROVIDED BY TANK SUPPLIER.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.

	UNIT	VALUE
VOLUME	m³	1.5
DESIGN PRESSURE	bar	6
TEST PRESSURE	bar	8
MAX. TEMPERATURE	°C	40
MIN. TEMPERATURE	°C	0
CORROSION ALLOWANCE	mm	0
CONTENTS		AIR&WATER
DESIGN MATERIAL		EN 1.4301
DESIGN STANDARD	ASME/PED	ASME
FLANGES	ANSI/EN	ANSI B16.5
SURFACE TREATMENT	INSIDE	Acid pickling
SURFACE TREATMENT	OUTSIDE	Acid pickling

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING: t1, a = 0,7*min*

GENERAL: t1, amin = 2mm, a = 0,5 - 0,7*min*

EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

REV	DATE	REVISION DESCRIPTION	MVa	NL	RM	SL
			DRN	CHK	ENG	APP
01	15-10-2021	ISSUED FOR APPROVAL				

DOCUMENT NUMBER | REFERENCE DOCUMENTS

Flootech

FLOOTECH DRAWING NUMBER: P06204M003.00 | REVISION: 3.00

LARSEN & TOUBRO DRAWING NUMBER: - | REVISION: 1

OUTOTEC DRAWING NUMBER: - - - - - | DOCUMENT ID | REVISION: 1

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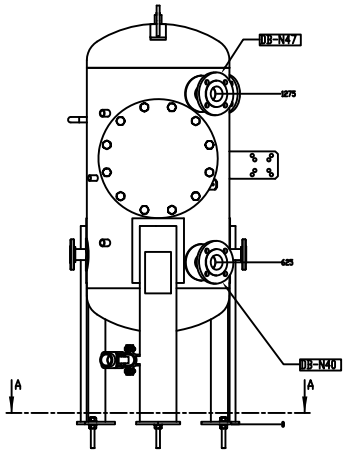
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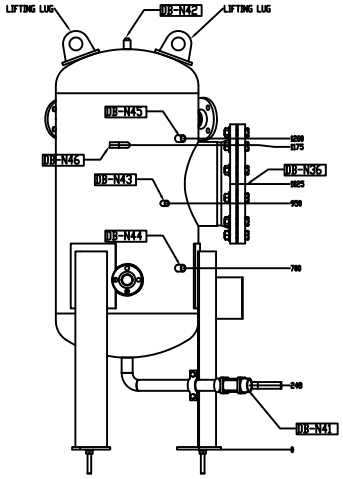
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CHECKED	Jlo	11.10.2021
ENGINEER	Jlo	11.10.2021
APPROVED	SL	11.10.2021
EMPLOYER		
SCALE	1:15	A1

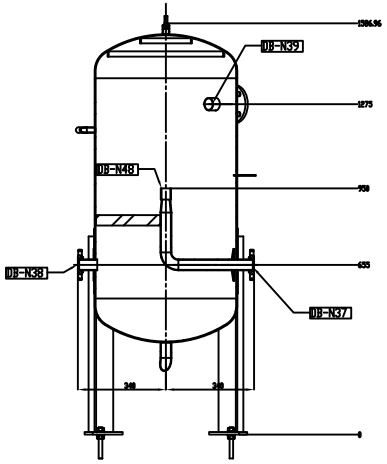
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
RAW WATER TREATMENT PLANT
49W1-RCV-001, Dispersion water Tank 1 DB-1500

DRAWING NUMBER: - - - - - - | REV: A01

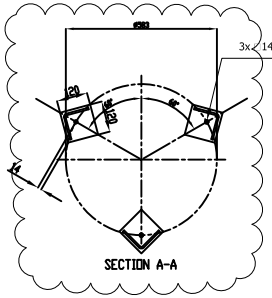
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REV: A01



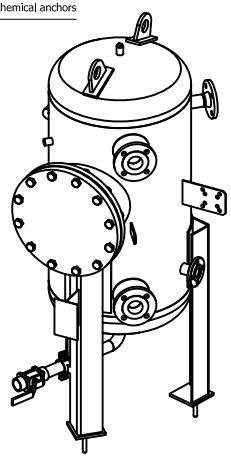


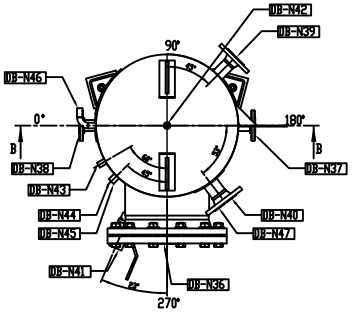


SECTION B-B



SECTION A-A





DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	49W1-RCV-002
DESCRIPTION	Dispersio water tank
DESIGN CODE	API 650 / EN14015
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS (TOTAL)	3 mm
BOTTOM THICKNESS (TOTAL)	3 mm
CORROSION ALLOWANCE: SHELL INSIDE	-
CORROSION ALLOWANCE: BOTTOM	-
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	680mmx680mmx1587mm
CAPACITY: FULL VOLUME	0.2m³
CAPACITY: LIVE VOLUME	0.2m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	500 kg
TOTAL WEIGHT (EMPTY)	300 kg
FINAL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	EN1.4404
STRUCTURALS	EN1.4404
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP2.0/4 - FE SA 2½
PAINTING	SEE MD-216-1000-EG-ME-SPC-1007

DB-N36	HANDHOLE	270°	12"(DN300) RF FLANGE	
DB-N37	INLET	180°	1 1/4"(DN32) RF FLANGE	#150
DB-N38	OUTLET	0°	1 1/4"(DN32) RF FLANGE	#150
DB-N39	SAFETY VALVE	135°	1½"(DN40) RF FLANGE	#150
DB-N40	LEVEL MEAS. LOWER	233°	2"(DN50) RF FLANGE	#150
DB-N41	DRAIN	293°	1 1/4"(DN32) RF FLANGE	#150
DB-N42	PRESSURE TEST	ABOVE MIDDLE	NPT 1/2"(DN 15) FEMALE THREAD	#150
DB-N43	PRESSURE GAUGE	330°	NPT 1/2"(DN 15) MALE THREAD	#150
DB-N44	SIGHT GLASS	315°	NPT 1/2"(DN 15) FEMALE THREAD	#150
DB-N45	SIGHT GLASS	315°	NPT 1/2"(DN 15) FEMALE THREAD	#150
DB-N46	COMPRESSED AIR IN	0°	NPT 1/4"(DN 15) MALE THREAD	#150
DB-N47	LEVEL MEAS. UPPER	233°	2"(DN50) RF FLANGE	#150
DB-N48	DISPERSION NOZZLE	CENTER OF VESSEL	BSP R1" FEMALE THREAD	#150

NOTE:
1. SUFFICIENT TEMPORARY ATTACHEMENT AND SUPPORTS FOR TRANSPORTATION TO BE PROVIDED BY TANK FABRICATOR
2. NAME PLATE WITH BRACKET AND EARTHING LUGS TO BE PROVIDED BY TANK SUPPLIER.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.

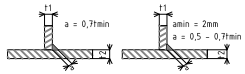
	UNIT	VALUE
VOLUME	m³	0.2
DESIGN PRESSURE	bar	6
TEST PRESSURE	bar	10
MAX. TEMPERATURE	°C	40
MIN. TEMPERATURE	°C	0
CORROSION ALLOWANCE	mm	0
CONTENTS		AIR&WATER
DESIGN MATERIAL		EN 1.4404
DESIGN STANDARD	ASME/PED	ASME
FLANGES	ANSI/EN	ANSI
SURFACE TREATMENT	INSIDE	Acid pickling
SURFACE TREATMENT	OUTSIDE	Acid pickling

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUCTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

REV.	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	ENP
01	15-10-2021	ISSUED FOR APPROVAL	NVa	NL	RM	SL	

DOCUMENT NUMBER | REFERENCE DOCUMENTS

Flootech

FLOOTECH DRAWING NUMBER: P06204M014.00 REVISION: 00

LARSEN & TOUBRO DRAWING NUMBER: - REVISION: 1

OUTOTEC DRAWING NUMBER: - DOCUMENT ID: - REVISION: 1

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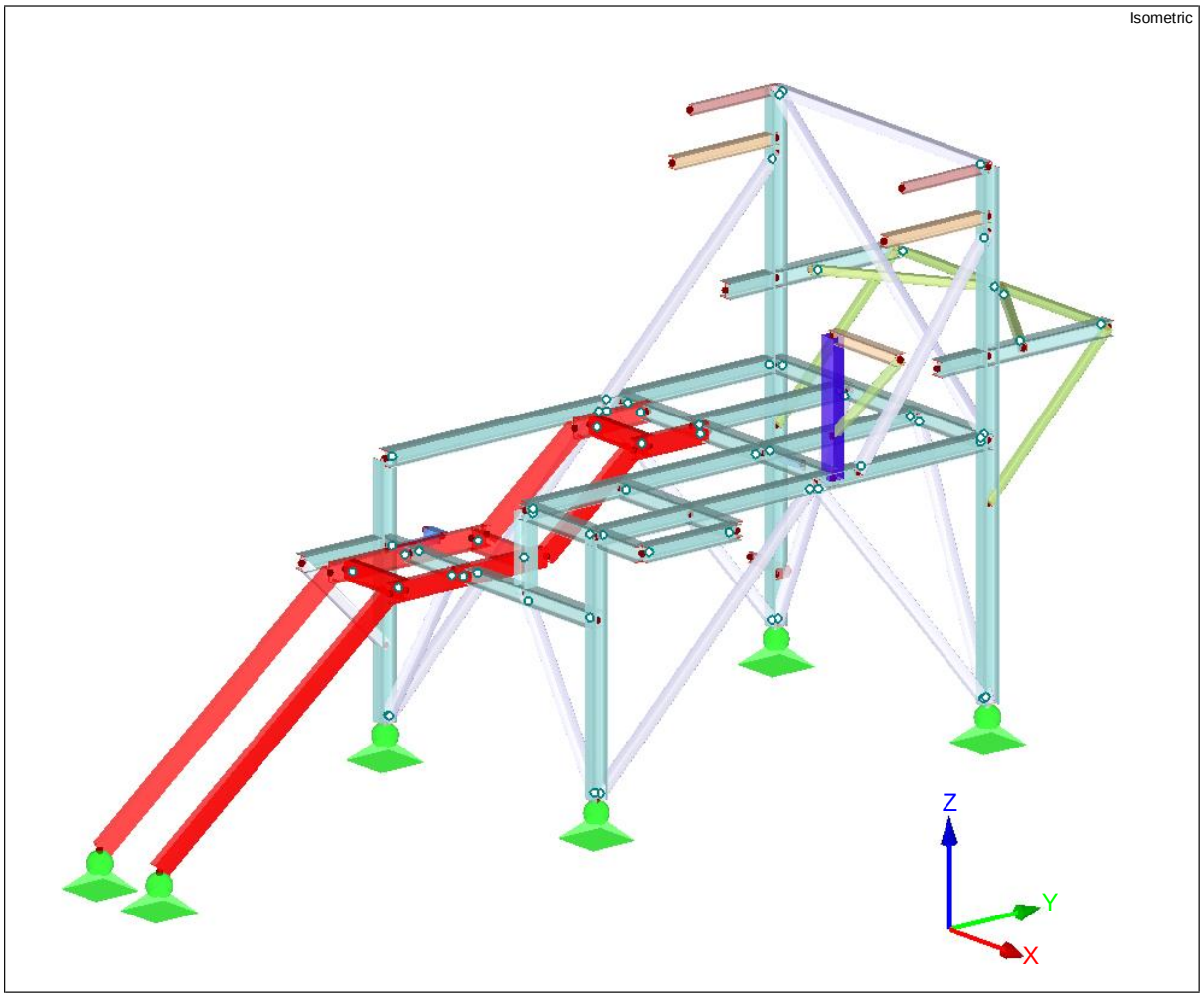
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DRAWN	NVa	26.2021
CHECKED	NVa	26.2021
ENGINEER	NL	26.2021
APPROVED	SL	26.2021
EMPLOYER		
SCALE	1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
RAW WATER TREATMENT PLANT
49W1-RCV-002, Dispersion water tank 2

DRAWING NUMBER: - REV: A01

ATTACHMENT 2 BACKWASH STAIRCASE LOAD INPUT



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Node		Support Forces [kN]			Support Moments [kNm]		
No.	LC/CO	$P_{X'}$	$P_{Y'}$	$P_{Z'}$	$M_{X'}$	$M_{Y'}$	$M_{Z'}$
5	LC1	-1.71	-0.47	-6.97	0.00	0.00	0.00
	LC2	-3.90	1.77	-10.27	0.00	0.00	0.01
	LC3	4.96	0.82	6.73	0.00	0.00	0.07
	LC4	-4.96	-0.82	-6.73	0.00	0.00	-0.07
	LC5	-0.36	7.75	5.47	0.00	0.00	-0.02
	LC6	0.36	-7.75	-5.47	0.00	0.00	0.02
	LC7	-0.13	5.54	8.10	0.00	0.00	0.02
	LC8	0.13	-5.54	-8.10	0.00	0.00	-0.02
16	LC1	-0.58	3.06	-10.81	0.00	0.00	0.00
	LC2	-1.64	5.59	-42.97	0.00	0.00	-0.01
	LC3	6.65	0.02	11.30	0.00	0.00	-0.03
	LC4	-6.65	-0.02	-11.30	0.00	0.00	0.03
	LC5	-0.05	3.56	-6.64	0.00	0.00	-0.05
	LC6	0.05	-3.56	6.64	0.00	0.00	0.05
	LC7	-0.07	3.10	-2.63	0.00	0.00	0.00
	LC8	0.07	-3.10	2.63	0.00	0.00	0.00
27	LC1	0.00	-2.18	-2.82	0.00	0.00	0.00
	LC2	0.00	-6.39	-8.53	0.00	0.00	0.00
	LC3	0.64	0.21	0.12	0.00	0.00	0.11
	LC4	-0.64	-0.21	-0.12	0.00	0.00	-0.11
	LC5	0.00	1.65	0.66	0.00	0.00	0.00
	LC6	0.00	-1.65	-0.66	0.00	0.00	0.00
	LC7	0.00	-7.85	-4.53	0.00	0.00	0.00
	LC8	0.00	7.85	4.53	0.00	0.00	0.00
40	LC1	0.00	-1.70	-2.52	0.00	0.00	0.00
	LC2	0.00	-4.40	-7.27	0.00	0.00	0.00
	LC3	0.65	0.98	0.57	0.00	0.00	0.11
	LC4	-0.65	-0.98	-0.57	0.00	0.00	-0.11
	LC5	0.00	2.25	1.00	0.00	0.00	0.00
	LC6	0.00	-2.25	-1.00	0.00	0.00	0.00
	LC7	0.00	-4.40	-2.54	0.00	0.00	0.00
	LC8	0.00	4.40	2.54	0.00	0.00	0.00
55	LC1	1.50	-2.25	-8.09	0.00	0.00	-0.02
	LC2	3.36	-8.30	-18.29	0.00	0.00	-0.04
	LC3	4.93	-1.64	-7.86	0.00	0.00	-0.03
	LC4	-4.93	1.64	7.86	0.00	0.00	0.03
	LC5	0.38	8.97	7.62	0.00	0.00	-0.02
	LC6	-0.38	-8.97	-7.62	0.00	0.00	0.02
	LC7	-0.40	1.87	3.52	0.00	0.00	-0.09
	LC8	0.40	-1.87	-3.52	0.00	0.00	0.09
67	LC1	0.79	3.53	-11.38	0.00	0.00	0.00
	LC2	2.18	11.73	-50.93	0.00	0.00	-0.01
	LC3	6.36	-0.39	-10.86	0.00	0.00	-0.03
	LC4	-6.36	0.39	10.86	0.00	0.00	0.03
	LC5	0.03	3.87	-8.11	0.00	0.00	0.00
	LC6	-0.03	-3.87	8.11	0.00	0.00	0.00
	LC7	0.60	1.74	-1.92	0.00	0.00	-0.03
	LC8	-0.60	-1.74	1.92	0.00	0.00	0.03

91	LC2	3.62	-0.18	-31.21	0.00	0.00	0.00
	LC3	3.13	0.17	16.13	0.00	0.00	0.21
	LC4	-3.13	-0.17	-16.13	0.00	0.00	-0.21
	LC5	-0.01	4.44	26.69	0.00	0.00	-0.41
	LC6	0.01	-4.29	-24.96	0.00	0.00	0.41
	LC7	-1.52	-1.48	0.03	0.00	0.00	0.00
	LC8	1.52	1.48	-0.03	0.00	0.00	0.00
99	LC2	0.32	0.02	-6.91	0.00	0.00	0.00
	LC3	2.60	0.44	-3.76	0.00	0.00	0.05
	LC4	-2.60	-0.44	3.76	0.00	0.00	-0.05
	LC5	0.38	6.44	-26.68	0.00	0.00	-0.13
	LC6	-0.36	-6.15	24.95	0.00	0.00	0.13
	LC7	-0.49	1.71	-0.03	0.00	0.00	0.00
	LC8	0.49	-1.71	0.03	0.00	0.00	0.00
112	LC2	-3.94	0.16	-16.24	0.00	0.00	0.00
	LC3	6.38	-0.62	-16.13	0.00	0.00	0.07
	LC4	-6.38	0.62	16.13	0.00	0.00	-0.07
	LC5	-0.25	5.70	9.83	0.00	0.00	0.27
	LC6	0.24	-5.70	-9.83	0.00	0.00	-0.27
	LC7	1.98	-0.24	-0.03	0.00	0.00	0.00
	LC8	-1.98	0.24	0.03	0.00	0.00	0.00
118	LC2	-0.01	0.00	-4.16	0.00	0.00	0.00
	LC3	0.10	0.00	3.76	0.00	0.00	0.06
	LC4	-0.10	0.00	-3.76	0.00	0.00	-0.06
	LC5	-0.11	0.04	-9.83	0.00	0.00	-0.10
	LC6	0.11	-0.04	9.84	0.00	0.00	0.10
	LC7	0.03	0.01	0.03	0.00	0.00	0.01
	LC8	-0.03	-0.01	-0.03	0.00	0.00	-0.01
122	LC2	0.00	-0.19	-4.01	0.21	0.01	0.00
	LC3	1.22	0.00	0.01	0.02	1.93	-0.04
	LC4	-1.22	0.00	-0.01	-0.02	-1.93	0.04
	LC5	0.00	2.50	3.26	-0.59	0.01	0.00
	LC6	0.00	-2.50	-3.26	0.59	-0.01	0.00
	LC7	-0.01	-0.41	-0.45	0.02	-0.03	0.02
	LC8	0.01	0.41	0.45	-0.02	0.03	-0.02
124	LC2	-0.14	0.18	-3.41	-0.21	-0.12	0.00
	LC3	0.32	-0.02	3.73	0.03	0.19	0.00
	LC4	-0.32	0.02	-3.73	-0.03	-0.19	0.00
	LC5	-0.01	0.63	-3.23	-0.56	-0.01	0.00
	LC6	0.01	-0.63	3.23	0.56	0.01	0.00
	LC7	-0.40	0.40	0.81	-0.73	-0.48	0.00
	LC8	0.40	-0.40	-0.81	0.73	0.48	0.00
127	LC2	-0.34	0.00	-4.72	0.00	-0.28	0.00
	LC3	0.61	0.00	4.29	0.00	0.54	0.00
	LC4	-0.61	0.00	-4.29	0.00	-0.54	0.00
	LC5	0.00	0.73	0.00	-1.00	0.00	0.00
	LC6	0.00	-0.73	0.00	1.00	0.00	0.00
	LC7	-0.54	0.00	0.64	0.00	-0.69	0.00
	LC8	0.54	0.00	-0.64	0.00	0.69	0.00
129	LC2	-0.34	0.00	-4.72	0.00	-0.28	0.00

	LC3	0.61	0.00	4.29	0.00	0.54	0.00
	LC4	-0.61	0.00	-4.29	0.00	-0.54	0.00
	LC5	0.00	0.73	0.00	-1.00	0.00	0.00
	LC6	0.00	-0.73	0.00	1.00	0.00	0.00
	LC7	-0.54	0.00	0.64	0.00	-0.69	0.00
	LC8	0.54	0.00	-0.64	0.00	0.69	0.00
131	LC2	0.14	0.00	-3.58	-0.01	0.14	-0.01
	LC3	2.58	0.01	-3.74	-0.03	0.50	0.00
	LC4	-2.58	-0.01	3.74	0.03	-0.50	0.00
	LC5	0.00	0.89	-0.03	-1.01	-0.04	-0.04
	LC6	0.00	-0.89	0.03	1.01	0.04	0.04
	LC7	0.41	0.02	-0.36	-0.06	0.04	-0.02
	LC8	-0.41	-0.02	0.36	0.06	-0.04	0.02
133	LC2	0.34	0.00	-5.08	0.00	0.11	0.00
	LC3	2.28	0.00	-4.29	0.00	0.62	0.00
	LC4	-2.28	0.00	4.29	0.00	-0.62	0.00
	LC5	0.00	0.87	0.00	-1.09	0.00	-0.06
	LC6	0.00	-0.87	0.00	1.09	0.00	0.06
	LC7	0.54	0.00	-0.64	0.00	0.11	0.00
	LC8	-0.54	0.00	0.64	0.00	-0.11	0.00
135	LC2	0.34	0.00	-5.08	0.00	0.11	0.00
	LC3	2.28	0.00	-4.29	0.00	0.62	0.00
	LC4	-2.28	0.00	4.29	0.00	-0.62	0.00
	LC5	0.00	0.87	0.00	-1.09	0.00	-0.06
	LC6	0.00	-0.87	0.00	1.09	0.00	0.06
	LC7	0.54	0.00	-0.64	0.00	0.11	0.00
	LC8	-0.54	0.00	0.64	0.00	-0.11	0.00

ATTACHMENT 2 COMMENT RESOLUTION SHEET

When comments are received from Employer, they need to be placed in a comment resolution table, and the filled table is always attached to the end of the document.

COMMENT TABLE	Document Name:	Document Number:	Revision No.	Comment Date:
General Comment:				

Sr. No	Chapter	Regarding	Comment	Resolution
1	6.2.1.2	load value	Please show how this reactions have been calculated	added, load acquired from steel structure designer
2	6.3	max settlement	Please indicate the reference where the criteria is.	added
3	7.3	max settlement	Please indicate the reference where the criteria is.	added

1 Remember also to always mark the made changes with a triangle to the left side of the document.

You can also reply to the comments on an excel template, then it will be coded as a separate document and submitted to Employer together with the revised document itself.

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You can find the instruction and the excel template from “Document Codes and Numbering Procedure” [1].

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